

Bangladesh University of Engineering and Technology

IPE 300: Industrial Practice

Kohinoor Chemicals Co. (Bd) Ltd.

Submitted to:

Dr. Abdullahil Azeem

Professor, Industrial and Production Engineering, BUET.

Dr. Nafisa Mahbub

Assistant Professor, Department of Industrial and Production Engineering, BUET.

Engr. Md. Hamidur Rahman

Vice President, Engineering Department, Kohinoor Chemicals Co. (Bd) Ltd.

Submitted by:

1808011-Arnab Kumar Debnath

1808013-Afshana Mimi Jhinuk

1808018-Mostafizur Rahman Khan

1808020-Rumeen Chowdhury

1808031-Anika Tahsin

180841-Md. Tanvir Ahmmmed Fahim

Industrial and Production Engineering, BUET.

CERTIFICATE OF APPROVAL

This report with the title “Industrial Practice at Kohinoor Chemicals Company (Bangladesh) Ltd” submitted by Arnab Kumar Debnath (1808011), Afshana Mimi Jhinuk (1808013), Mostafizur Rahman Khan (1808018), Rumeen Chowdhury (1808020), Anika Tahsin (1808031) and Md. Tanvir Ahmmed Fahim (1808041) has been deemed acceptable under the course “Industrial Practice” having the course number “IPE 300”.

Approved by:

Academic Supervisors:

Dr. Abdullahil Azeem

Professor, Department of IPE, BUET.

Dr. Nafisa Mahbub

Assistant Professor, Department of IPE, BUET.

Industrial Supervisor:

Md. Hamidur Rahman

Vice President, Engineering Department,

Kohinoor Chemical Company (Bangladesh) Limited.

DECLARATION

We, hereby, state that this report titled “Industrial Attachment at Kohinoor Chemical Company (Bangladesh) Limited” is produced using the information collected during the industrial attachment on the factory premises from 6 May 2023 to 8 June 2023. The elements that are included in this report are solely for the purpose of research and we, the authors, pledge not to use this data without the prior permission of the company.

Authors:

Arnab Kumar Debnath

Student ID: 1808011

Afshana Mimi Jhinuk

Student ID: 1808013

Mostafizur Rahman Khan

Student ID: 1808018

Rumeen Chowdhury

Student ID: 1808020

Anika Tahsin

Student ID: 1808031

Md. Tanvir Ahmmed Fahim

Student ID: 1808041

FORWARDING LETTER

12 August 2023

Dr. Abdullahil Azeem

Professor

Dr. Nafisa Mahbub

Assistant Professor

Department of Industrial and Production Engineering

Bangladesh University of Engineering and Technology.

Subject: Report submission on “Industrial Practice at Kohinoor Chemical Company (Bangladesh) Limited”

Dear Teachers,

We are honored to present the report of our industrial practice experience at Kohinoor Chemical Company (Bangladesh) Limited. We got the opportunity to gain knowledge and understanding of the various facets of a prestigious industry and increase our practical expertise.

Therefore, we would like to express our gratitude for your generous support and guidance throughout the visit and the preparation of this report. We sincerely apologize for any unintentional errors.

Sincerely,

Arnab Kumar Debnath (Student ID: 1808011)

Afshana Mimi Jhinuk (Student ID: 1808013)

Mostafizur Rahman Khan (Student ID: 1808018)

Rumeen Chowdhury (Student ID: 1808020)

Anika Tahsin (Student ID: 1808031)

Md. Tanvir Ahmmed Fahim (Student ID: 1808041)

Department of Industrial and Production Engineering

Bangladesh University of Engineering and Technology

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ABSTRACT

In this report, we present the knowledge that we gained from the industrial practice in detail. We could learn how to apply our theoretical knowledge in an industry. In the initial chapters, we discussed the various processes and equipment of the industry. We analyzed the processes and their management, maintenance of machines, sampling, collection of data, inventory management etc. to understand how efficient the processes are and if they could be improved. The consequences of the analysis demonstrate the opportunity to increase inventory space by utilizing the vertical space in their warehouse. We have designed a platform that will utilize vertical space and help in loading/unloading the goods efficiently. Moreover, errors in manual loading/unloading can be decreased by using different colors for different categories of inventory. Additionally, we have created graphs that will allow them to see whether the various important parameters are in control.

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Chapter 1: Introduction

1.1 History of Kohinoor Chemical Company (Bangladesh) Limited

Kohinoor Chemical Company (Bangladesh) Limited (KCCL) is the oldest and largest national soap, cosmetics and toiletries manufacturer and marketer in Bangladesh. KCCL started its business in 1956. After the independence of Bangladesh, KCCL became a fully government-owned company under the direct control of Bangladesh Chemical Industries Corporation (BCIC) up to May 05, 1988. It was transformed into a Public Limited Company under the Companies Act 1913 and was named as Kohinoor Chemical Company (Bangladesh) Limited from May 05, 1988. It is currently situated at 36, Shahid Tajuddin Ahmed Sarani Tejgaon I/A, Dhaka-1208.

1.2 Mission, Vision, and Values of Kohinoor Chemical Company (Bangladesh) Limited

1.2.1 Mission

The goal of Kohinoor Chemical Company (Bangladesh) Limited is to enhance the lives of the people through the highest quality protocol. They place a high importance on society in general and their customers in particular. They continue to roll out newer technologies while maintaining the most beneficial incentive and benefit programs for all retail chains. Additionally, they are developing a price strategy that embodies a special balance between affordability and quality. Moreover, their comprehensive strategy framework concerning the supply chain across the entire country enables them to be naturally conferred with the prestige and privilege of leadership.

1.2.2 Vision

Through innovation, commitment, and ethics, Kohinoor seeks to create a better, healthier, and cleaner way of life and also to ensure that the inherent qualities of life are sustained. This keeps inspiring them to improve the world by making everything they do more valuable so that people can look better, feel better, and live better. Kohinoor consistently works to demonstrate the greatness of their products and care for their valued customers and stakeholders in an effort to win their unwavering allegiance.

1.2.3 Values

Quality of the products, reliability, and strong distribution channel are the core values of the organization and are the basis on which they do business.

1.3 Business Ethics & Code of Conduct of Kohinoor Chemical Company (Bangladesh) Limited

Business ethics are moral principles that guide the way a business behaves. A code of ethics specifies the ethical rules of operation. It is applicable from members of the Board of Directors to all employees of Kohinoor Chemical Company (Bangladesh) Limited. Ethics in the workplace sensitizes employees to how they should act and helps retain a strong moral compass when leaders and managers are struggling due to crisis and confusion. In compliance with the new Corporate Governance Code imposed by BSEC, the existing code of conduct for the Board members including the Chairman and the Managing Director/Chief Executive Officer of the company is under review by the newly formed Nomination & Remuneration Committee to conform to the above-mentioned Corporate Governance Code. After finalizing the said code of conduct, the same shall be posted on the website of the company.

Compliance with laws, rules & regulations:

Complying with the prevailing laws, rules and regulations and being in conformity with social norms are the basics of their business. Nationally and internationally wherever business is done, they obey the relevant laws and regulations and avoid any action that can discredit the company.

- **Accountable & responsible**

KCCL takes full responsibility for all their business decisions, actions and conduct and is accountable to the stakeholders. Their accountability and responsibility show the most effective way to achieve high standards.

- **Keeping and showing accurate data**

The business information that they work with is vital to their success. They keep the data in a well-organized and easily accessible manner so that it can give them important knowledge for prudent business decisions, playing a crucial role in business growth strategies and financial reporting. They use all of this information wisely and carefully to get the best value from it.

- **Protection of data and assets**

Information is a vital asset of a company. Information and data which is not in the public domain is considered confidential and proprietary information. Customers, suppliers and others also may share their confidential and proprietary information with us. They are conscious and committed to other assets of their company to protect them from misappropriation, loss and destruction. Moreover, they consider it their duty to properly use, and safeguard business information belonging to and shared with the company.

- **Business integrity and honesty**

Integrity and honesty are key roles in building a strong, solid and successful business. They try to be truthful both in communication and in conduct.

- **Respecting people and upholding their interest**

People are the greatest asset if they can be utilized properly. They provide new ideas and concepts that can be given in practical form. So KCCL maintains their workplace so that the employees can feel valued and respected. They do not engage in any conduct that could be considered disrespectful, harassing, intimidating and aggressive.

- **Best sales and marketing practices**

KCCL believes in fair competition and avoids unhealthy rivalry with competitors which might lead to any unethical conduct. Marketing and promotion are carried out for the well-being of the society.

Chapter 2: Product Portfolio

Kohinoor Chemicals produce a variety of products, namely soap, detergent as well as various cosmetics for summer, winter etc. They are mentioned below:

2.1.1 Soap

Soap is sodium or potassium salt of higher fatty acid. In this factory, 2 types of soap are produced, and they are as follows:

2.1.1.1 Toilet soap

Toilet soap is used for bathing and washing oneself. This type of soap contains more TFM (Total Fatty Matter) and less moisture compared to laundry soap. They produce various toilet soaps, as given below:

Sandalina

Sandalina is one of the most in-demand luxury soaps of this company. It comes in 2 variants - Sandal & Moisturizer and Sandal & Rose. They come in different sizes - Sandal & moisturizer comes in 75gm, 100 gm and 125 gm sizes whereas Sandal & Rose comes in 75gm, 100gm and 150gm. Sandalina soap is blended with 100% pure quality sandalwood extracts. It is a grade-1 soap that has a TFM of 80% and is enriched with balanced moisturizer. This soap is purely made of vegetable fat. It helps in removing facial stains, freckles, pimples, pimple marks and protects skin from itching & irritation due to dryness. It also makes the skin soft and smooth and keeps the skin fresh.

Tibet Luxury Soap

This comes in 2 variants- Pink (Natural Glow) enhanced with Jojoba Oil & Vitamin E and White (Smooth Care) comprising Milk Cream with Chamomile & Sweet Almond. They come in 4 different sizes: 35 gm, 50 gm, 75 gm & 100 gm. It is a high-quality soap made of 100% vegetable fat and enriched with a moisturizing complex. It helps enhance skin smoothness and glow.

Bactrol Soap

This soap is formulated with Active Symdeo Plus and is made from 100% vegetable fat. It has a TFM content of 74%. It is enhanced with antibacterial agents, chilling menthol, moisturizing complex, and fragrance. It ensures great protection against bacteria. It is effective against body odor, cuts, scratches, and pimples. Bactrol Antibacterial Hand Wash works as a defense against the spread of many diseases starting from common cold to more serious diseases such as meningitis, bronchitis, influenza, hepatitis A and most types of infectious diarrhea caused by hidden germs in hands. It has 2 variants which are the Fresh variant and the Cool & Antibacterial variant. It is found as 35 gm and 100 gm soaps, and as hand-wash of 180 ml and 200 ml pouches.



Figure 2.1.1.1: Different types toilet soap

2.1.1.2 Laundry Soap

Laundry soaps are used to wash clothes. They have a lower TFM, and higher moisture content compared to Toilet Soaps. They are as follows:

Tibet 570 Laundry Soap

This is one of the most in-demand laundry soaps of this company. It comes in one size - 125 gm. It is glycerin and EDTA enriched and free of silicate. Glycerin is a humectant which retains moisture and hence keeps skin moisturized and smooth. EDTA is an anti-cracking agent which prevents soap from being cracked after washing several times. It is good for clothes.

Tibet Laundry Soap

It is available in 2 variants - green and blue variant. It comes in 1 size - 125 gm. It is glycerin and EDTA enriched. Glycerin is a humectant which retains moisture and hence keeps skin moisturized and smooth. EDTA is an anti-cracking agent which prevents soap from being cracked after washing several times. Free alkali is balanced here. This soap makes clothes bright and there is no need for separate whiteners.

Tibet Ball Soap

It is glycerin, EDTA and color fixative enriched. Glycerin is a humectant which retains moisture and hence keeps skin moisturized and smooth. EDTA is an anti-cracking agent which prevents soap from being cracked after washing several times. Free alkali is balanced here. It has a blend of optical brighteners which are used to brighten and protect clothes. It contains anti-cracking agents which keep the soap compact. It is available in 125 gm units.



Tibet 570 Laundry Soap



Tibet Laundry Soap



Tibet Ball Soap

Figure 2.1.1.2: Different types of laundry soap

2.1.2 Cosmetics

KCCL produces various cosmetics for skin care, hair care, oral care, etc. They are shown below:

2.1.2.1 Oral Care

AM.PM Toothpaste: This comes in 3 sizes - 45 gm, 100 gm and 200 gm. It is enriched with herbal ingredients like Basil, Eucalyptus, Clove, orange, and anise etc. Herbal ingredients keep the gum healthy and strengthen the enamel. It provides strong protection against cavities and fights germs, plaque & tooth decay.



Figure 2.1.2.1: Oral care products

2.1.2.2 Summer Care

Tibet Luxury Talcum Powder

It is enriched with sun-blocking agents and has antiperspirant agents. Besides, it contains a world-famous fragrance from Switzerland. It keeps the body cool, protects the skin from harmful UV rays of the sun and provides protection against body odor. It comes in 3 different sizes: 50 gm, 100 gm and 200 gm bottles.

Tibet Prickly Heat Powder

It is enriched with Irgason which is a deodorant and Boric powder. Besides, it contains antiperspirant agents. It helps treat and prevent rashes and chafing. Moreover, it soothes heat irritation quite instantly by absorbing excess perspiration. It keeps one away from bacteria, fungal agents and body odor all day long. It is sold in 100 gm bottles.

Ice Cool Prickly Heat Powder

It is enriched with Irgason which is a deodorant and Boric powder. Besides it contains antiperspirant agents and natural cooling agents. It gives one relief from the itching heat rash and skin irritation in hot weather. The triple action formula provides a cooling sensation to refresh and relax one's body and mind from unbearable heat and stress. It protects one from bacteria, fungal agents, and body odor all day long.



Tibet Luxury Talcum Powder



Prickly heat powder



Ice Cool Prickly Heat

Figure 2.1.2.2: Different types of Summer Care products

2.1.2.3 Home Care

Xpert Dishwash

It comes in different variants and SKUs: Xpert Dishwash Liquid- 250 ml & 500 ml. Xpert Dishwash Bar- 100 & 300 gm. Xpert Dishwash Powder- 200, 500 & 750 gm. It has antibacterial and deep cleansing agents. It gives faster cleaning action against grease, oil, fat, stains and other deposits without any scratches. It kills bacteria on contact and leaves crockery hygienically clean.

Clean Master Liquid Toilet Cleaner

This toilet cleaner is enriched with Ultra Power Gel which removes tough stains, kills germs & deodorizes the toilet within a very short time. It comes in 500 ml and 750 ml containers.

Wiper Glass Cleaner

It can be used in both spray and stream form. It contains antibacterial agents. It cleans dust and also kills germs. Moreover, it ensures clear visibility. It comes in 350 ml bottles (pump and refill).



Xpert Dishwash



Clean Master Liquid Toilet Cleaner



Wiper Glass Cleaner

Figure 2.1.2.3: Different types of Home Care products

2.1.2.4 Skin Care

Beautina Face Wash

It comes in 2 sizes: 50 ml and 100 ml. It is made up of 100% herbal actives. It is enriched with Neem and Aloe Vera extracts. It is good for pimple control, oil control and deep cleansing. The antibacterial properties of Beautina face wash ensure healthy glowing skin by protecting from pimples, oiliness, and sunburn.

Beautina Body Lotion

It is enriched with Aloe Vera extract and Cocoa Butter. Aloe Vera helps to make the skin soft, tender, and fresh. Cocoa Butter gently helps to keep body moisture all day long. It has a sweet fragrance. It comes in 2 sizes: 100 ml and 200 ml.

Tibet Snow

It comes in 50 gm bottles and 50 gm tubes. It is enriched with anti-bacterial, antifungal agents and Emolin. It contains an extra whitening agent. It is an ideal skin care cream for all seasons. It keeps skin soft and smooth and makes skin bright and fair. It can also be used as a makeup base.



Beautina Face Wash



Beautina Body Lotion



Tibet Snow

Figure 2.1.2.4: Different types of Skin Care products

2.1.2.5 Perfumery

It is 100% alcohol-free and Halal. It is made specially from Sandalwood extracts. Its fragrance lasts all day long. Besides, it doesn't contain allergic elements. It comes in 3.5 ml bottles.



Tibet Chandan Attar

Figure 2.1.2.5: Perfume products

2.1.2.6 Winter Care

Tibet Petroleum Jelly

This product is made from high quality Jojoba oil, Vitamin E and anti-cracking agent. It also contains fragrance and moisturizer. It prevents dry and cracked skin and keeps skin fresh. It is sold in 15 ml and 50 ml bottles.

Tibet Pomade

It is made from premium quality rose oil, pure petroleum jelly, mineral oils and a refreshing fragrance. It protects the skin from dryness and cracking. It also cures cracked lips, ankles and feet. It is available in 50 ml bottles.

Fruity Lip Balm

This balm contains cocoa butter, jojoba oil, and vitamins C and E. It is enhanced with a fresh fruity fragrance. Fruity lip balm protects lips from dry and cold weather. The botanical extracts prevent collagen loss and flatten fine lines on lips, which keeps one's lips smooth. It is sold in 4.5 gm containers.

Tibet Lip Gel

This product is made of pure petroleum jelly, mineral oils, cocoa butter, extra moisturizing and emollient agents. It prevents the drying of lips in the winter. It is available in 10 gm tubes.

Tibet Glycerin

It prevents skin from dryness and improves complexion, resulting in skin softness and smoothness. It is enriched with Emolin and anti-cracking agents. It is sold in 60 gm and 125 gm bottles.



Tibet Petroleum Jelly



Tibet Lip Gel



Fruity Lip Balm



Tibet Pomade



Tibet Glycerin

Figure 2.1.2.6: Different types of winter care products

2.1.2.7 Hair Care

Tibet Pumpkin Hair Oil

It is one of the most demanded cosmetic products of the company. It consists of 7 herbal oils. It is enriched with Drago Botania Plus and multivitamins. It nourishes hair and keeps the head cool. It is available in 50 ml, 100 ml and 200 ml bottles.

Beautina Hair Oil

This oil is enriched with Vitamin E, natural and herbal extracts of Amla, Almond, Olive, Jojoba, Tea Tree and Coconut. It ensures complete hair therapy by giving 5 benefits to the hair. It controls hair fall, treats the scalp, controls dandruff, and makes hair light, soft, shiny & silky.



Beautina Hair Oil



Tibet Pumpkin Hair Oil

Figure 2.1.2.7: Different types of Hair Care products

2.1.2.8 Personal Care

Tibet Lather Shaving Cream

It has a unique combination of antiseptic and smoothening cream that guards the facial skin against infections and generates non-irritating foam to provide enough moisture for smooth shaving. It is enriched with antiseptic, moisturizing and cooling agents. It is available in 50 gm tubes.

Genstar Lather Shaving Cream

It is enriched with antiseptic, moisturizing and cooling agents. It has a special beard-softening formula that gives the coolest shave and leaves one's face fresh for hours after shaving. It reduces friction to make the shave comfortable. Genstar Lather Shaving Cream improves the performance of one's razor/blade. It is sold in 100 gm tubes.



Genstar Lather Shaving Cream



Tibet Lather Shaving Cream

Figure 2.1.2.8: Different types of personal Care products

2.1.3 Detergent

A detergent is a surfactant (surface-active agent) which, in a dilute solution, demonstrates cleansing properties. The detergents produced by KCCL are as follows:

2.1.3.1 Fast Wash Detergent Powder

Fast wash consists of white granules and multicolor speckles. It is enriched with optical brighteners and color guards. It removes tough stains and keeps clothes shiny and brilliant white. Even after washing several times, clothes look new. The Color Guard formula of the detergent protects the color and fabric of the clothes. The average time of cleaning is reduced. Besides, it does not cause skin irritation. It comes in 35 gm, 200 gm, 500 gm, 1 kg and 2 kg packets.

2.1.3.2 Tibet Detergent Powder

This product is enhanced with whitening agents, brightening agents, foam booster and lemon fragrance. It helps with deep cleansing. Moreover, whitening and brightening agents help to brighten the fabric. It is available in 200 gm, 500 gm, 1 kg, and 2 kg packets.



Fast Wash Detergent Powder



Tibet Detergent Powder

Figure 2.1.3.1: Different types of Detergent powders

Chapter 3: Panshed

3.1 Introduction

The primary and first vital stage of soap manufacturing is performed in the panshed department. The Saponification reaction takes place here. This department consists of eighteen pans (Soap boilers) inside which the raw materials are heated, and neat soap is produced. The method that Kohinoor Chemicals Co. (Bd) Ltd. uses in neat soap production is termed the full boil process. Neat soap is sent to the Soap department where additives are added, and further processing is performed to produce the final product.

The production can take up to seven days, so the panshed department runs 24x7. To run the machinery and other maintenance procedures, the operators work in Two shifts.

Panshed is the most critical department when it comes to soap production as the quality of the finished product largely depends on the quality of the neat soap. After the production of each batch, the sample is sent to the QC department and the neat soap is not dispatched until the QC department approves the batch.

The role of the panshed department in the production of soaps is very crucial because:

- 1. Soap Formation:** The soap boiling section is where the chemical reaction known as saponification takes place. Saponification involves combining fats or oils with an alkali, such as sodium hydroxide, to produce soap. This reaction occurs at elevated temperatures in the soap boiling section, allowing the conversion of raw materials into soap.
- 2. Quality Control:** The soap boiling section provides an opportunity to closely monitor and control the soap-making process. Precise control of temperature, mixing, and timing in this section ensures consistent quality and properties of the soap produced. This is important to meet customer expectations and maintain product standards.
- 3. Soap Composition:** The soap boiling section allows for the incorporation of additives, such as fragrances, colors, moisturizers, and exfoliants, into the soap formulation. These additives can enhance the appearance, scent, and skin benefits of the soap, making it more appealing to consumers.

- 4. Removal of Impurities:** During the soap boiling process, impurities and undesirable substances present in the raw materials, such as fatty acids, glycerol, and unsaponifiable matter, can be removed. This helps improve the purity and cleanliness of the soap, ensuring a high-quality product.
- 5. Texture and Consistency:** The soap boiling section plays a crucial role in determining the texture and consistency of the soap. Through precise control of factors like temperature, stirring, and cooling, manufacturers can achieve the desired soap texture, whether it is solid, liquid, transparent, or opaque. Consistency is important for ease of use, packaging, and customer satisfaction.
- 6. Process Efficiency:** The soap boiling section contributes to the overall efficiency of soap production. By optimizing the process parameters and equipment used, such as agitators, heat exchangers, and reactors, manufacturers can increase productivity, reduce energy consumption, and minimize waste generation.

3.2 Layout of Panshed Department

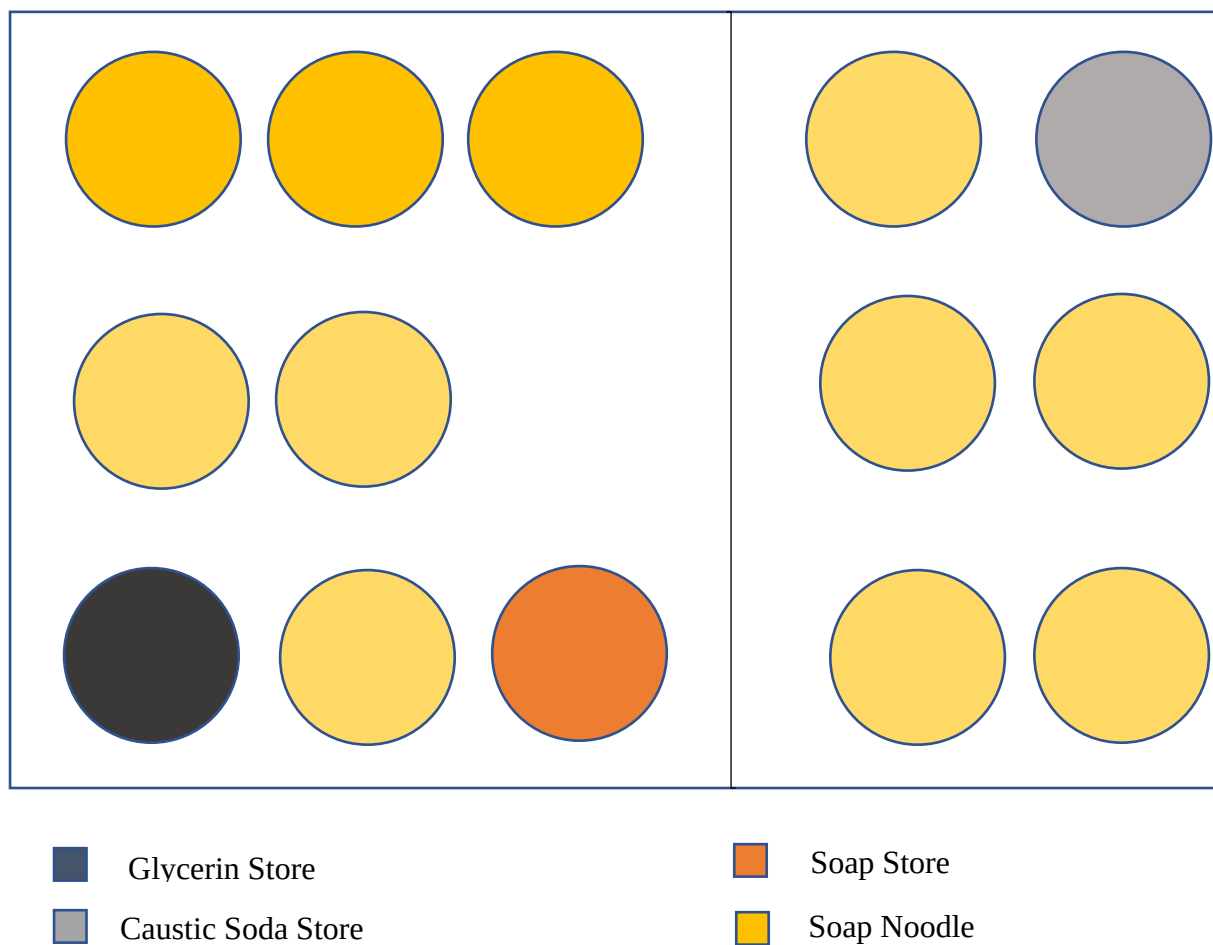


Figure 3.2.1: Layout of the panshed department

3.3 Raw Materials in the Panshed Department

Raw material requirements for Toilet soaps and laundry soaps are different. The main raw materials for both kinds are:

1. Fats and Oils:
 - RBD Plum Stearate
 - Plum Kernel
 - Coconut Acid Oil
2. Caustic Soda
3. Salt (NaCl)

The percentage of fat in toilet soaps is significantly greater than that of laundry soaps.

3.3.1 Raw Material Procurement

- **Supplier Relationships:** Raw materials are mainly imported from Indonesia and Malaysia. Kohinoor Chemicals Co. (Bd) Ltd. does not have a fixed supplier, rather they collect the raw materials from suppliers who can offer quality materials at the lowest price.
- **Safety Stock:** One of the main categories of raw materials required in the panshed department is plum oils (RBD Plum Stearate, Plum Kernel oil, etc.). The demand for Plum oils is highly elastic. So, some safety stock is kept to guard against price hedges. Again, some safety stock is kept avoiding material shortage.

3.3.2 SQC of raw materials

- **Pre-Shipment Test:** After the price, policies, etc. are finalized from a preliminary discussion with the supplier, the sample is sent to a third-party testing laboratory that tests the sample and the results are sent to Kohinoor Chemicals Co. (Bd) Ltd. Based on the result, the contract with the supplier is signed.
- **Lot Acceptance Sampling:** After the raw material lot arrives at the factory, 20% of the barrels of total lot size (in numbers of barrels) are opened, and samples are collected and tested in the QC department.

3.4 Maintenance Activities in the Panshed Department

- The maintenance facility of Kohinoor Chemicals Co. (Bd) Ltd. Is decentralized, which means the panshed department has its maintenance staff.
- Maintenance of Klinger valves requires the shutdown of the entire plant, so the scheduling is critical.
- Yearly periodic maintenance program is performed in the panshed department.

3.5 Process of Chemical Formation of Toilet Soap

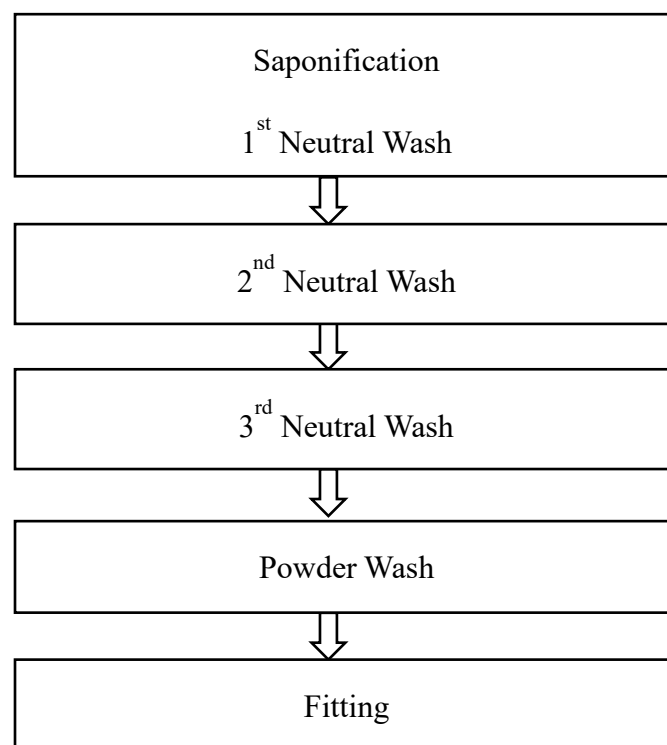


Figure 3.5.1: Toilet soap process flow chart

Saponification is the alkaline hydrolysis of fats. Here, RBD (refined, bleached, and deodorized) fats and oils are reacted with caustic soda (30-35%) to produce soap. Glycerin is produced as a by-product.

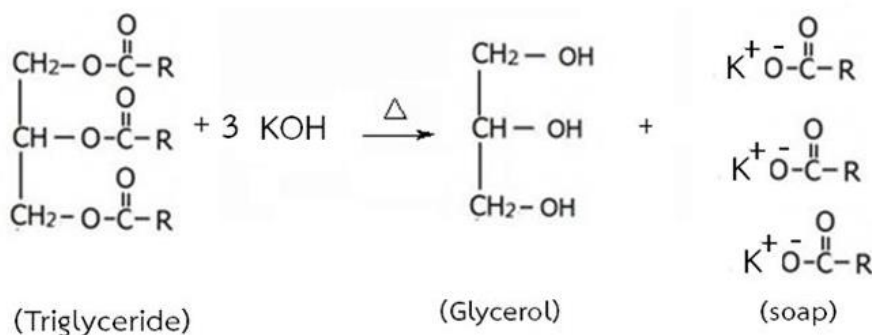


Figure 3.5.2: Soap formulation

Brine (21%) is used to precipitate the soap, which floats at the top. Afterward, glycerin is gradually recovered from soap lye in 3 stages, where the amount of caustic soda is maintained and hence, this process is called neutral wash. During powder wash, caustic soda is added (0.4-0.5%) to convert the unreacted fat to soap. Venol powder is then used to separate the iron, which becomes a part of the soap lye. Finally, free fat and sodium chloride levels are controlled in the last stage. In this way, toilet soap is produced.

3.6 Process of chemical formation of laundry soap

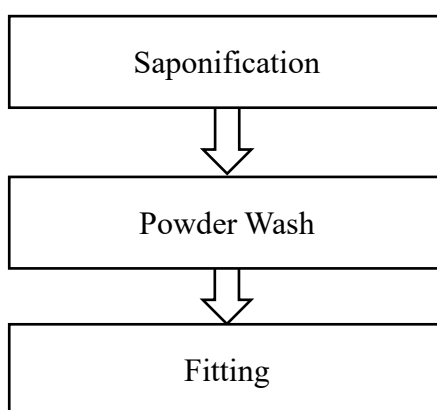


Figure 3.6.1: Process flowchart of chemical formation of laundry soap

First, saponification is carried out by reacting RBD fats and oils with caustic soda (30-35%) to produce soap. Glycerin is produced as a by-product. During powder wash, caustic soda is added to convert the unreacted fat to soap. Venol powder is then used to separate the iron, which becomes a part of the soap lye. Finally, free fat and sodium chloride levels are controlled in the last stage. Thus, laundry soap is formed.

3.7 Machinery used in the Panshed Department

3.7.1 Pan

The pan is very important in the soap making process, especially when making neat soap. The pan's features and specifics are as follows:

1. **Clean Production of Soap:** The container is where the creation of a slick cleanser happens. The final soap product that is free of impurities and has the desired composition is referred to as "neat soap."
2. **Pan size:** In the panshed area, each pan has a height of 19 feet and a diameter of 16 feet. These dimensions enable the formation of two distinct layers within the pan and provide sufficient space for the soap production process.
3. **Formation with Two Layers:** During the cleanser creation process in the container, two layers are made: soap lye and clean soap. Glycerin, sodium chloride, and impurities make up the soap lye layer. To ensure a high-quality finished product, it is essential to separate the soap lye from the soap itself.
4. **Removal of the Soap Lye:** A cock valve is used to separate the soap lye from the clean soap. The valve takes into consideration the controlled partition and moves the cleanser lye to one more segment of the creation region. Glycerin recycling and further processing would not be possible without this separation procedure.
5. **Recycling of Glycerin:** The separated soap lye with glycerin, impurities, and NaCl is sent to a separate section for recycling glycerin. The extraction and recovery of glycerin from the soap lye for reuse or further processing is the primary focus of this section.
6. **The Making of Clean Soap:** The neat soap, which is the desired end product, is obtained through the separation process. Neat soap has gone through the necessary processes to achieve the desired composition, texture, and quality and is free of impurities.



Figure 3.7.1.1: Neat soap in a pan

3.7.2 Pump used in panshed

There are two types of pumps that are used. One is the centrifugal pump and the other is the rotary pump.

1. **Centrifugal pump:** This pump is used for other sections like pumping Salt, Fat, Caustic Soda etc. in the pan.
2. **Rotary pump:** The rotary pump is made to specifically pump soap throughout the production area. It is a type of positive displacement pump that moves the soap through the system by creating suction by rotating. There are valves used in this pump that are described below:
 - a. **Initial Valve:** The rotary pump's first valve directs the soap's flow from various pans. The soap flow can be controlled by opening or closing this valve.
 - b. **Second Valve:** The rotary pump's second valve is in charge of deciding where the pumped neat soap will go. If this valve is open at the same time as the first valve, neat soap is carried to the 'Mazzoni' floor.
 - c. **Third Valve:** The neat soap is kept in a storage pan by the rotary pump's third valve. If this valve is open at the same time as the first valve, neat soap is carried to the storage tank.

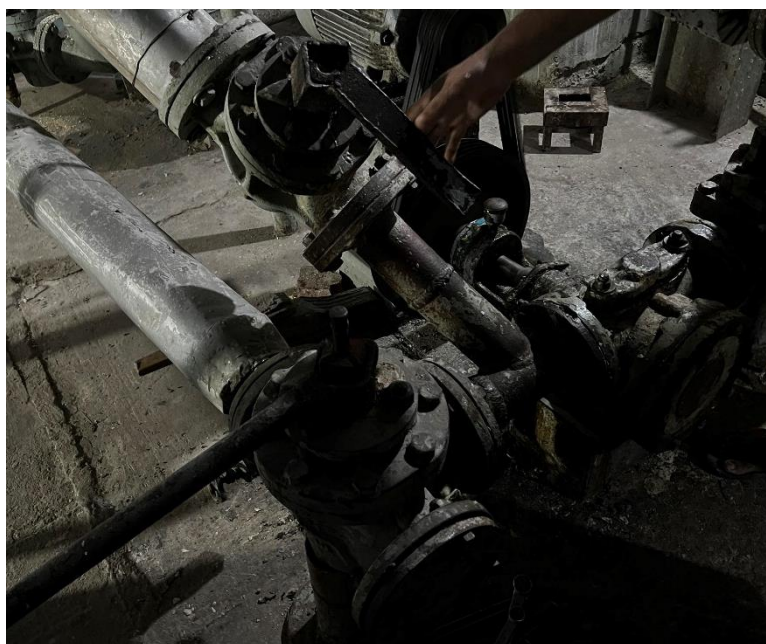


Figure 3.7.2.1: Rotary Pump with three valves

3.7.3 Pipe inside panshed

There are many pipes carrying different materials to the pans. Pipes are also colored and labeled for easy recognition. They are mentioned below:

Table 3.7.1: Pipes and their specifications

Color of pipe	Material Carried	Valve used
Grey + Red	Steam	Klinger
Red	Caustic soda	Saunders
Off white	Fat	
Green	Salt/ Brine	
Grey	Rough soap	Gate
Violate	Soap lye	
Indigo	water	



Figure 3.7.3.1: Color coded pipes with certain materials

Pipe used for steam:

1. **Direct flow:** Steam goes inside the pan and agitates, melts, and mixes raw materials of the soap and goes out. There are four pipes connected to four concentric circle coils.

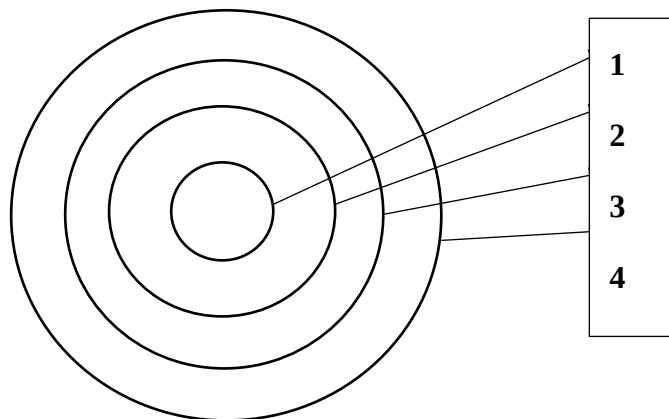


Figure 3.7.3.2: Concentric Coils inside the pan

2. **Indirect flow:** Steam does not come out of the pan. It adds moisture to soap. Only one pipe is used for this purpose.



Figure 3.7.3.3: Direct and Indirect pipelines of steam

3.7.4 Pipe throughout the plant

The plant transports various materials throughout the plant and to various departments via a pipe network. In order to guarantee safe handling and identification, these pipes are color-coded. The plant heats and generates energy with the steam carried by the gray pipes. Off-white pipes are designed to transport fat in its liquid form, a common cosmetic ingredient. The red lines contain compressed water, used for different purposes like fire safety and cleaning. The indigo pipes are in charge of delivering the raw water, which is used in subsequent processing and treatment. The variety coded framework helps to keep up with productivity and security all throughout the plant, guaranteeing that the right materials arrive at their planned objections.

Table 3.7.2: Direct and Indirect pipelines of steam

Color	Material carried
Grey	Steam
Off White	Fat (liquid form)
Red	Pressurized water
Indigo	Raw Water

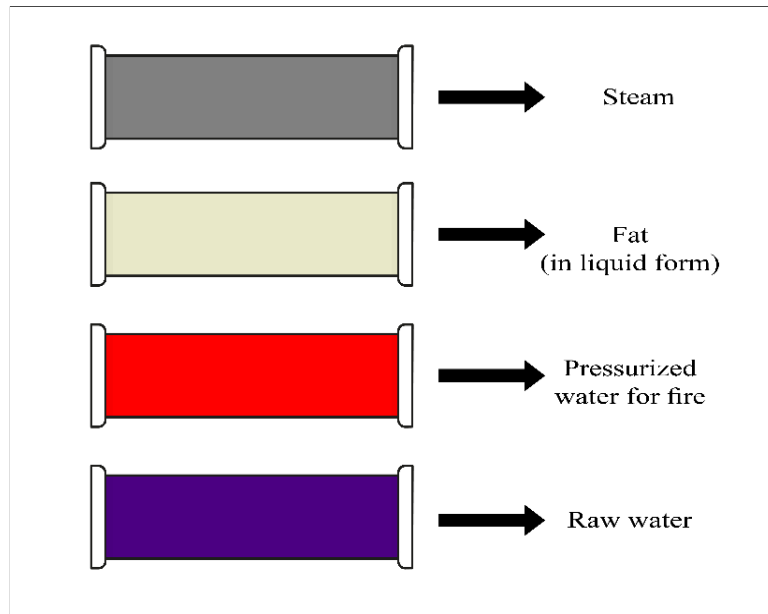


Figure 3.7.4.1: Pipelines throughout the plant

Chapter 4: Soap Section

4.1 Introduction

KCCL mainly focuses on soap production and employs a continuous production model. The soap production process works nonstop in three shifts, guaranteeing a consistent stock of soap. Two kinds of soap are made in this part of the manufacturing plant: Toilet soap and laundry soap. The panned is where soap noodles are made, which are the most important ingredient in making soap. Though, for Sandalina, the cleanser noodles are obtained remotely from a provider. This is finished to meet the particular prerequisites and detailing of Sandalina. Sandalina, Tibet luxury soap, Tibet 570 soap, Tibet ball soap, and other soap products are available soaps that are produced by the KCCL. KCCL is able to effectively and efficiently meet the demand for soap products for its continuous production strategy.



Figure 3.7.4.1: Toilet and laundry soap produced in soap department

4.2 Maintenance activities in the soap section

The soap production area also receives weekly maintenance and cleaning, just like other departments do. This guarantees that the gear, hardware, and creation lines stay in ideal condition and cleanliness norms are kept up with. Notwithstanding the planned weekly cleaning, any cleaning or support undertakings that are expected at different times are quickly done by the specialists. This proactive methodology guarantees that the cleanser creation process chugs along as expected and effectively, limiting any possible disturbances and guaranteeing the nature of the eventual outcomes.

4.3 Raw material used in the soap section

Two kinds of soap are made in this part of the manufacturing plant: Toilet soap and laundry soap. The panned region is where soap noodles, the main ingredient in making soap, are made. Notwithstanding, for the Sandalina soaps, the cleanser noodles are obtained remotely and brought into the office. To meet the particular requirements and formulation of this particular soap variant, Sandalina noodles are obtained from outside suppliers.

4.4 Machinery used in the soap section

Different types of machinery are used here. They are described below:

4.4.1 Crutcher

Crutcher is used to make liquid soap. The following are some specifics about how it works are given below:

1. **Mixing Container:** The Crutcher includes a substantial kettle or mixing vessel for the production process. The vessel can carry between 2 to 2.5 tons.
2. **Ingredients:** The vessel is filled with perfume, color, and filler during the production of liquid soap. These fixings are important to upgrade the appearance, fragrance, and surface of the last fluid cleanser item.

3. **Inner Screw:** An inner screw in the mixing vessel aids in the mixing of the liquid soap, perfume, color, and filler. The screw moves the vessel's contents, ensuring that they are thoroughly blended and distributed evenly.
4. **Capacity Decomposition:** The Crutcher can carry 2-2.5 tons in total. The vessel can hold 2 tons of neat soap, the primary component of the liquid soap, within this capacity. The remaining capacity of 0.5 tons is used for other materials like perfume, color, and filler.
5. **Time for Mixing:** The Crutcher's mixing process typically takes between 20 and 30 minutes. The inner screw continuously blends and mixes the ingredients to form a homogeneous mixture during this time.

The Crutcher facilitates the mixing and blending of ingredients to produce a consistent and appealing product, which is crucial to the production of liquid soap. The vessel's capacity allows for large-scale production, and the mixing time ensures that all of the components are thoroughly mixed and incorporated.



Figure 4.4.1.1: Crutcher

4.4.2 Filter

The mixture of perfume, filler, color, and liquid soap is moved to a feed tank following the Crutcher's mixing process. The mixture goes through a filter system before reaching the feed tank. Details about the filter are given below:

1. **System of filters:** The filter system is made to get rid of relatively large materials like paper and plastic scraps and any other contaminants that might be in the mixture. This helps ensure the quality of the finished product and prevents the inclusion of any undesirable ingredients in the liquid soap.
2. **Removal of Particles:** The larger materials are effectively separated and removed by the filter, allowing only the liquid soap mixture that has been filtered to pass through. This keeps up with the ideal consistency and neatness of the fluid cleanser.
3. **Storage:** The filter system filters the mixture and then it moves from the Crutcher to the feed tank.



Figure 4.4.2.1: Filter

4.4.3 Feed-tank

A tank is used to store liquid soap until it is needed for the soap-making process. The following are some essential points regarding liquid soap storage:

1. **Tank for storage:** The liquid soap is kept in a separate tank. The tank has a maximum capacity of 5 tons, which allows for a significant amount of soap to be stored at once.
2. **Holding the soap:** The storage tank's primary function is to store the liquid soap until it is needed for further processing. This guarantees that the production line will always have soap on hand.
3. **Prepared to Atomize:** The soap can be moved directly from the storage tank to the atomizer whenever it is required for the production process. Passing the liquid soap to the

atomizer does not necessitate any additional preparation or waiting time, unlike other steps in the process. The soap already meets the requirements for further processing.



Figure 4.4.3.1: Feed Tank

4.4.4 Heat exchanger

By bringing the soap mixture to a certain temperature, the heat exchanger plays a crucial role in the soap production process. The heat exchanger's most important aspects are as follows:

1. **Purpose:** The primary function of the heat exchanger is to achieve the desired soap mixture temperature. This temperature is important for a number of things, like starting certain chemical reactions, making the soap more viscous, or making sure the ingredients are mixed correctly.
2. **Medium of Heating:** The heating medium in the heat exchanger is typically steam or hot water. Within the heat exchanger, this fluid moves through a single set of tubes or channels.
3. **Heat Conversion:** There are many small tubes and through the tubes, steam flowing as a result heat exchange occurs. Conduction is the method by which heat energy from the steam is transferred into the soap mixture through the tubes' walls. The temperature of the soap mixture rises as a result.
4. **Temperature Management:** In order to guarantee that the soap mixture reaches and maintains the desired temperature, the heat exchanger is equipped with temperature control mechanisms. The temperature sensors, control valves, and feedback loops that monitor and adjust the heating medium's flow rate and temperature are examples of these mechanisms.

5. **Safety and effectiveness:** The heat exchanger's design aims to ensure safety while maximizing heat transfer efficiency. To ensure the equipment's safe operation and reduce heat loss, adequate insulation and heat exchanger sizing are essential.



Figure 4.4.4.1: Heat Exchanger

4.4.5 Atomizer

The process of making soap relies heavily on the atomizer. The following are some specifics about its features and functionality:

1. **Mixture of Soap:** A nozzle or spraying mechanism is included in the atomizer to break up the soap mixture into smaller droplets. The forces exerted by the soap mixture as it moves through the atomizer nozzle break it up into tiny droplets. A consistent and fine spray of soap particles is made possible by this atomization process.
2. **Making a Vacuum:** The atomizer works under a vacuum climate for a few purposes. The atomizer's vacuum makes it easier for the soap mixture to cool down. Lower pressures are applied to the soap droplets as they are dispersed, facilitating the cooling process. Moreover, any vapor or dust particles in the soap mixture can be separated thanks to the vacuum environment. By drawing these impurities away from the soap droplets, a cleaner and purer final product is produced.
3. **Scrubber:** The atomizer has a scrubber that makes it easier to get rid of any soap that might build up on the inside surface. This ensures consistent atomization of the soap mixture and helps to prevent buildup.



Figure 4.4.5.1: Atomizer

4.4.6 Plodder

The plodder is an important component of the soap-making process, particularly in the refinement and homogenization of the soap mixture. Its function is described in detail below:

1. **Function:** The plodder is answerable for packing and refining the cleanser blend as it moves along the machine. The soap will have a more uniform texture and composition as a result of this process, which helps get rid of air bubbles, lumps, and other impurities.
2. **Mechanism:** The plodder receives the soap mixture, and a screw or auger moves it forward from there, moving the soap mixture through the machine.
3. **Process of Refining:** A process of refinement takes place as the soap mixture moves through the plodder. A series of screens or dies are used to pass the mixture through for this purpose. These screens or dies aid in the breakdown of any larger particles, resulting in a soap texture that is smoother and more consistent.

There are two plodders respectively for better soap quality. They are described below:

- a. **The Initial Plodder:** There are rotating parallel or series screws on this kind of plodder. The compression and refining processes are made easier by the parallel movement of the screws, which ensures that the soap mixture is thoroughly mixed and homogenized.
- b. **The final Plodder:** A single, substantial screw characterizes the second kind of plodder. This larger screw has a greater capacity for compression and refinement and is designed to handle greater quantities of soap mixture. There is also a heater on the mouth of the plodder so that there will be no extra bubbles or vapor inside the solid soap bar.



Figure 4.4.6.1: Plodder

4.4.7 Cutter

There are two types of cutters used in this soap section. One is a transverse cutter and other is an electric cutter. They are:

- **Transverse cutter:** During soap production, a continuous soap bar is cut with a cross cutter into individual soap pieces or beads. It has some following important characteristics:
 1. **Cutting mechanism:** The TV cutter has horizontally arranged blades. When activated, the cutting mechanism moves horizontally to cut the soap strands.
 2. **Cutting method:** As the continuous soap bar passes through the TV cutter, horizontal blades separate it into individual soap pieces.
 3. **Capacity:** The cutter can produce 80 soap bars or pellets per minute at maximum capacity. This shows how the cutting process can be carried out quickly and efficiently, allowing large quantities of soap to be produced.

- **Electric cutter:** In the production of soap, an electric soap cutter is utilized to cut soap bars into the desired sizes. An electric soap cutter has the following essential characteristics and functions:
 1. **Energizing Power:** Electricity is used to power the machine, which gives the cutting mechanism the power it needs.
 2. **Cutting Edges:** Sharp stainless steel or other long-lasting cutting blades are included with the electric soap cutter. These blades are made to slice through soap bars effortlessly and precisely.
 3. **Customizable Cutting Settings:** The machine has cutting settings that can be changed to fit different sizes and shapes of soap.
 4. **Capacity:** The electric soap cutter cuts one soap at a time.
 5. **Security Highlights:** It should be covered that way so that no harm can occur to the workers.



Figure 4.4.7.1: transverse cutter and electric transverse cutter

4.4.8 Stamper

After the soap bars have been cut into the desired shape and size during the manufacturing process, they are typically stamped with a logo, brand name, or other markings. There are two types of stamper machine according to the mechanism of applying pressure. One is pneumatic stamping machine and other is mechanical stamping machine. A few specifics about the stamping machines are as follows:

- **Pneumatic stamping machine:**

1. **Die and stamp:** The stamper is the gadget that applies pressure to the cleanser bar. An automatic stamping machine keeps stamping at a given rate set up by the controller.
2. **Cooling:** The soap bars go through a cooling phase during the stamping process to solidify and harden the soap. This cooling is done by chiller. Cooling assists with guaranteeing that the stepped markings stay in salvageable shape and don't smirch or blur.
3. **Application of Pressure:** During the stamping process, the pneumatic valve is used to control the pressure. The die is pressed against the soap bar's surface by using pneumatic power to exert force on it. By vacuum pressure stamped soap are collected and kept on the conveyor for further processing.
4. **Capacity:** The stamper can be intended to stamp two, four, or six cleanser bars all the while, contingent upon the particular setup of the stamper and the limit of the cleanser producing plant.

- **Mechanical stamping machine:**

1. **Die and stamp:** The stamper is the gadget that applies pressure to the cleanser bar. An automatic stamping machine keeps stamping at a given rate set up by the controller.
2. **Cooling:** The soap bars go through a cooling phase before the stamping process to solidify and harden the soap. This cooling is done by dryer. So before stamping in mechanical stamper they are cooled separately in a machine and then these soaps are collected manually and kept in the stamping machine conveyor. Cooling assists with guaranteeing that the stepped markings stay in salvageable shape and don't smirch or blur.
3. **Application of Pressure:** During the stamping process, mechanical pressure is applied on the soap. This system is powered by electricity and after stamping soaps are taken manually.
4. **Capacity:** The stamper can be intended to stamp two, four, or six cleanser bars all the while, contingent upon the particular setup of the stamper and the limit of the cleanser producing plant.



Figure 4.4.8.1: Stamper (Plant 6)



Figure 4.4.8.2: Stamper (plant 7,8,9)

4.4.9 Packaging

The stamped soap bars are typically inspected for quality and consistency. They are then packaged. It robotizes the most common way of wrapping cleanser bars in bundling materials, like plastic film, paper, or containers. The following details are about soap packaging machines:

1. **Manual input:** The cleanser bars are taken care of into the bundling machine either physically or naturally, contingent upon the particular machine design. They are typically placed on conveyor belt during the packaging process.
2. **Material for the Wrap:** The packaging machine is capable of handling a variety of wrapping materials, paper, and plastic film. The requirements of the soap product and the style of packaging that is desired determine the particular material that is used.
3. **Mechanism for Sealing:** The bundling machine utilizes a fixing system to safely close the bundling material around the cleanser bar. Depending on the chosen plastic packaging material, this can be accomplished through heat sealing.
4. **Speed and Limit:** Depending on the model and the needs of the production line, soap packaging machines can operate at different speeds. The number of soap bars packaged per minute or hour is typically used to measure the machine's capacity.
5. **Adjustments and Controls:** Soap packaging machines are outfitted with control boards or connection points that permit administrators to change settings like conveyor speed, fixing temperature, cutting length etc.



Figure 4.4.9.1: Packaging

4.4.10 Chiller

Chillers are an essential part of the soap production process. Here are some specifics of it:

1. **Temperature Management:** To guarantee efficient cooling, the chiller maintains a specific temperature range. So, the temperature is ordinarily set at around - 18 degrees Celsius.
2. **Cooling Agent:** The chiller involves a combination of Glycol and Water as the cooling medium. Water is used for the actual cooling process because it is a great heat conductor. However, in order to keep the water from freezing at low temperatures, glycol is added to it. This makes sure that the chiller works well and smoothly even in colder places.
3. **Temperature Observing and Control:** Chillers are furnished with temperature checking and control frameworks to guarantee exact temperature guidelines. The chiller can adjust its operation in response to the temperature of the cooling medium and molds.



Figure 4.4.10.1: Chiller

4.4.11 Dryer

The soap bars are dried using a drying machine in plants 1, 2, and 4 of the soap production facility. This machine consolidates an inner fan arrangement that makes a controlled wind current to help with the drying system. The internal fan moves air through the machine as the soap bars go through it, making it easier for moisture to evaporate from the soap's surface. For each soap bar, the drying machine runs for about 18 minutes, giving the soap enough time to dry to the desired level and release moisture. By successfully eliminating dampness, the drying machine guarantees the cleanser bars keep up with their quality, forestalling issues like delicateness, shape development, and debasement. It contributes to the overall texture, durability, and quality of the final product by producing uniformly dried soap bars that are ready for subsequent processing and packaging.



Figure 4.4.11.1: Dryer

4.5 Flow chart of soap section

4.5.1 Flow chart for Plant 1,2 & 4

Plant 1 and plant 2 are used to make “Tibet ball soap”. The production plan starts from a Crutcher, flows through a filter and then the liquid soaps are poured into feed tank. After that the liquid soap flows through atomizer, where perfume is mixed. Next the liquid soap is extruded from the atomizer and a TE cutter cuts the soap according to the design. After that the soap is heated in a heater for about 18 minutes to release its moisture. Finally using a packaging machine, it is packaged.

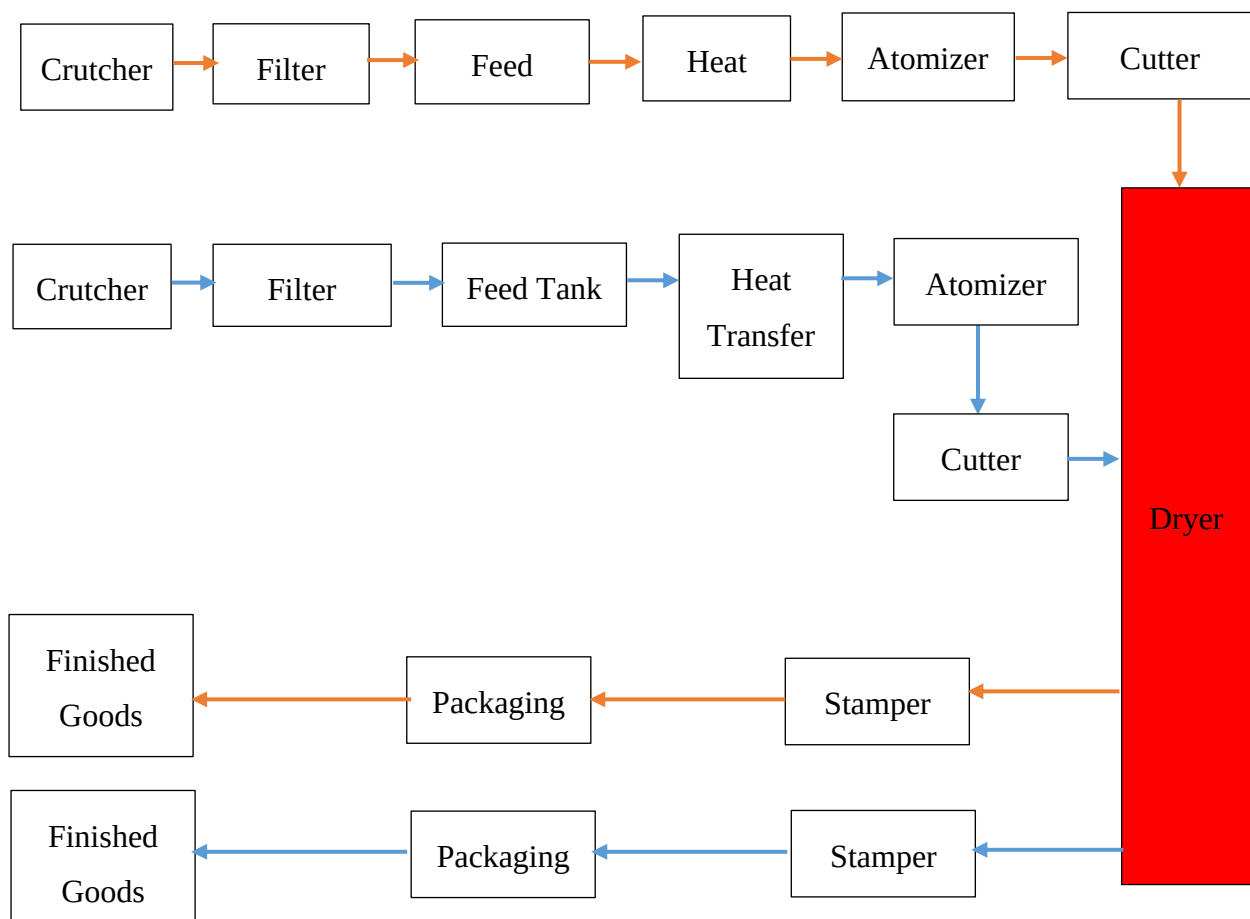


Figure 4.5.1.1: Flowchart of the sequence of processes performed in plant no.1,2 and 4.

4.5.2 Flow chart of Plant 5

Plant 5 is used to make the noodles, which works as raw material for plant 7,8,9. Same as plant 1 and 2, net soap is flown through Crutcher, filter to the feed tank. After that the atomized noodles are extruded from the plant and are packaged in sacks to send it to plant 7,8,9.

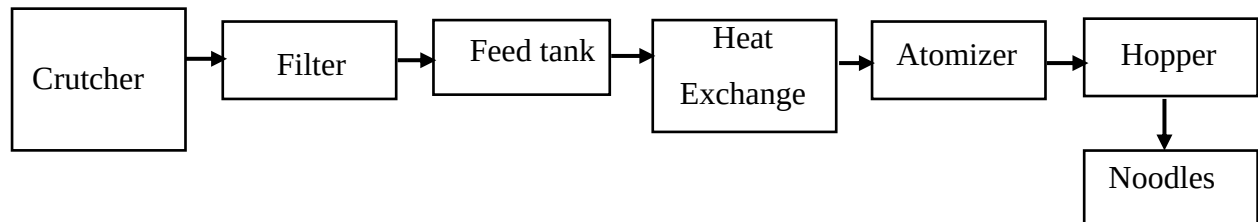


Figure 4.5.2.1: Flowchart of the sequence of processes performed in plant no. 5

4.5.3 Flow chart of Plant 6

Plant 6 is used to make “Tibet Lundry Soap” And “Tibet 570”. The mechanism is almost as same as the plants discussed before. After extruding from atomizer, the semi solid soap is cut into bars using a TV cutter. Then it is stamped using a stamper. The temperature of the die in the stamper is kept -18 degree Celsius so that the soup does not stick to the die. The soap is rested for 3 to 5 hours to release moisture. Finally, it is packaged using packaging machines.

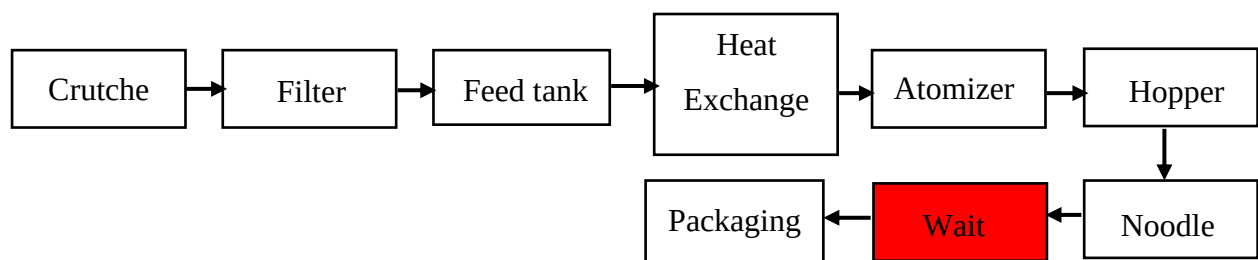


Figure 4.5.3.1: Flowchart of the sequence of processes performed in plant no. 6

4.5.4 Flow chart of Plant 7,8,9

Plant 7,8,9 has the same mechanism and layout. The only difference is in the capacity. Among them plant 7 is used to make “Sandalina soap”. And plant 8,9 are used to make other toilet soaps including “Bactrol”, “Tibet” etc. First noodles, color and perfume are added in the mixture machine. Then, this mixture is passed through a plodder. The plodder net has a diameter of 0.5 cm. It basically mixes everything more properly. The plodder mixes everything to a granular level. To get even a fine mixing and fine particles, the output from the first plodder is passed through a second plodder. The second plodder’s net has a diameter of 0.3 cm. From the second plodder the soap is extruded and cut into long bars. This bar is then stamped using suction pressure and a die to give the desired shape. Soaps except “Tibet Luxury Soap” are left open for several hours to let them Dry. Finally, they are sent to the packaging section.

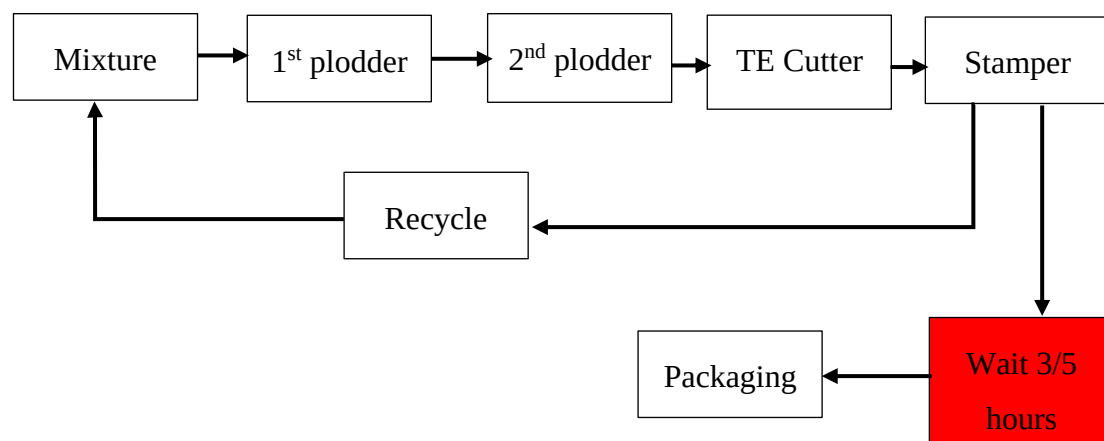


Figure 4.5.4.1: Flowchart of the sequence of processes performed in plant no. 7.8 &9

Chapter 5: Cosmetics

5.1 Introduction

There are two floors on which the cosmetic manufacturing plant operates, each with its own distinct function. The ground floor centers around the development of beauty care products that are appropriate for seasons. These items are planned to be flexible and versatile, taking care of clients.

Seasonal cosmetic products are the focus of production on the second floor of the facility. These things are explicitly intended to line up with the prerequisites and inclinations of clients during explicit seasons. For instance, the production of cosmetics with lightweight formulas and sun protection features may be the primary focus during the summer, while hydrating and nourishing products may be the primary focus during the winter to combat dryness and harsh weather.

By assigning the creation cycle across two stories, cosmetic department can actually satisfy the needs of clients over time. The company is able to offer a wide range of cosmetics that meet the needs and preferences of different seasons thanks to this strategic approach. Whether it's all year basics or occasional needs, the plant guarantees that clients approach great beauty care products that address their one-of-a-kind necessity.

5.2 Layout of the cosmetics department

5.2.1 Layout of the ground floor of the cosmetics department

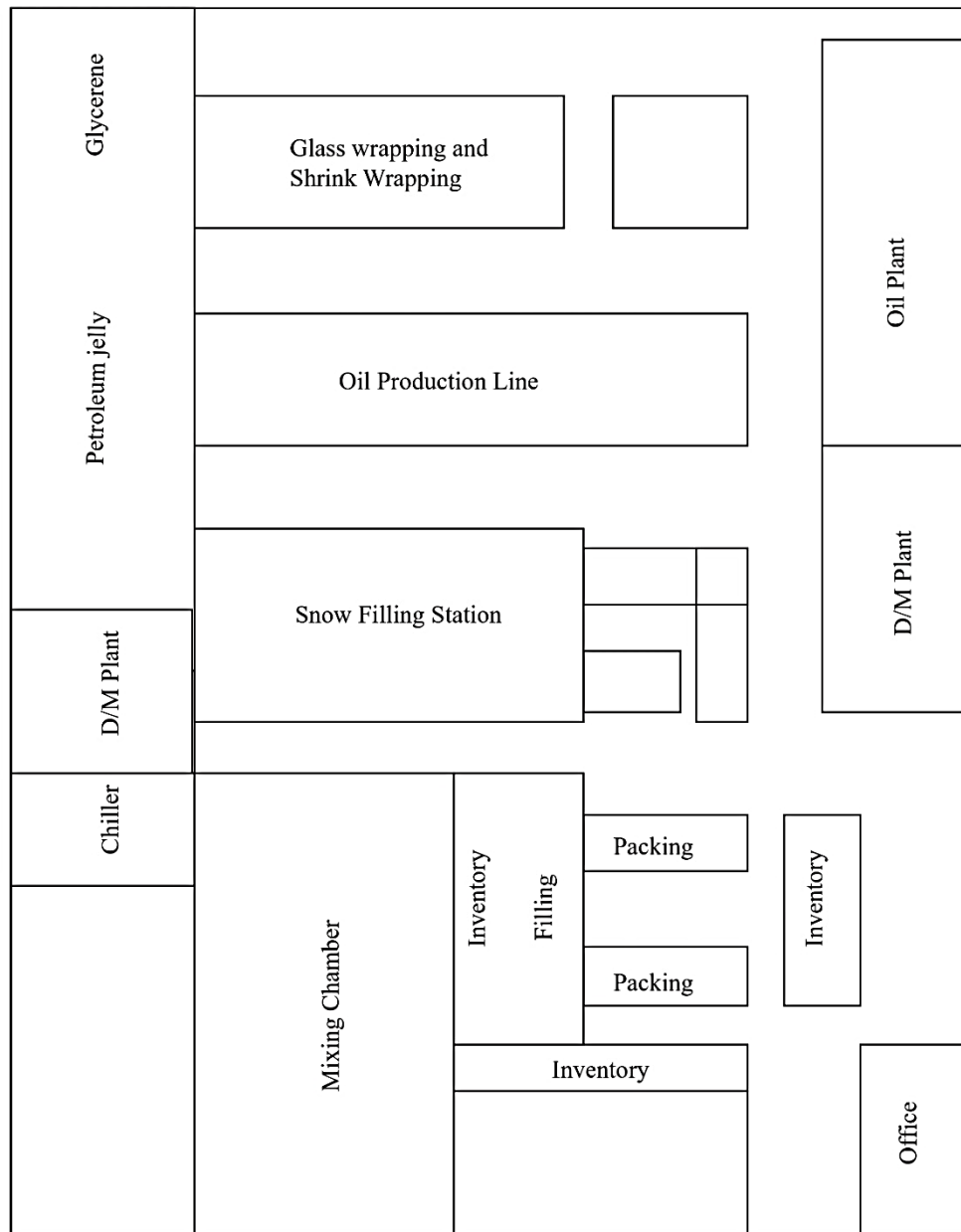


Figure 5.2.1.1: Layout of cosmetic ground floor

5.2.2 Layout of oil production of cosmetics department

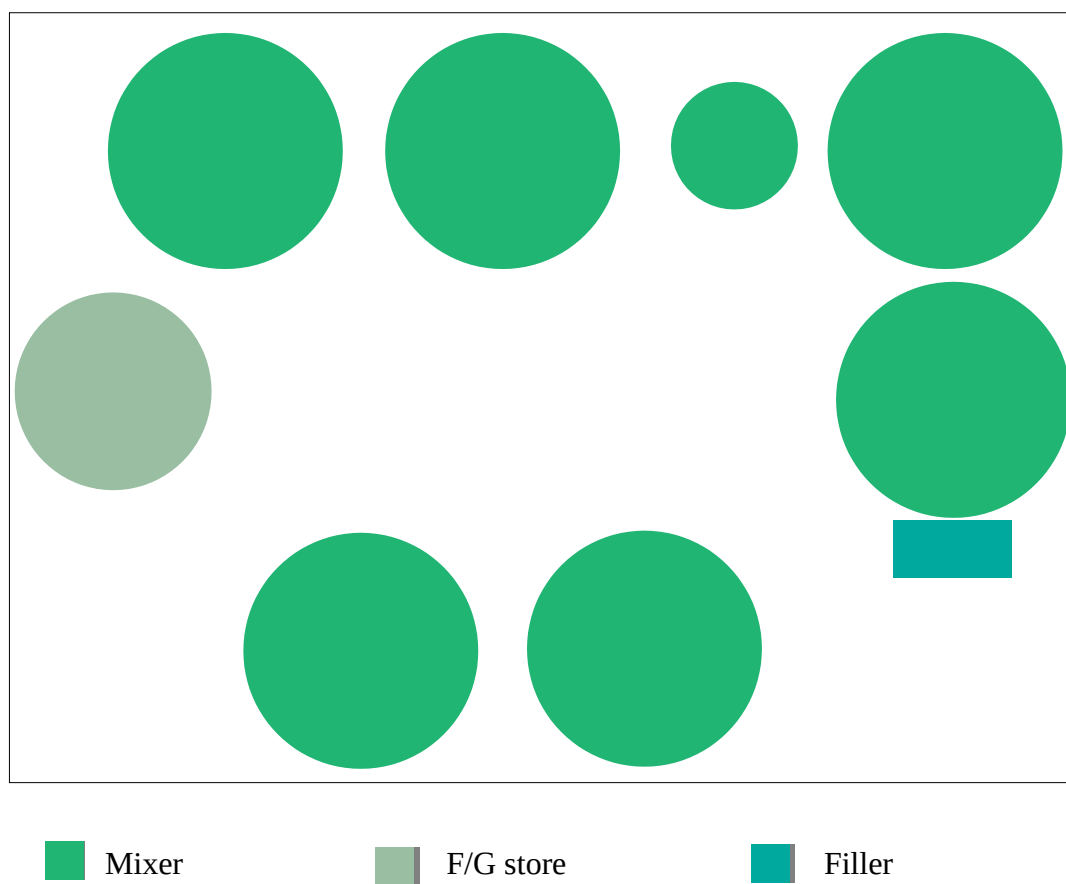


Figure 5.2.2.1: Layout of oil production of cosmetic ground floor

5.2.3 Layout of the first floor of cosmetics department

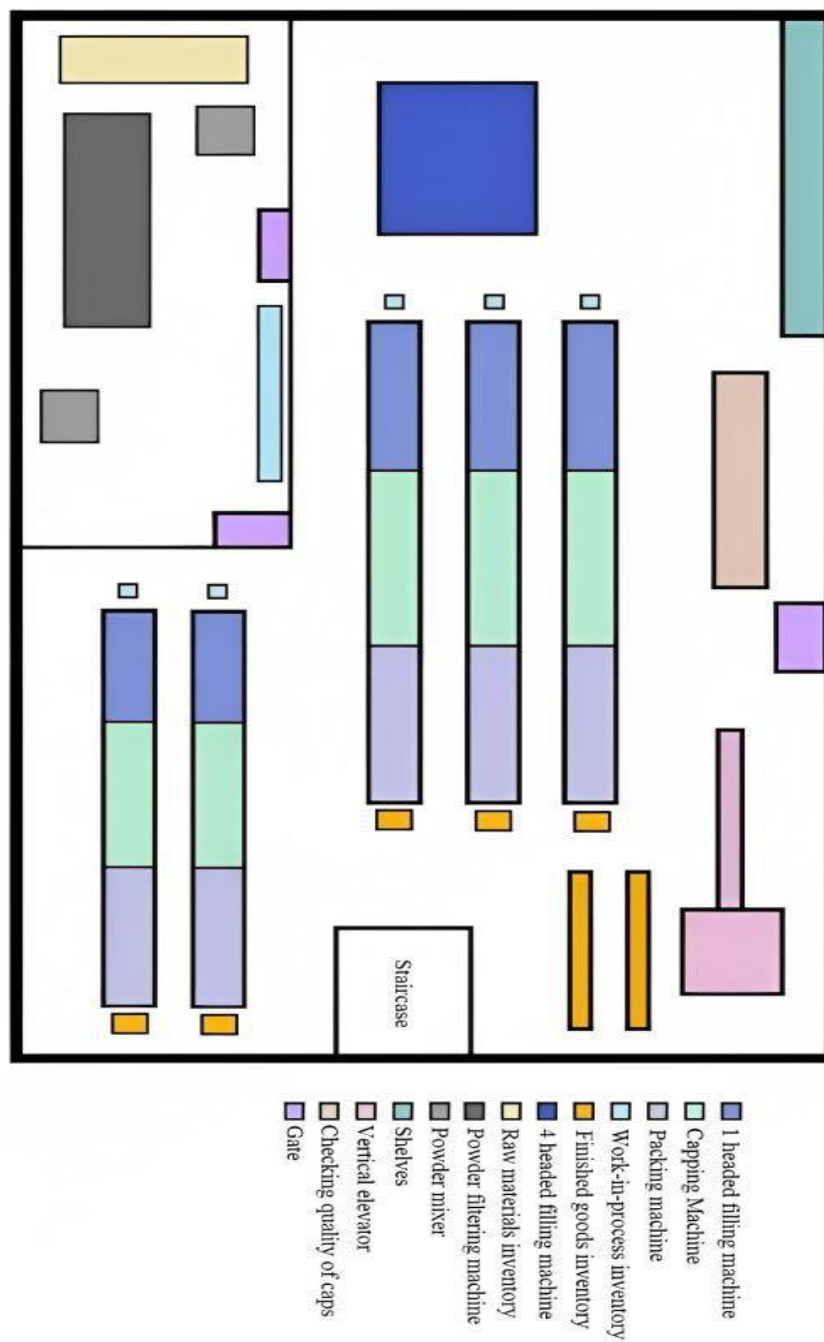


Figure 5.2.3.1: Layout of cosmetic first floor

5.3 Maintenance activities in cosmetic department

The plant produces cosmetics using a batch production method, which means that products are made in batches instead of being produced based on demand from the market. After producing a particular bunch of a specific item, the machines are entirely cleaned and ready for the next production. This guarantees that there is no cross-pollution between various items and keeps up with the quality and honesty of each group.

The workers at the plant carry out routine maintenance and cleaning tasks on a weekly basis in addition to the routine machine cleaning that they do after each batch. The machinery is cleaned, inspected, and maintained more thoroughly as part of this to keep them in working condition. The workers reduce the likelihood of breakdowns and production disruptions by maintaining the machines on a regular basis, ensuring their durability, effectiveness, and dependability.

In general, the plant's batch production method and routine cleaning contribute to the production of high-quality cosmetics that meet market demands and adhere to stringent quality and safety standards. Each batch's integrity is protected by meticulous cleaning and maintenance procedures, ensuring that customers receive safe cosmetics.

5.4 Raw materials used in cosmetics department

The various cosmetic products produced by the cosmetic production department need various raw materials. Base fixings like oils, waxes, emulsifiers, and surfactants structure the underpinning of the items, while dynamic fixings like cancer prevention agents, lotions, and natural concentrates convey explicit advantages. Preservatives and stabilizers ensure the product's safety and longevity, while fragrances, colorants, and essential oils enhance the sensory experience. To get better results you can use thickeners, defoliants, sunscreens, pH adjusters, and other ingredients. So various items are needed in this cosmetics section in order to satisfy customer needs.

5.5 Machineries used in cosmetics department

There are lots of machines used in the cosmetic department. Both the machineries used in ground and first floor in this department is described below:

5.5.1 Machines used in ground floor

In this cosmetic production section, various types of machines are utilized, with the majority of them being mixing, filling, shrinking, packaging, image printing machines. These machines play crucial roles in the different stages of the production process.

- Mixing machines are responsible for combining and blending various ingredients to create the desired cosmetic formulations. They ensure that the ingredients are thoroughly mixed to achieve a consistent and uniform product.
- Filling machines are used to fill the mixed cosmetic formulations into containers or packaging. These machines are designed to accurately dispense the desired amount of product into each container, ensuring proper measurement and minimizing waste.
- A shrinking machine, also known as a shrink wrap machine or shrink tunnel, is a device used to apply heat to shrink plastic film tightly around a product or a group of products. It is commonly used in the packaging industry to provide a secure and professional finish to various items.
- A key piece of machinery used in the cosmetics industry to automate and streamline the packaging of cosmetic items is the packaging machine. These devices are made to accurately and efficiently package the goods into different containers or packaging materials. They are capable of handling a variety of package types, including bottles, tubes, sachets, and blister packs.
- Image Printing machines are employed for applying labels, codes, or imprints onto the packaging of cosmetic products. They can add important information such as product names, ingredients, manufacturing dates, or batch numbers, helping with product identification, tracking, and compliance.

By utilizing these different types of machines, the cosmetic production facility can automate and streamline the manufacturing process, improving efficiency, accuracy, and productivity. Each machine plays a specific role in the overall production line, contributing to the creation of high-quality cosmetic products.

5.5.1.1 Mixing Machine

Mixing machines are equipped with rotating blades or agitators that ensure thorough blending of ingredients. They can have various sizes and capacities, depending on the production volume and the viscosity of the cosmetic formulations. These machines may include temperature controls to maintain the desired temperature during the mixing process, as some cosmetic ingredients require specific temperature conditions for optimal blending. Some mixing machines employ vacuum or emulsifying capabilities to create stable emulsions or to remove air bubbles from the formulation. High-quality mixing machines help ensure consistent product quality and homogeneity, resulting in a uniform texture and appearance of the final cosmetics.

There are several mixing machines. They are described below with specification:

- **Local Machine -1:** This Mixing machine is designed and manufactured in Bangladesh to facilitate the mixing process and ensure the desired formula is obtained accurately and efficiently.
 1. **Mixing Capability:** The machine utilizes rotating blades to ensure proper mixing of ingredients. However, it is not suitable for sticky materials. Therefore, it is primarily used for mixing Xpert Dish Wash, which likely has non-sticky ingredients.
 2. **Heating and Cooling System:** The machine incorporates both heating and cooling systems. Heating is essential to facilitate proper ingredient mixing and reaction. Steam is used as a heating medium. On the other hand, Cooling is needed for fragrances, flavor, preservatives at low temperature and prevent any degradation of the product. Water is used as a cooling medium.
 3. **Two-Chamber Design:** The machine consists of two chambers. The inner chamber is dedicated for mixing the ingredients at high temperature and so there is a steam line there. On the contrary, the outer chamber facilitates water flow for cooling the mixture.
 4. **Reaction:** The mixing process requires a temperature of around 80 degrees Celsius to promote proper ingredient mixing and reactions. Preservatives, flavors, or fragrances are added after a certain period of mixing. To maintain their properties, these additives need to be added at lower temperatures, typically between 70 to 50 degrees Celsius.
 5. **Insulation:** The machine is insulated to prevent excessive heat transfer to the surrounding environment. This insulation helps maintain the desired temperature within the machine and minimizes energy loss. The machine is carefully insulated such that there is no contamination or mixing with other formulas.

6. **Product:** Mainly Xpert dish is made.
7. **Origin:** The machine is manufactured in Bangladesh.
8. **Capacity:** The machine has a capacity of producing 1 ton of Xpert Dish Wash.



Figure 5.5.1.1: local Machie-1

- **Local Machine -2:** This Mixing machine designed and manufactured in Bangladesh, serves the purpose of facilitating the precise and efficient mixing process to obtain the desired formula.
 1. **Mixing Capability:** The machine employs rotating blades to ensure thorough mixing of ingredients, but it is not suitable for sticky materials. It is primarily utilized for non-sticky ingredients, making it well-suited for mixing Glass Cleaner formulations.
 2. **Heating and Cooling System:** The machine incorporates both heating and cooling systems. Heating is crucial for promoting ingredient mixing and reactions, with steam being the heat source. Cooling is necessary to maintain the integrity of fragrances, flavors, and preservatives, and water is used as the cooling medium.
 3. **Two-Chamber Design:** Similar to before, the machine consists of two chambers. The inner chamber is dedicated to high-temperature ingredient mixing and includes a steam line, while the outer chamber facilitates the flow of water for cooling the mixture.
 4. **Reaction:** The mixing process for Glass Cleaner typically requires a temperature of approximately 80 degrees Celsius to ensure proper ingredient mixing and reactions. After a certain period of mixing, additives such as preservatives, flavors, or fragrances are incorporated at lower temperatures, usually between 70 to 50 degrees Celsius.
 5. **Insulation:** The machine is carefully insulated to prevent excessive heat transfer to the surrounding environment. This insulation helps maintain the desired temperature within the machine and minimizes energy loss. Moreover, it is designed to prevent contamination or mixing with other formulas.
 6. **Product:** Glass Cleaner is mainly produced.
 7. **Origin:** The machine is proudly manufactured in Bangladesh.
 8. **Capacity:** The machine has a production capacity of 1 ton, specifically for manufacturing Glass Cleaner.

- **GANSONS:** This machine is made in India and capable of mixing all kinds of ingredients. Its features are given below:
 1. **Mixing ability:** The rotating blades of the machine ensure even mixing of the ingredients. It has also vacuum system so both sticky and non-sticky materials can be mixed here properly. So, all kinds of products can be mixed in this machine.
 2. **Vacuum system:** The machine integrates a vacuum system, allowing to mix the adhesive materials evenly. This characteristic is especially beneficial when manufacturing products, which often contain sticky ingredients. The vacuum system helps to rotate the blades properly and thus homogeneous mixture of these materials can be obtained.
 3. **Heating and cooling system:** The machine is equipped with both heating and cooling system. The heating system uses steam as the medium to raise the temperature in the mixing chamber. This high temperature promotes proper mixing and reaction of the ingredients. The cooling system, on the other hand, uses water as the medium to cool the mixture, especially when adding aromas, flavors and preservatives. Cooling prevents degradation of these additives and helps maintain their properties.
 4. **Dual chamber design:** The machine is designed with two chambers to facilitate the mixing process. The inner chamber is dedicated to high-temperature mixing and contains a steam line to inject heat. In contrast, the outer chamber allows water to flow through to cool the mixture.
 5. **Reaction:** Mixing in the machine takes place at an elevated temperature. An optimum temperature was chosen to ensure efficient mixing of the components and to facilitate reactions between the components. To maintain the properties of additives such as preservatives, flavors or fragrances, they are added at a lower temperature.
 6. **Insulation:** The machine is carefully insulated to minimize heat transfer to the surroundings. Insulation is important for maintaining a work environment. In addition, isolation is designed in a proper way that no unintended contamination or mixing with formulations can take place.
 7. **Product:** Although the machine is versatile and capable of producing a wide variety of products, it is mainly used for the production of sticky formulations such as lotions and toothpastes.
 8. **Origin:** The machine is made in India.
 9. **Capacity:** The machine has a production capacity of 400 kg.

- **FRYMA:** This machine has so many advanced features. This is also an automated machine. Its features are given below:
 1. **Blending Capability:** The machine offers progressed blending capabilities and is reasonable for mixing a wide extend of fixings. Its flexibility permits it to handle different sorts of mixings, both sticky and non-sticky ingredients.
 2. **Vacuum system:** The machine integrates a vacuum system, allowing to mix the adhesive materials evenly. The vacuum system helps to rotate the blades properly and thus homogeneous mixture of these materials can be obtained.
 3. **Chamber:** The machine highlights a primary blending chamber where the essential blending process takes place. It is planned with a proper heating and cooling system.
 4. **Additional Chambers connecting main chamber:** In exception to the most blending chamber, the machine build with partitioned chambers for purposes. One chamber is committed to boil and add water, which guarantees exact control over the temperature amid the blending handle. Another chamber is planned for adding ingredients which are small in proportion like flavors, fragrances, or preservatives. For this mechanism, there is no need to open the main chamber cover and add ingredients which helps to keep proper temperature at main chamber and gives workers safety.
 5. **Warming System:** Heating for the blending handle is produced utilizing steam, which gives a productive and controlled source of warmth. This permits for exact temperature control and encourages legitimate mixing and reaction of raw ingredients.
 6. **Cooling system:** There are two types of cooling system in this machine. They are:
 7. The machine consolidates a water-cooling system to preserve typical temperature conditions when required.
 8. The machine consolidates a chiller to cool at very low temperature like three to four degrees Celsius.
 9. **Blending Devices:** The machine is manufactured with different blending devices to guarantee careful mixing of the ingredients. These instruments regularly incorporate a scraper, dissolver, and homogenizer. The scrapper viably rubs the sides of the blending chamber, guaranteeing that all mixings are equitably blended. The dissolver helps in breaking down bigger particles or agglomerates, advancing consistency in the mixture. The homogenizer assists upgrading the mixing handle by accomplishing a homogeneous consistency.

- 10. Control:** This is an automated machine and is prepared with a natural control board that permits workers to set and screen different parameters such as blending speed, temperature, and timing. This guarantees exact control and productive operation. The machine's operation is encouraged by pneumatic pressure, for example moving one part, opening, and closing valves.
- 11. Perception and Security:** The machine is outlined with security highlights, counting visual markers such as lights or show boards. There is a green button. If one switches on the button, the light inside the mixing chamber is on and workers can observe the mixing from outside. This increases accuracy and workers' safety.
- 12. Product:** All types of mixing can be done but mainly sensitive mixing is done here, because there are available machines for normal mixing.
- 13. Origin:** Made in
- 14. Capacity:** This machine has a capacity of 300 kg.



Figure 5.5.1.2: FRYMA machine



Figure 5.5.1.3: Different typrs of blades of FRYMA machine

- **PETZHOLE:** There are two 'Petzhole' machines and they are used for mixing mainly Tibet snow, hand wash or other sticky materials. Here are some specifications of it.
 1. **Mixing:** This machine can mix all kinds of things, but it's especially good for things that are sticky like Snow, Shaving cream, or Hand Wash.
 2. **Vacuum system:** The vacuum system helps mix sticky ingredients. It works for things that stick and things that don't stick.
 3. **Heating and cooling system:** The machine has a heating system that uses steam to make the mixing chamber hotter, which helps mix the ingredients and make them react correctly. It uses water to cool down, especially when you're putting in stuff like smells, flavors, and preservatives.
 4. **Dual Chamber design:** This machine has two parts called the inner chamber and outer chamber. The inside room is for mixing things at very hot temperatures. It has a pipe that sprays hot steam to make it even hotter. The outside part helps water flow to cool the mixture.
 5. **Reaction:** Mixing in the machine takes place at an elevated temperature to ensure efficient mixing of the components and to facilitate reactions between the components. To maintain the properties of additives such as preservatives, flavors, or fragrances, they are added at a lower temperature.
 6. **Insulation:** Machine is insulated to control the temperature around it. The insulation is designed in such a way mixed up with other things.
 7. **Product:** This machine can make many things, but it works best with sticky products like Snow, Shaving cream, and Hand Wash.
 8. **Origin:** The machine is made in India.
 9. **Capacity:** This machine can make up to 400 kilograms of product.

- **Oil Mixing Machine:** This is a simple machine with a motor and blades for mixing oil. Some specification of it is given below:
 1. **Container size:** Oil blenders may have large containers or tanks to hold large amounts of oil and other additives. Container sizes vary depending on the specific application and desired batch size.
 2. **Mixing blades:** Blades in the container serve to stir and mix the oil and additives. These blades are usually attached to a motor or drive shaft that rotates the blades.
 3. **Mixing at ambient temperature:** The mixing of oils and additives does not require special temperature conditions. So, there is no temperature rise and as a result no insulation is needed.
 4. **Viscosity considerations:** The viscosity of the oil being mixed can affect the design and performance of the mixing blade. If the oil is highly viscous, the mixing blades can be specially designed to efficiently handle and thoroughly mix such liquids.

5.5.1.2 Filling Machine

The mixed cosmetic formulations are dispensed into containers by filling machines, which are created to do so precisely and effectively. Depending on the consistency and viscosity of the product being filled, they can be fitted with a variety of filling mechanisms, including piston fillers, peristaltic pumps, or gravity fillers. Filling machines frequently have settings that can be changed to regulate the amount of product that is dispensed, enabling accurate and reliable filling.

In the cosmetic section, filling machines are used for snow filling and oil filling. They are described below:

- **Snow filling:** A semi-automatic machine is used for snow filling. The machine features are given below:
 1. **Snow store:** Workers manually fill a large container with snow. The container serves as a reservoir for the snow. A convex mirror is installed to visually check the height or volume of snow in the container. This ensures that the container is properly filled.
 2. **Snow nozzle:** The machine is equipped with a snow distribution hose connected to the snow storage tank. A time sensor controls the nozzle operation, activating it to deliver

snow every 15 seconds. The hose discharges a controlled amount of snow into the snow jar below.

3. **Rotary disc:** The snow pots or jars are manually placed by the worker on a rotating disc. The speed of rotation of the disc is controlled so that each snow jar passes through the hose after a fixed time, in this case 15 seconds. This controlled rotation ensures that each snow pot receives a consistent amount of snow from the nozzle.
 4. **Manual handling:** The process of placing the snow jars on the rotating disc and taking them out after it is full is done manually by the workers. Workers load empty snow jars onto rotating discs at designated locations. After the snow pots are passed through the nozzle and filled, workers manually remove them from the plate.
- **Oil filling:** There are two types of oil filling machine. One is automatic and the other is semi-automatic. They are described below:
 - **Semi-automatic:** This machine is used to fill ‘Pumpkin heir oil’. It has a huge capacity. Its nature is described below:
 1. **Time sensor:** A time sensor is installed to control the operation of the filling process. After a set amount of time, in this case 15 seconds, the timing sensor activates the oil delivery mechanism.
 2. **Oil distribution hose:** The machine is equipped with an oil distribution nozzle connected to the tank. When activated by the time sensor, the nozzle dispenses oil into the bottle.
 3. **Employee role:** A worker is responsible for manually placing the bottles under the oil dispenser nozzle. After one bottle is filled, the worker places another bottle under the nozzle for sequential filling.
 4. **Number of bottles:** This process allows for the filling of two bottles at once, with workers ensuring that the bottles are in place for filling.
 5. **Finishing:** After the bottles are filled, workers remove them from the filling area and prepare them for further processing or packaging. Filled bottles may undergo additional steps, such as capping, labeling or sealing, depending on specific requirements.
 - **Automatic:** This machine is used for filling ‘Beutina heir oil’. It has the capability of filling oil with any waste. Its nature is given below:

1. **Position sensors:** The machine is equipped with a position sensor that detects the presence of empty oil bottles on the conveyor.
2. **Conveyor:** The automatic conveyor moves the empty bottles to the filling station.
3. **Oil reservoir:** The machine is connected to an oil tank where the oil is stored. Oil flows from the storage tank to the filling station without manual intervention.
4. **Oil nozzles:** The filling station is equipped with one or more nozzles for the simultaneous filling of several bottles.
5. **Filling process:** When the position sensor detects an empty bottle on the conveyor, it activates the filling process. The machine activates the respective nozzles, and the oil is poured into the bottle. The filling process takes place automatically without requiring bottle placement or manual labor.
6. **Handling multiple bottles:** The machine is designed to process two bottles at the same time. The conveyor simultaneously moves two empty bottles to the filling station. The respective nozzles distribute the oil to both bottles at the same time.
7. **Completion and further processing:** After the bottles are filled, they continue along the conveyor belt for further processing or packaging. This may include capping, labeling or additional steps depending on specific requirements.

5.5.1.3 Capping Machine

There are different types of capping machine. They can be automatic or semi-automatic machines. According to capping mechanism, two types of capping machine were used in cosmetic section. One machine uses a pressing mechanism and the other uses a turning mechanism which finally cuts threads on the lid. They are described below:

- **Capping machine with pressing mechanism:** It is a semi-automatic capping machine. Its features are described below:
 1. **Manual Cap Arrangement:** Laborers physically put caps on top of cans or bottles. The caps are picked up by hand and situated on the holder opening. Caps used in this machine are threaded before.
 2. **Capping or Squeezing Machine:** Once the cap is physically put on the holder, a pressing or squeezing machine is utilized to secure the cap in place. This guarantees

that the cap is safely secured, giving a tight seal to preserve item keenness and anticipate spillage. The capping machine can be semi-automatic, meaning that it requires manual input from the laborer to start the capping process.

3. **Number of Bottle Capping:** Some capping machines are outlined to cap multiple bottles at the same time. These machines can cap two bottles, four bottles, or more in a single operation. This includes increments efficiency by diminishing the time required to cap numerous bottles.
 4. **Product:** This machine is used for 'Tibet pumpkin oil' and 'Butina hair oil' container capping.
- **Capping machine with turning mechanism:** It is a semi-automatic capping machine. Its features are described below:
 1. **Manual Cap Arrangement:** Laborers physically put caps on top of cans or bottles. The caps are picked up by hand and situated on the holder opening. Caps used in this machine are not previously threaded so they cannot be capped by pressing.
 2. **Turning Machine:** Once the caps are physically put on the holder and placed under the handle, the machine applies rotational force to tighten the cap. Rotational motion is created by cam mechanism. As the cap is twisted and cut threads on the cap for precise alignment with bottles, it engages with the container's threads and creates a secure closure.
 3. **Number of Bottle Capping:** In this machine, several number of bottles can't be capped at a time. Capping is done one by one.
 4. **Product:** This machine is used for 'Tibet snow' container capping.

5.5.1.4 Shrinking Machine

A shrinking machine or shrink-wrapping machine is a device used to package products by tightly wrapping them with a plastic film and shrinking the film using heat. This process creates a sealed and protective covering around the product.

Here are some key features and functionalities of shrinking machines:

1. **Packing and sealing:** Workers manually wrap plastic stickers and labels around containers and pack products. The label contains product and company information.
2. **Heat shrinking:** The shrinking machine heats the plastic film wrapped around the product. A heat tunnel or heat gun is commonly used for this process. The heat causes the film to shrink and fit the shape of the product.
3. **Conveyor speed:** Conveyor speed is adjusted to control product movement in the heat tunnel or in front of the heat gun. Proper synchronization ensures even heat exposure and effective shrinkage.
4. **Thermal control:** Temperature on the control panel is set according to the material to be wrapped. Low temperature for plastics and high temperature for materials such as glass and metals.
5. **Additional features:** Some machines have adjustable fans or cooling mechanisms for faster cooling after deflation. These features help maintain production efficiency.
6. **Packaging:** In this cosmetic section, packaging is done manually. Some details about this hand packaging process are given below:
7. **Manual boxing:** Workers manually stack bottles, cans or other items into individual bundles or boxes. Packages or boxes can be pre-shaped or self-assembled by workers. Goods are carefully placed inside the individual package or box, ensuring they are stored safely and properly.
8. **Seal with adhesive tape:** After the items are placed in individual bundles or boxes, workers seal the package or boxes with adhesive tape. Tape provides a secure closure to prevent items from falling or being damaged during transit.
9. **Packed in cartons:** After sealing the bundles or boxes, workers pack them into larger containers, usually cartons. The number of goods packed in each carton may vary depending on the specific product and packaging requirements. Workers ensure that bins are efficiently and safely filled, optimizing space utilization while maintaining the integrity of packaged items.

10. Loading and unloading: After the bins are filled, workers handle and stack them properly. Cartons can be stacked on pallets or placed on shelves for further storage or transport.



Figure 5.5.1.4: Heat shrinking machine

5.5.1.5 Packaging

In this cosmetic section, packaging is done manually. Some details about this hand packaging process are given below:

1. **Manual boxing:** Workers manually stack bottles, cans or other items into individual bundles or boxes. Packages or boxes can be pre-shaped or self-assembled by workers. Goods are carefully placed inside the individual package or box, ensuring they are stored safely and properly.
2. **Seal with adhesive tape:** After the items are placed in individual bundles or boxes, workers seal the package or boxes with adhesive tape. Tape provides a secure closure to prevent items from falling or being damaged during transit.
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4. **Loading and unloading:** After the bins are filled, workers handle and stack them properly. Cartons can be stacked on pallets or placed on shelves for further storage or transport.

5.5.1.6 Image Printing

An image printing machine is a gadget utilized to duplicate computerized pictures or photos onto different surfaces such as paper, canvas, texture, or other materials. These machines utilize progressed printing advances to convey high-quality and dynamic prints.

Here are a few key highlights and functionalities of picture printing machines:

1. **Printing Technology:** Picture printing machines utilize different printing advances, counting inkjet, laser, or dye-sublimation, depending on the show and application. Inkjet printers utilize minuscule ink beads that are splashed onto the printing surface. They are flexible and able to create high-resolution prints. Laser printers utilize laser bars to exchange toner onto the printing surface, coming about in sharp and exact prints. Dye-sublimation printers utilize warm to exchange color onto uncommon papers or movies, which are at that point exchanged onto the printing surface. This strategy is frequently utilized for high-quality photo prints.

2. **Print Resolution:** Picture printing machines come in several sizes and offer different print resolutions. They can create prints. The print resolution determination decides the level of detail and sharpness within the printed picture.
3. **Print Media Compatibility:** These machines are planned to work with a wide run of print media, counting distinctive sorts and weights of paper, photo paper, canvas, vinyl, texture, and more.
4. **Print Speed:** The print speed shifts depending on the printer show, determination, and estimate of the prints. A few machines can create prints in seconds, whereas others may require more time for bigger or higher-quality yields.

5.5.2 Machines used in first floor of cosmetics department

Production on the first floor focuses on seasonal products. In winter, they produce ‘Tibet petroleum jelly’, Pomet, and other related products. In the summer, production is switched to ‘Tibet talcum powder’ and ‘Ice cool powder’. The production process involves the use of various equipment including:

- A shrinking machine, also known as a shrink wrap machine or shrink tunnel, is a device used to apply heat to shrink plastic film tightly around a product or a group of products. It is commonly used in the packaging industry to provide a secure and professional finish to various items.
- Mixing machine is responsible for mixing and combining the necessary ingredients for seasonal products. It consists of a large container in which ingredients are mixed together using motors.
- Filling machines are used to fill the mixed cosmetic formulations into containers or packaging. These machines are designed to accurately dispense the desired amount of product into each container, ensuring proper measurement and minimizing waste.
- A key piece of machinery used in the cosmetics industry to automate and streamline the packaging of cosmetic items is the packaging machine. These devices are made to accurately and efficiently package the goods into different containers or packaging materials. They are capable of handling a variety of package types, including bottles, tubes, sachets, and blister packs.

With the help of these machines, seasonal products are made.

5.5.2.1 Shrink rapping machine:

A shrinking machine or shrink-wrapping machine is a device used to package products by tightly wrapping them with a plastic film and shrinking the film using heat. This process creates a sealed and protective covering around the product.

Here are some key features and functionalities of shrinking machines:

1. **Packing and sealing:** Workers manually wrap plastic stickers and labels around containers and pack products. The label contains product and company information.
2. **Heat shrinking:** The shrink machine heats the plastic film wrapped around the product. A heat tunnel or heat gun is commonly used for this process. The heat causes the film to shrink and fit the shape of the product.
3. **Conveyor speed:** Conveyor speed is adjusted to control product movement in the heat tunnel or in front of the heat gun. Proper synchronization ensures even heat exposure and effective shrinkage.
4. **Thermal control:** Temperature on the control panel is set according to the material to be wrapped. Low temperature for plastics and high temperature for materials such as glass and metals.
5. **Additional features:** Some machines have adjustable fans or cooling mechanisms for faster cooling after deflation. These features help maintain production efficiency.

5.5.2.2 Powder Mixing Machine:

A powder mixer is used to mix and blend the raw materials used in the production of powders such as 'Tibet powder' and 'Ice cool powder'. The machine is described below:

1. **Addition of raw materials:** The raw materials required for the powder formula are poured into the mixing chamber of the machine. These materials may contain ingredients depending on the desired powder formula.
2. **Mixing Chamber and Motor:** After adding the raw material, the mixing chamber is sealed, and the machine's engine is turned on. The motor rotates the mixing chamber horizontally, providing a thorough mixing of the raw materials. Rotation ensures even distribution of the ingredients and promotes efficient mixing.

3. **Filtration for ‘Ice cool powder’:** ‘Ice cool powder’ has an additional filtration step integrated into the mixing process. This is because ‘Ice cool powder’ often contains substances such as menthol and fragrance, which require additional filtration to remove impurities or oversized particles.
4. **No filtration for ‘Tibet powder’:** On the other hand, ‘Tibet powder’ usually does not undergo filtration during the mixing process. The ‘Tibet powder’ formula should not contain any ingredients that require filtration. Instead, the focus is on the thorough mixing of raw materials to obtain a homogeneous mixture.

The powder mixer is designed to mix different powder ingredients for efficient and consistent results. It ensures that the raw materials are evenly distributed and mixed to produce high-quality powders.

5.5.2.3 Powder filling machine:

A powder mixer is used to mix and blend the raw materials used in the production of powders. There are two types of powder filling machines. Their features are described below:

- **One head powder filling machine:** This machine can fill one bottle powder at a time, so it is called one head powder filling machine. The key features are given below:
 1. **Manual powder placement:** The workers fill the machine container with powder by scooping from large powder storage.
 2. **Manual powder bottle placement:** A worker manually places a container, such as a bottle or can, under the filler neck. The bottle is to hold the powdery substance.
 3. **Fixed amount of powder:** The worker sets the desired amount of powder to be sprayed into the bottles. This can be done through changing settings on control panel on the machine. Thus, the operator determines the weight of the filled powder.
 4. **Time sensor:** With the help of time sensor, after a fixed time nozzle disperses the powder.
 5. **Powder nozzle:** After placing the container and setting the desired amount, the nozzle of the machine sprays or pushes the powder substance into the container. The powder is given according to the amount determined by the worker.

6. **Worker Check:** After the container is filled, another worker checks the weight of the filled box using a digital weight machine. This step ensures that the correct amount of powder has been dispensed and that it meets the required weight or volume specifications.
- **Four head powder filling machines:** This machine can fill four powder bottles at a time, so it is called four head powder filling machines. The key features are given below:
 1. **Manual Powder Placement:** The workers fill the machine container with powder by scooping from large powder storage.
 2. **Manual powder bottle placement:** A worker manually places a container, such as a bottle or can, under the filler neck. The container serves to hold the powdery substance.
 3. **Fixed amount of powder:** The worker sets the desired amount of powder to be poured into the container. This can be done using the controls or settings on the machine. The worker determines the weight of the powder to be filled.
 4. **Time sensor:** With the help of time sensor, after a fixed time nozzle disperses the powder.
 5. **Powder measuring:** After the required amount is set, the machine's nozzle will spray or disperse the powdered substance into the bottle. The powder is dosed according to the specified amount.
 6. **Worker Check:** After filling the bottles, another worker checks the weight of the filled bottles using a digital weight machine. This ensures the correct amount of powder has been dispensed and meets the required weight.

5.5.2.4 Capping Machine:

It is a semi-automatic capping machine. Its features are described below:

1. **Manual Cap Arrangement:** Labourers physically put caps on top of cans or bottles. The caps are picked up by hand and situated on the holder opening. Caps used in this machine are threaded before.

2. **Capping or Squeezing Machine:** Once the cap is physically put on the holder, a pressing or squeezing machine is utilized to secure the cap in place. This guarantees that the cap is safely secured, giving a tight seal to preserve item keenness and anticipate spillage. The capping machine can be semi-automatic, meaning that it requires manual input from the labour to start the capping process.
3. **Number of Bottle Capping:** Some capping machines are outlined to cap multiple bottles at the same time. These machines can cap two bottles, four bottles, or more in a single operation. This includes increments efficiency by diminishing the time required to cap numerous bottles.
4. **Product:** This machine is used for both 'Tibet powder' and 'Ice cool powder' container capping.

5.5.2.5 Packaging machine:

In this cosmetic section, packaging is done manually. Some details about this hand packaging process are given below:

1. **Manual boxing:** Workers manually stack bottles, cans or other items into individual bundles or boxes. Packages or boxes can be pre-shaped or self-assembled by workers. Goods are carefully placed inside the individual package or box, ensuring they are stored safely and properly.
2. **Seal with adhesive tape:** After the items are placed in individual bundles or boxes, workers seal the package or boxes with adhesive tape. Tape provides a secure closure to prevent items from falling or being damaged during transit.
3. **Packed in cartons:** After sealing the bundles or boxes, workers pack them into larger containers, usually cartons. The number of goods packed in each carton may vary depending on the specific product and packaging requirements. Workers ensure that bins are efficiently and safely filled, optimizing space utilization while maintaining the integrity of packaged items.
4. **Loading and unloading:** After the bins are filled, workers handle and stack them properly. Cartons can be stacked on pallets or placed on shelves for further storage or transport.

5.5.3 Demineralized water plant

A demineralized water plant, moreover, known as a demineralization plant or a deionization plant, may be a process outlined to deliver high-purity water with negligible mineral substance. Demineralized water is commonly utilized in different businesses such as cosmetics, pharmaceuticals, hardware fabricating, and research facility applications where water quality is vital. Few key points of a demineralized water plant are given below:

1. **Purpose:** The most reason of a demineralized water plant is to expel pollutions, minerals, salts, and other broken up solids from the water supply, coming about in exceedingly decontaminated water.
2. **Pre-treatment:** The water entering the demineralization plant experiences pre-treatment to evacuate bigger particles, suspended solids, natural matter, and certain contaminants. Pre-treatment forms ordinarily incorporate filtration, sedimentation, coagulation, and in some cases sanitization to guarantee the water is free from any potential microbial contamination. For this purpose, there is a process for iron filtration and then the water is filtered in activated carbon. When the water is fully pre-treated then it goes through other process for demineralising.
3. **Demineralization Process:** After pre-treatment, water goes through two anion and cation plant. Basically, there are two main processes for removing ions from water in a water purification system: cation exchange and anion exchange. They are described below:
 - a. **Cation Exchanger:** This process uses a cation exchange resin such as R-H resin. The resin is usually in the form of small beads or granules containing hydrogen ions (H^+). When water containing cations such as Na^+ , Ca^{2+} or Mg^{2+} comes into contact with the resin, the cations bind to the resin and displace hydrogen ions. The cations are effectively replaced with hydrogen ions, leaving the water cation-free.
 - b. **Anion exchanger:** Similarly, an anion exchanger involves the use of an anion exchange resin such as R-OH resin. The resin is also made up of small beads or granules, but this time they contain hydroxide ions (OH^-). When water containing anions such as SO_4^{2-} or PO_4^{3-} comes into contact with the resin, the anions are attracted to the resin and replace the hydroxide ions. The anions are replaced with hydroxide ions, removing anions from the water and creating water without anions.

By combining cation and anion exchange processes in a water treatment system, both cations and anions can be effectively removed, resulting in demineralized water. Compared to the original water, this water has a significantly reduced mineral content and a significantly lower ion concentration. In general, cation and anion exchange processes play a crucial role in the production of high-quality deionized water by selectively removing cations and anions from water, as water used in cosmetic production must be high.

4. Regeneration: Actually, ion exchange resins have a limited ability to remove ions. Over time, the resin becomes saturated with trapped ions and must be regenerated or replaced to restore its ion exchange capacity. During regeneration, specific regenerating solutions are passed through the resin to release trapped ions and recharge the resin for later use. Regeneration of cation and anion is given below:

- a.** Regeneration by cation exchange: During cation exchange, if the cation exchange resin is saturated with cations such as Na^+ , Ca^{2+} or Mg^{2+} , it must be regenerated to restore its ion exchange capability. The regeneration process involves the use of a regeneration solution, usually a strong acid such as hydrochloric acid (HCl). The regenerating solution flows through the resin bed, displacing trapped cations and replacing them with hydrogen ions (H^+). This regeneration step effectively rejuvenates the resin and makes it available for further cation exchange.
- b.** Anion Exchange Regeneration: Similarly, with an anion exchange resin, if the anion exchange resin is saturated with anions such as Cl^- , SO_4^{2-} or PO_4^{3-} , regeneration is required. The anion exchange regeneration process uses a regeneration solution that contains a strong base such as sodium hydroxide (NaOH). The regenerating solution is passed through the resin bed, displacing the trapped anions and replacing them with hydroxide ions (OH^-). This regeneration step restores the anion exchange capacity of the resin.

Recycled resins can be reused to remove cations and anions from water in subsequent cycles of the ion exchange process. The spent regeneration solution containing trapped ions is normally handled or disposed of properly to avoid environmental pollution. Regeneration of cation and anion exchange resins is an important step in maintaining the efficiency and longevity of the ion exchange process in water treatment systems.

- 5. Filtration:** The demineralization of water mainly involves the removal of minerals such as calcium, magnesium, sodium, etc. However, microorganisms such as bacteria, viruses or other microorganisms are not effectively removed from the water. Additional disinfection measures are necessary to resist the destruction of microorganisms. A commonly used method is ultraviolet (UV) filtration. UV filtration utilizes ultraviolet light to disinfect water by targeting and disrupting the DNA of microorganisms, rendering them inactive and unable to reproduce. This process helps to ensure that the water is free from harmful microorganisms. UV filtration systems typically consist of UV lamps that emit UV light and a flow chamber through which the water passes. As the water flows through the chamber, it is exposed to the UV light, which effectively neutralizes or eliminates microorganisms present in the water. UV filtration is a widely used method for disinfecting water in various applications, including drinking water treatment, wastewater treatment, and industrial processes. Besides, UV filtration is primarily effective against microorganisms and does not remove other types of contaminants such as chemicals, heavy metals, or particles. Therefore, in a complete water treatment system, UV filtration is often combined with other filtration and disinfection methods to ensure a safe and clean water supply.
- 6. Quality Monitoring:** Demineralized water quality is closely observed to guarantee it meets the specified guidelines and specifications. Parameters such as conductivity, pH, add up to broken up solids (TDS), and particular particle concentrations are routinely measured to evaluate the water's purity.
- 7. Storage and Distribution:** The demineralized water is put away in committed tanks or stores to preserve its purity. There are two storage tanks with capacity 2000kg. Conveyance systems, including pumps and pipelines, are utilized to convey the demineralized water to the specified focuses of utilize inside the facility.



Figure 5.5.3.1: DM Water plant

5.6 Process flow chart of cosmetic products

Flow charts of some cosmetic products are given below:

5.6.1 Tibet Snow

Raw Materials: Stearic acid, Cetyl alcohol, Aqua, Perfume, Preservatives.

Process Flow chart:

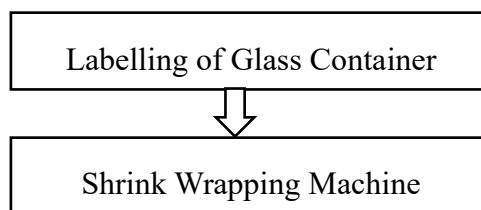


Figure 5.6.1.1: Flowchart of preparation of glass containers for Tibet Snow

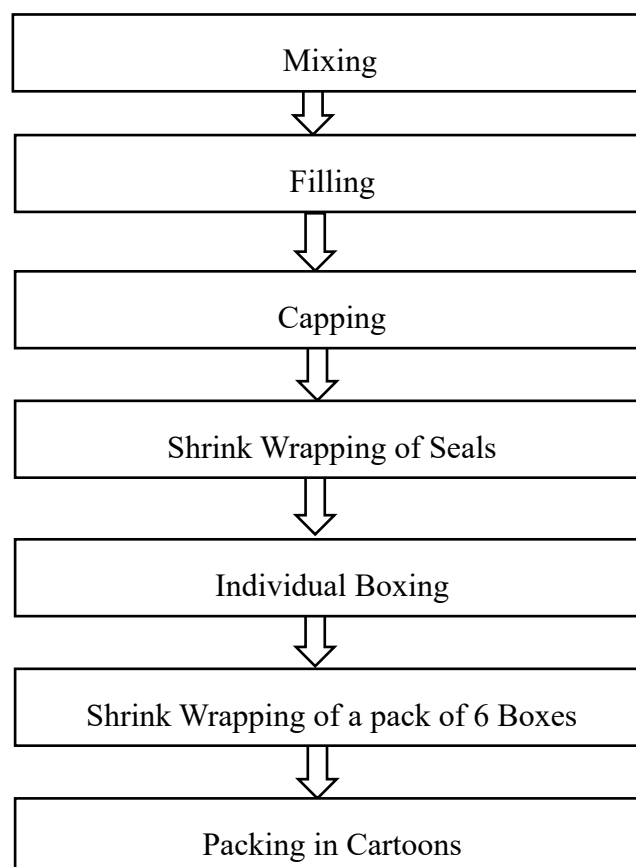


Figure 5.6.1.2: Flowchart of production process of Tibet Snow

Process: Glass containers are prepared by putting the label on them and passed into a shrink-wrapping machine at 190°C where the labels are shrink wrapped. Besides, they are also sterilized in this process. The solid raw materials are melted and then mixed together in a mixer. At the later stage of the mixing, fragrance is added so that its intensity remains. Preservatives are also added. Then a predetermined volume is filled in the glass containers automatically using a filling machine. Afterwards, caps are manually put on the glass containers and an automatic capping machine is used to fix the caps on the glass containers. Safety seals are put on and shrink wrapped on the glass containers and each are packed in boxes. A pack of 6 boxes are shrink wrapped with a plastic wrap and are finally stacked in cartons.

Layout of tibet snow process:

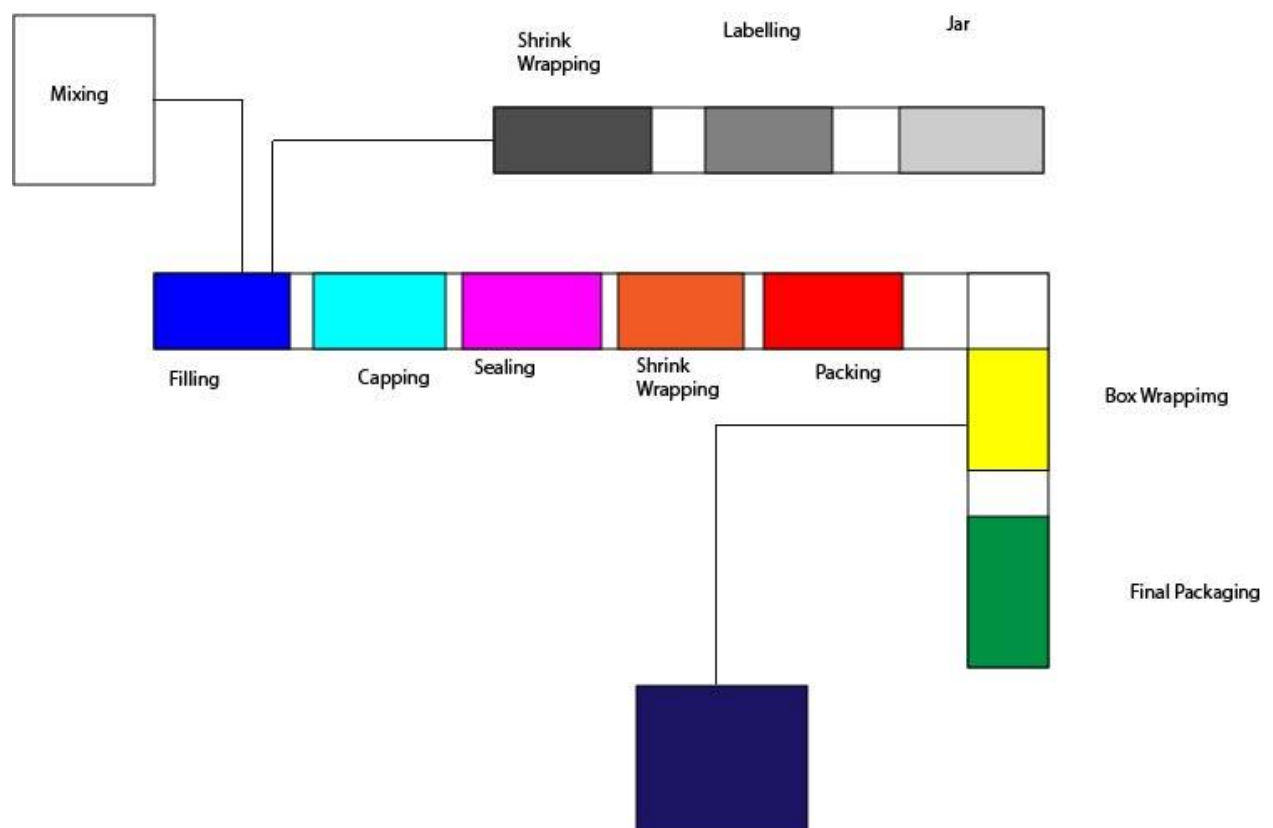


Figure 5.6.1.3: Process layout of tibet snow

5.6.2 Tibet Pumpkin Hair Oil

Raw Materials: Paraffin oil, Perfume, Color, Macadamia, Avocado, Jojoba, Wheat, Calendula, Chamomile, Seaweed, Vitamin.

Process flow chart:

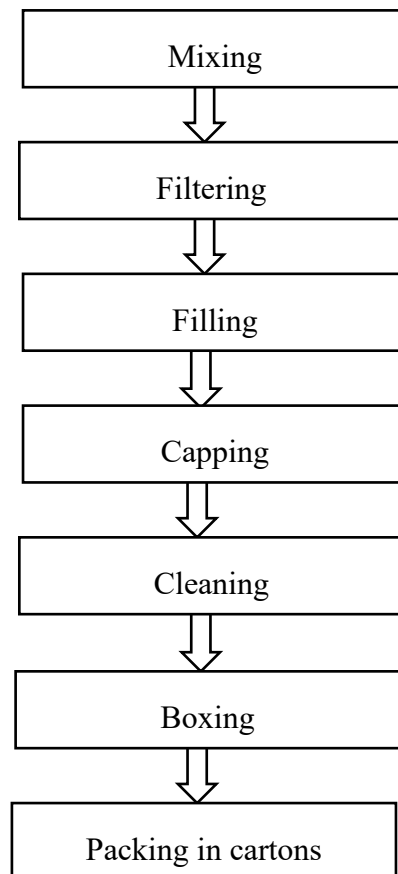


Figure 5.6.2.1: Flowchart of production process of Pumpkin hair oil

Process: Firstly, the bottles are labeled. Then paraffin oil is mixed with other additives in a mixer. Fragrance and color are added at a later stage for retaining their intensity. Preservatives are also added. The mixture is then filtered using a specialized filtering machine. Then a filling machine is used to fill 2 bottles at a time. Then caps are put on the bottles manually and the caps are then attached firmly to the bottles. Afterwards, the bottles are wiped to get rid of the excess oil. Each bottle is then inserted in a box and is then stacked in cartons.

5.6.3 Beautina Facewash

Raw Materials: Aqua, Sodium laureth sulfate, Sodium acrylates co-polymer, CAPB, Decyl glucoside, Sodium lauryl glucose carboxylate, De-panthenol, Lauryl glucoside, Phenoxy ethylene, Disodium EDTA, Citric Acid, Neem extract, Aloe Vera extract, CI 47005, CI 42090, Perfume.

Process flow chart:

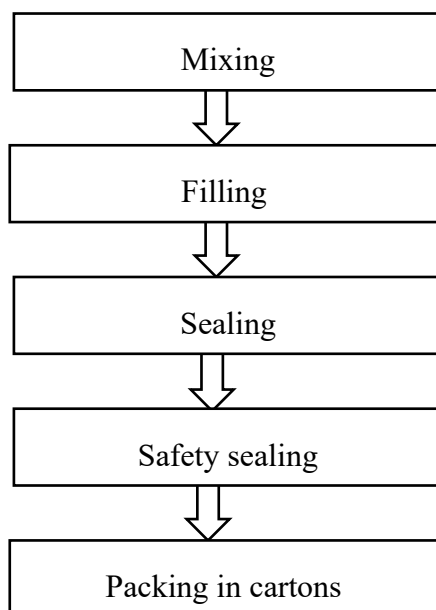


Figure 5.6.3.1: Flowchart of Beautina facewash production process

Process: The raw materials are mixed in a mixer. Fragrance and color are added at a later stage for retaining their intensity. Then a certain volume is filled in the containers. Next, the opening is sealed using 4 heaters at 225°C. A safety seal is applied over the product and shrink wrapped. Then they are packed in a honeycomb type carton to prevent them from being damaged.

5.6.4 Genstar lather shaving cream

Raw Materials: Potassium stearate, Glycerin, Aqua, perfume.

Process flow chart:

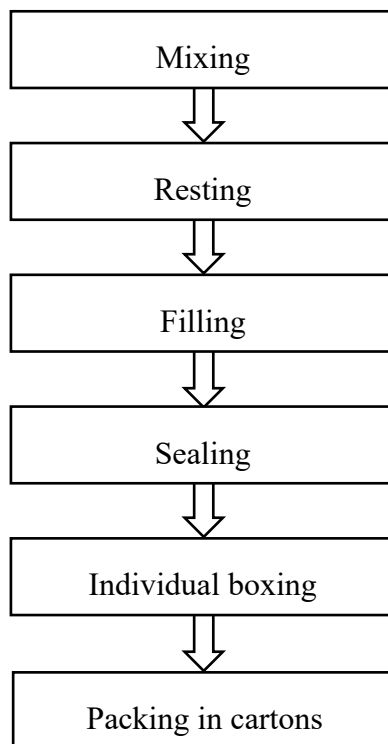


Figure 5.6.4.1: Flowchart of production process of Genstar lather shaving cream

Process: The raw materials are mixed in a mixer. Fragrance is added at a later stage for retaining its intensity. Preservatives are also added. Then the mixture is allowed to rest for about 1 week so that the neutralization reaction becomes complete. Then a certain volume is filled in the tubes. Next, the opening of the tube is sealed using heaters at high temperature. Each tube is inserted in boxes. They are then packed in cartons.

5.6.5 Xpert Dishwash

Raw Materials: Sodium LAS, SLAS, Concentrated lime juice, Disodium EDTA, Color, Aqua.

Process flow chart:

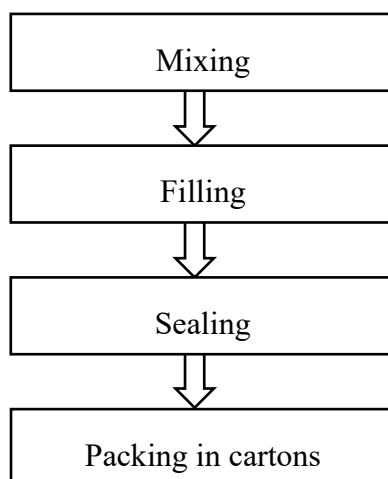


Figure 5.6.5.1: Flowchart of production process of Xpert dishwash

Process: The raw materials are added according to the temperature in the mixer and are mixed. Fragrance and color are added at a later stage for retaining their intensity. Preservatives are also added. Then a certain volume is filled in the containers. Next, the opening is sealed using heaters at high temperature automatically. They are then packed in cartons.

5.6.6 Am.Pm. Toothpaste

Raw Materials: Calcium carbonate, Sorbitol, Sodium lauryl sulphate, Cellulose gum, Silica, Sodium Mon fluorophosphate, Preservative, Flavor, Aqua.

Process flow chart:

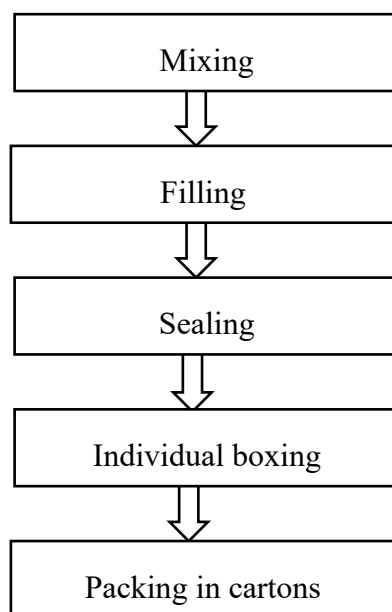


Figure 5.6.6.1: Flowchart of production process of Am Pm Toothpaste

Process: The raw materials are added in the mixer and are mixed. Fragrance is added at a later stage for retaining their intensity. Preservatives are also added. Then a certain volume is filled in the tubes. Next, the opening of the tube is sealed using heaters at high temperature automatically. Each tube is inserted in a box. They are then packed in cartons.

5.6.7 Bactrol Handwash

Raw Materials: Sodium lauryl ethyl sulphate, Cocodiethyl amide, TEA, Glycerin, Anti-microbial agent - decyl glycol and phenoxyethanol, perfume, CA-45100, preservatives, Aqua.

Process flow chart:

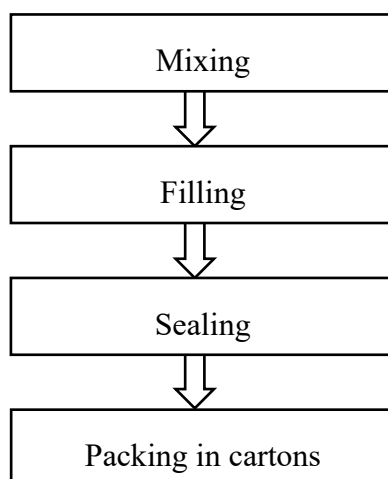


Figure 5.6.7.1: Flowchart of Bactrol handwash production process

Process: The raw materials are mixed in a mixer. Fragrance and color are added at a later stage for retaining their intensity. Preservatives are also added. Then a certain volume is filled in the containers. Next, the opening is sealed using heaters at high temperature. They are then packed in cartons.

5.6.8 Powders

Products: Kohinoor Chemicals Ltd. Bd produces three types of powders namely:

1. Tibet Luxury Talcum Powder
2. Tibet Prickly Heat Powder
3. Ice Cool Powder

Raw material: Magnesium Silicate, Magnesium Carbonate, Zinc Oxide

Process Flow Chart:

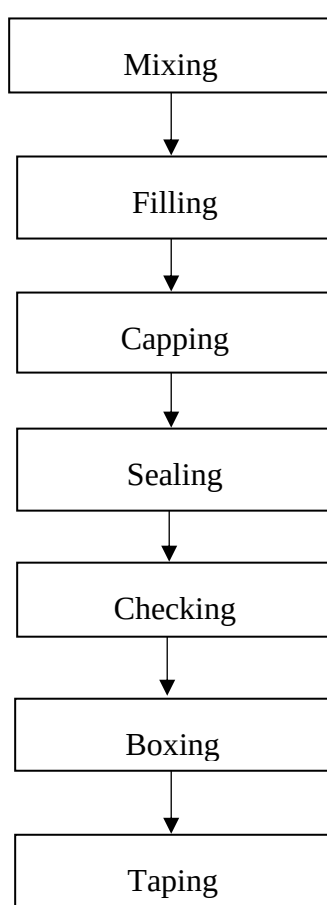


Figure 5.6.8.1: Process flow chart of powders

Process: Dry raw materials are mixed in mixing chambers. Finished Powders are stored in barrels and sent to packaging stations where filling machines are used.

Chapter 6: Quality Control Department

6.1 Introduction

The Quality Control (QC) Department of Kohinoor Chemicals Co. (Bd) Ltd. plays a crucial role in ensuring the products' quality and consistency. The primary function of the QC department is to implement and maintain the quality of the products' regulatory requirements and customer expectations.

Three sub-departments work under the QC department to ensure the quality of the products. Namely:

- QC (Testing)
- QA (Quality Assurance)
- R&D (Research and Development)

KCCL's QC department is one of its most sophisticated departments with modern testing equipment and trained QC professionals and chemists. The QC department runs 24/7 in three shifts. The quality control operations of KCCL's sister concern *Reedisha Foods and Beverages* are also performed in the QC department of KCCL.

6.2 Quality control

In KCCL testing is performed through a combination of analytical techniques and experimental procedures to assess the quality, composition, and safety of chemical products in the QC(Testing) sub-department. The testing process involves collecting representative samples from different production batches or raw materials and subjecting them to rigorous analysis using methods such as chromatography, spectroscopy, and titration. Inspection of packaging materials, sealing of the finished goods and aesthetics is also performed in this section.

Major functions performed in the QC sub-department are:

- **Raw Material Inspection:** The QC Department is responsible for thoroughly inspecting and testing all incoming raw materials, such as active ingredients, additives, and packaging materials, to ensure they meet the required standards and specifications. This includes

conducting various tests, such as microbiological analysis, chemical analysis, and physical testing.

- **Finished Product Testing:** Before any cosmetic product leaves the factory, it undergoes comprehensive testing by the QC Department. This involves analyzing samples to assess their overall quality, stability, appearance, and compliance with regulatory standards.
- **Compliance with Regulations:** The QC Department ensures that all the critical parameters of the cosmetic products comply with local and international regulations, such as those set by the Bangladesh Standard and Testing Institution (BSTI) and other relevant regulatory bodies.
- **Quality Documentation:** The QC Department is responsible for maintaining detailed documentation related to quality control activities. This includes recording test results, specifications, batch records, and other relevant information.

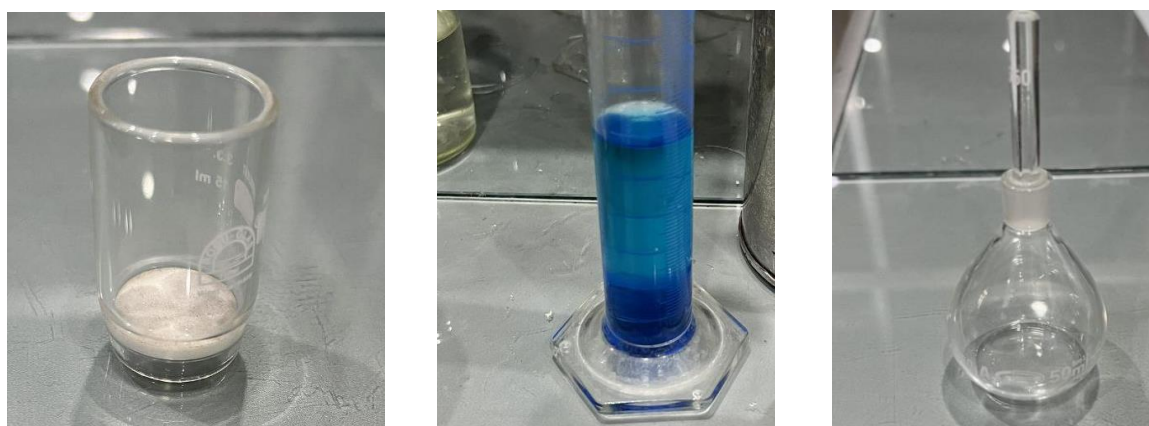


Figure 5.6.8.1: Some components used to check quality

6.3 Quality Assurance

The QA sub-department regularly inspects and tests the products' quality throughout the manufacturing process to ensure that they are consistent and of high caliber. This includes examining process attributes, confirming measurement accuracy, and assuring compliance with standard operating procedures (SOPs) and good manufacturing practices (GMP).

The QA professionals in KCCL are responsible for In-Process Quality Control (IPQC). in-process quality control (IPQC) is very important for the following reasons:

- **Early Defect Detection:** During the production process, IPQC assists in identifying and addressing possible faults or problems. It enables prompt intervention and remedial actions before the product reaches its final stage by monitoring crucial quality metrics at several stages. By doing this, waste, rework, and the creation of faulty or non-compliant items are reduced.
- **Consistency and Compliance:** IPQC makes ensuring that goods are consistently made in accordance with established quality standards and specifications. Throughout the production process, it aids in maintaining the required quality level by continuously monitoring and managing crucial elements. This is crucial in sectors with stringent regulatory requirements where adherence to quality standards is necessary.
- **Cost and Time Efficiency:** IPQC enables KCCL to save money and time. Early quality problem detection and correction helps to avoid expensive rework, product recalls, and customer complaints. By resolving problems quickly, it also reduces production downtime and delays, leading to increased effectiveness and productivity.



Figure 5.6.8.1: Quality assurance department

6.4 Research and Development

The R&D department's main goal is to create innovative products that satisfy evolving consumer wants and needs. In KCCL the Marketing and Branding departments are responsible for studying demand patterns and conceptualizing new product ideas. This entails staying current with fashion trends, researching consumer tastes, and performing market research to find market gaps. The R&D department then performs extensive research and formulates the product, so it meets the requirements set by the marketing department. The R&D team strives to develop cutting-edge products that offer higher performance, quality, and safety by staying at the forefront of scientific and technological breakthroughs.

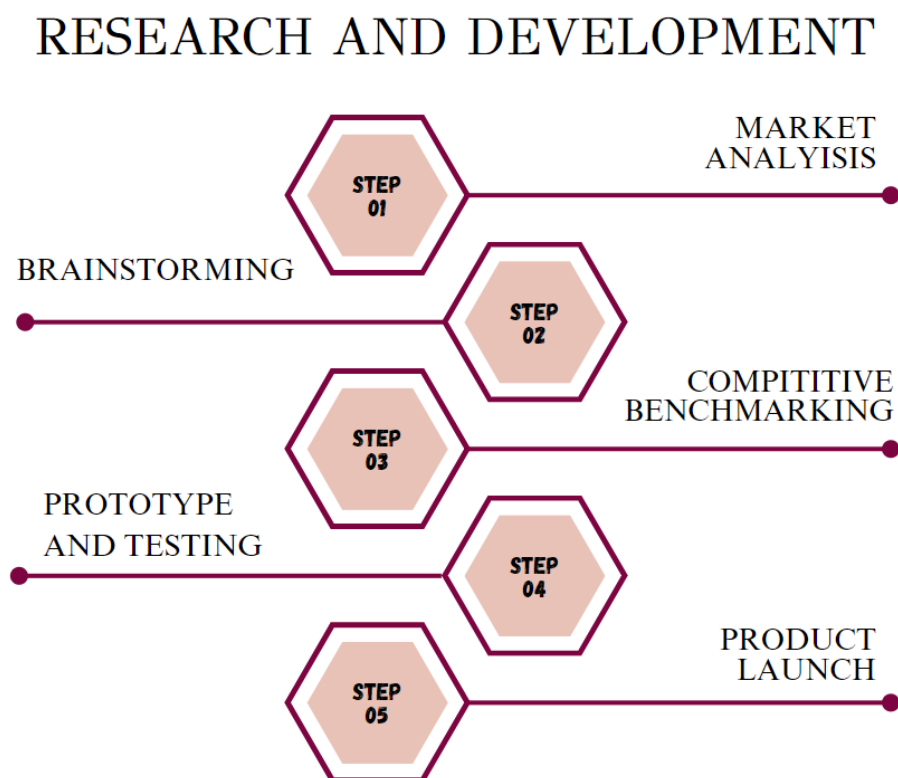


Figure 5.6.8.1: R&D process in Kohinoor Chemicals Co. (Bd) Ltd.

In KCCL, the R&D department collaborates with various professionals, including chemists, biologists, pharmacists, dermatologists, and engineers. This multidisciplinary approach allows for a comprehensive understanding of the diverse factors involved in cosmetic formulation,

such as chemical composition, biological effects, skin compatibility, and manufacturing processes.

The R&D department of KCCL uses competitive benchmarking and reverse engineering along with other product development techniques to formulate new products/improve existing products.

In KCCL, the marketing department and R&D professionals work hand in hand to develop new products and improve the existing to sustain in the highly competitive market.

6.5 Sampling Strategies

Sampling strategies in QC (Quality Control) are the systematic selection of representative samples from a larger population or batch to get an assumption about its quality and compliance to the standards. Both Stratified and Random sampling are common approaches used in the QC department of KCCL. In stratified sampling, the population is divided into homogeneous subgroups, or strata (batches, shifts, etc.) and samples are selected from each subgroup. Random sampling is another widely employed strategy, where samples are chosen randomly.

KCCL's QC department follows sampling strategies stated in *The British Pharmacopoeia (BP)*, which is less conservative and hence ideal for chemical products made for external usage only.

6.6 Test parameters

6.6.1 Soap

Seven important chemical parameters along with some physical properties are tested in soaps to make sure that the quality of a batch is acceptable. The seven chemical parameters are:

1. M&V (Moisture and Volatile material)
2. TFM (Total Fatty Matter)
3. FA (Free Alkali as NaOH)
4. FFA (Free Fatty Acid)
5. pH
6. Glycerin Content
7. Chloride as NaCl

Apart from these chemical properties, some physical aspects such as color, shape, weight etc. are also tested by visual inspection. The chemical parameters are tested for a few samples from each batch as all the soaps of a certain batch are made from the same neat soap/soap noodles. And for the physical aspects, a sample size of 10% of the total batch size is tested.

6.6.2 Cosmetic Products

The parameters tested vary from product to product as a very diverse range of cosmetic products are produced in KCCL. However, the most common chemical properties that are tested to ensure the quality of almost all cosmetic products are:

1. pH
2. FFA (Free Fatty Acids)
3. Density
4. Moisture Content
5. Color
6. Scent

The sampling scheme to test these properties is the same as the sampling scheme used for the QC of soaps. However, liquid final products are tested for leakage where a sample size of 2% of the total lot size is taken for testing.

6.6.3 Detergents

The important chemical parameters that are tested to make sure that the quality of a batch of detergent powder is acceptable are:

1. AD (Active Detergent)
2. Moisture Content
3. pH
4. Foam Volume

6.6.4 Packaging Materials

Customers first look at the external appearance of the product before they actually get to know about the product's quality. So, the quality of the packaging is a crucial factor in a product's success. In KCCL, it is the QC department's responsibility to check these in packaging materials:

1. Color
2. Print Quality
3. Packaging Material Quality

6.7 Machinery used in Quality control department

6.7.1 Moisture Oven

A moisture oven, often referred to as a drying oven or a moisture balance, is a laboratory tool used to dry a sample and then measure its moisture content. To make it easier to dry out the sample, the oven offers a controlled atmosphere with adjustable temperature and airflow. Moisture balances provide more accurate results than IR moisture balances.



Figure 6.7.1.1: Gallenkamp Moisture Oven

6.7.2 PH Meter

The pH of cosmetic products can significantly impact their compatibility with the skin. The skin's natural pH is slightly acidic, typically around 5.5. Deviations from this pH level can disrupt the skin's protective barrier and lead to irritation, dryness, or other adverse reactions.



Figure 6.7.2.2: PH meter

6.7.3 Microbiology Lab

The QC department is well-equipped with its own microbiology laboratory to ensure that the products are free from any kinds of harmful microorganisms. Cosmetics come into direct contact with the skin, making it crucial to ensure they are free from the presence of bacteria, yeast, mold, and other microorganisms that may cause infections or adverse reactions in consumers. Microbiology testing also helps to assess the effectiveness of preservatives in cosmetic formulations. By monitoring microbial growth over time, the chemists can determine the product's shelf life and ensure it remains stable and safe throughout its intended use.

KCCL's microbiology lab consists of: Biobased Biosafety Cabinet, Labtech oven and Incubator, and an autoclave.

The steps followed by chemists of KCCL for microbiology testing is illustrated below:

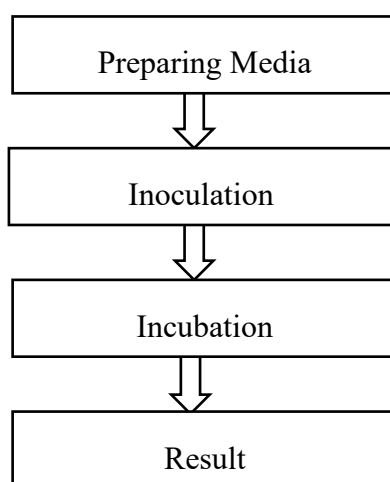


Figure 6.7.3.1: Steps involved in Microbiology test



Figure 6.7.3.2: BioBase Bio-Safety Chamber

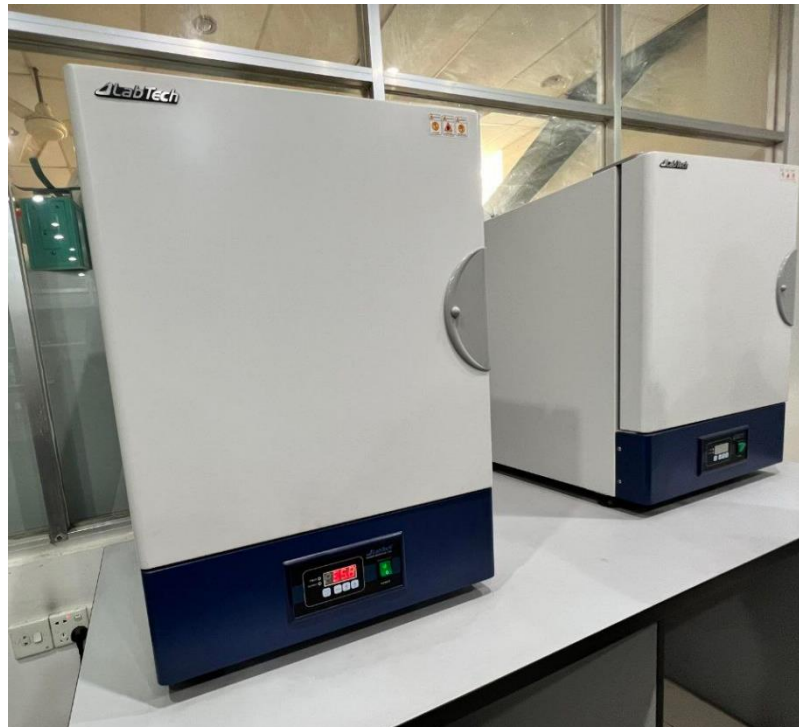


Figure 6.7.3.3: Labtech chamber and incubator

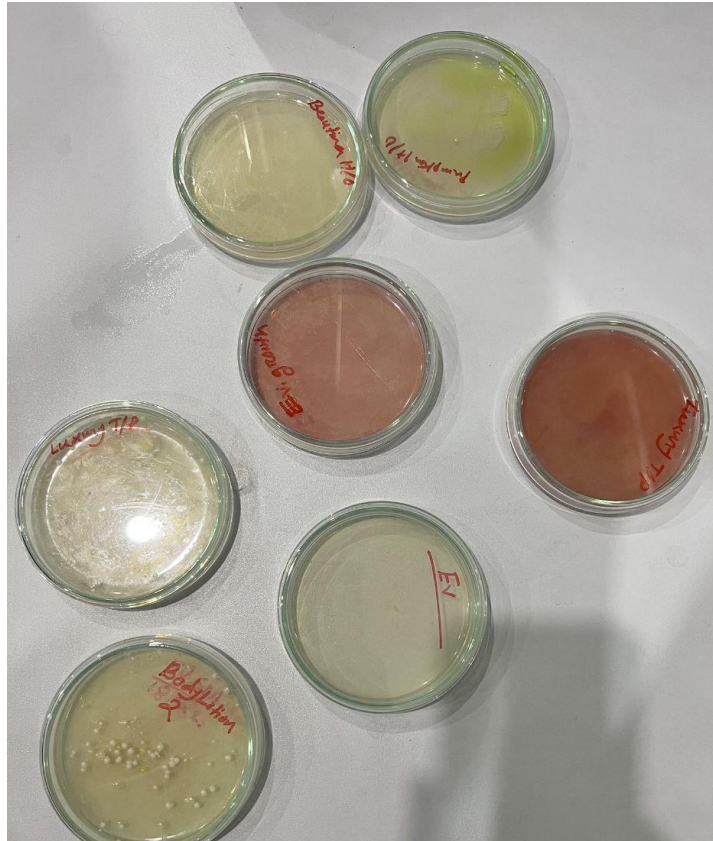


Figure 6.7.3.4: Samples of microbiology lab

6.8 Cause Effect Diagram

Quality problems that QC professionals face most frequently in KCCL are:

- pH level in Cosmetic products
- M&V in soaps

During our attachment, we observed that the above-mentioned quality problems were occurring most frequently and decided to draw cause-effect diagrams for those effects to potentially find the root causes behind those problems.

Cause-effect diagrams, also known as *fishbone diagrams* or *Ishikawa diagrams*, are made to visually represent the relationship between a specific problem or effect and its potential causes. We decided to generate cause-effect diagrams for the three most frequently observed quality problems (Effects) as C&E diagram is a very useful tool for the following:

- **Problem Analysis:** Cause-effect diagrams help identify and analyze the root causes of a problem or effect. By visually mapping out the various potential causes, they facilitate a structured approach to problem-solving and enable teams to delve deeper into the underlying factors.
- **Identifying Interconnections:** These diagrams help reveal the interdependencies and relationships between various causes. By organizing causes into categories such as people, process, equipment, environment, and materials (commonly referred to as the 5 M's), they provide a holistic view of the factors influencing the problem or effect.
- **Focus and Prioritization:** Cause-effect diagrams assist in focusing attention on the most significant causes contributing to the problem or effect. Through discussions and analysis, teams can prioritize the potential causes that require immediate attention and resources for resolution.

6.8.1 Cause Effect Diagram for Noncompliance (M&V)

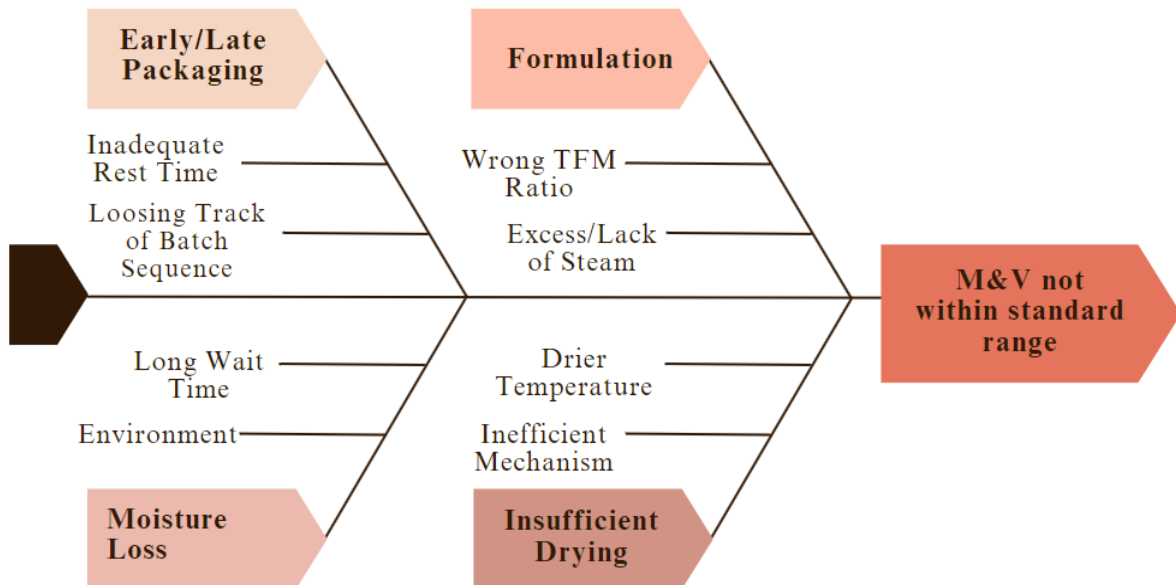


Figure 6.8.1.1: Fishbone Diagram/Ishikawa Diagram/C&E Diagram for M&V level beyond control limits

6.8.2 Cause Effect Diagram for pH nonconformance

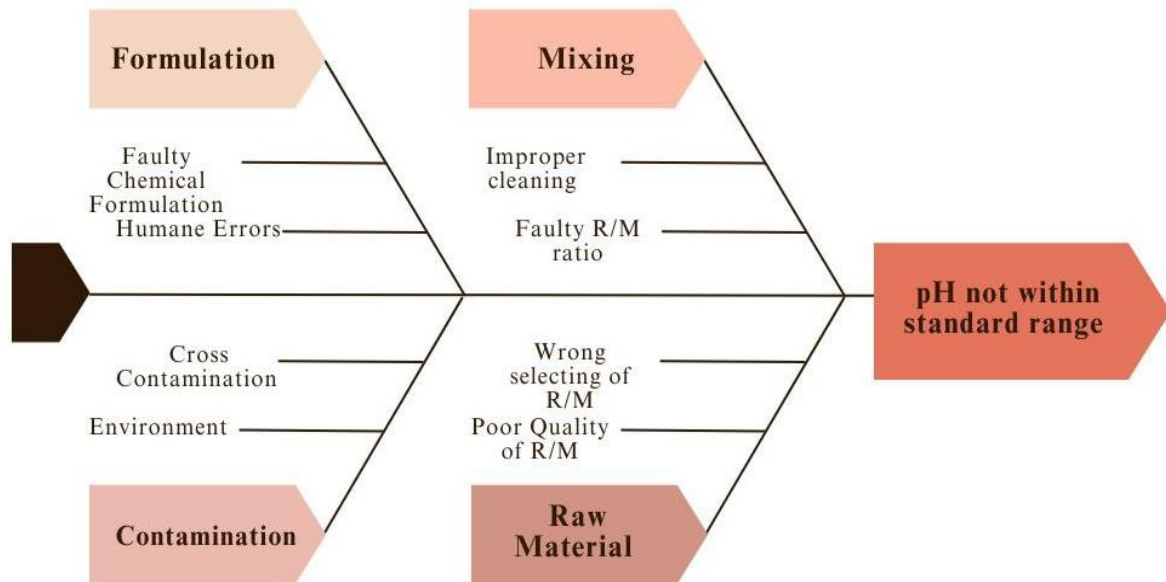


Figure 6.8.2.1: Fishbone Diagram/Ishikawa Diagram/C&E Diagram for M&V level beyond control limits

Chapter 7: Engineering Department

KCCL's maintenance division is in charge of assuring the efficient running of its facilities, machinery, and equipment. A maintenance department's major duties may involve:

- **Preventive Maintenance:** To avoid breakdowns and extend the lifespan of equipment and machinery, the department develops and conducts a preventive maintenance program that involves routinely inspecting, servicing, and maintaining these items.
- **Equipment Repairs:** Equipment, machinery, and system repairs are handled by the maintenance division. They carry out required repairs or arrange with outside service providers for repairs after troubleshooting, diagnosing, and carrying out issues.
- **Routine Maintenance:** To guarantee optimum performance and avoid unplanned breakdowns, the maintenance staff regularly inspects, cleans, lubricates, and calibrates machinery and equipment.
- **Complex Maintenance:** This division is in charge of preserving the general structural integrity, plumbing, HVAC (heating, ventilation, and air conditioning), lighting, and other utilities of the industrial complex.
- **Management of Spare Parts:** The maintenance division keeps an inventory of consumables and essential spare parts required for repairs and replacements. To guarantee that the necessary parts are available, they collaborate with teams from the supply chain or procurement.
- **Safety Inspections:** To make sure that machinery and equipment are in line with safety norms and standards, the department performs safety inspections. They recognize potential risks, put safety controls in place, and instruct staff on how to use equipment safely.
- **Documentation and Record-Keeping:** The maintenance division keeps thorough records of all maintenance tasks, including service plans, repairs, inspections, and equipment histories. This aids in keeping track of maintenance expenses, spotting reoccurring problems, and organizing upcoming maintenance tasks.
- **Emergency Response:** The maintenance department responds quickly to emergencies or equipment failures to reduce downtime. Critical repairs are given priority, and they work with other departments to guarantee a speedy conclusion.

- **Continuous Improvement:** To increase efficiency, dependability, and cost-effectiveness, the maintenance department takes part in monitoring equipment performance, finding areas for improvement, and recommending upgrades or adjustments.
- **Training and Development:** The department may offer instruction to staff members on how to operate equipment, how to do maintenance tasks, and how to follow safety regulations. To improve their performance, they stay current with technical developments and industry best practices.

Chapter 8: Projects

8.1 Project 1: Mezzanine Floor

8.1.1 Introduction

A mezzanine floor is an intermediate floor structure built within an existing space. It is typically constructed between the main floor and the ceiling, effectively utilizing vertical space to create additional usable floor area.

Some key points about mezzanine floors:

1. **Purpose:** Mezzanine floors are used to increase the available floor space in a building without the need for major renovations or expansions. They provide a cost-effective solution for optimizing vertical space.
2. **Construction:** Mezzanines are typically constructed using different types of steel or aluminum structural components. The component is chosen so that the structure is strong, durable, and capable of supporting heavy loads. However, in rare cases, the floor surface can be made of various materials such as plywood and wood. Mezzanines made of metal grating, or concrete are also very common.
3. **Types:** Mezzanine floors can be designed in different configurations according to the requirements. Some common types of mezzanines are free-standing mezzanines, rack-supported mezzanines, and shelving-supported mezzanines. The choice of type depends on factors such as available space, load-bearing capacity, and integration with existing structures.
4. **Uses:** Mezzanines are used in various sectors, including industrial warehouses, retail stores, offices, and even residential buildings. Often prefabricated mezzanines are assembled on-site, minimizing disruption to ongoing operations. Mezzanine floors offer versatile space utilization options. They can be used for additional storage, office space, production areas, showrooms, or retail displays. Mezzanines are also commonly used to create multi-level platforms in venues like theaters or stadiums. By constructing mezzanines, the need for new construction can be eliminated. They also offer flexibility for future modifications. Mezzanines can improve workflow efficiency by separating different functions or providing easy access to stored items.

5. **Safety Considerations:** Safety is a crucial factor for designing and using mezzanine floors. Guardrails or handrails are typically installed around the edges to prevent falls. Adequate lighting, fire protection measures, and proper load calculations are essential for maintaining a safe environment.
6. **Professional Design:** It is advisable to involve experienced architects, engineers, or specialized mezzanine manufacturers when planning and installing a mezzanine floor. There are 3rd party construction firms who specifically construct mezzanine floors in different industries assessing the structural requirements, providing customized designs, and ensuring compliance with building codes.
7. **Maintenance:** Regular maintenance is important to ensure the longevity and safety of a mezzanine floor. Inspections should be conducted to identify any structural issues, damaged components, or potential hazards. Proper care and maintenance can extend the lifespan of the mezzanine and keep it in optimal condition.

Overall, mezzanine floors offer an effective way to maximize vertical space, increase usable floor area, and optimize the layout of a building.

8.1.2 Our finding

We have observed that KCCL currently has an untapped opportunity to optimize their storeroom by utilizing the vertical space effectively. Properly utilizing the vertical space in the storeroom can improve inventory management, allowing for better tracking of stock levels and reducing the chances of stockouts or overstocking. By making better use of the vertical space, the need for costly expansions or relocation to larger facilities can be potentially avoided. Utilizing the vertical space will also allow the company to store more inventory or equipment, leading to increased efficiency and productivity. It will also improve accessibility, making it easier for employees to locate and retrieve items quickly.

The current barriers behind the underutilization of the vertical space are:

- According to BSTI, stacking of more than 10 boxes vertically is not allowed as that may affect the quality of the products inside the boxes of the bottom level.
- Retrieving boxes from the top would be difficult if the height is significantly high.

To solve this problem, we have assessed the feasibility of implementing vertical storage solutions and developed a plan to optimize the storeroom, taking into consideration factors such as cost, space utilization, and operational requirements. By implementing vertical storage solutions such as racks, shelves, or mezzanines, the company can maximize the available space without expanding the storeroom's footprint.

8.1.3 Our Proposed idea

The Mezzanine Floor we proposed will be equipped with the following parts:

Inclined Steel Pathway: An inclined steel path will be used for unloading the boxes. This will employ gravity so no external power source will be needed to unload the boxes from the mezzanine floor.

Vertical Elevator: The soap store of KCCL is already equipped with a one-way vertical elevator. The elevator is used to transfer the packages from the production floor (2nd floor) to the storeroom (1st floor). We plan to use a stopper in the existing elevator to transfer packages from the production floor to the mezzanine floor. In this setup, Soaps produced on the 2nd floor (Toilet soaps) will be stored on the mezzanine floor and soaps produced on the 1st floor (Laundry soaps) will be stored on the main storage floor.

Spiral Stair: A spiral stair will be attached to the mezzanine floor for the workers. Spiral stairs are compact and thus will maximize available storage space.

8.1.4 Solid Work Design of Mezzanine Floor



Figure 8.1.4.2: Side view

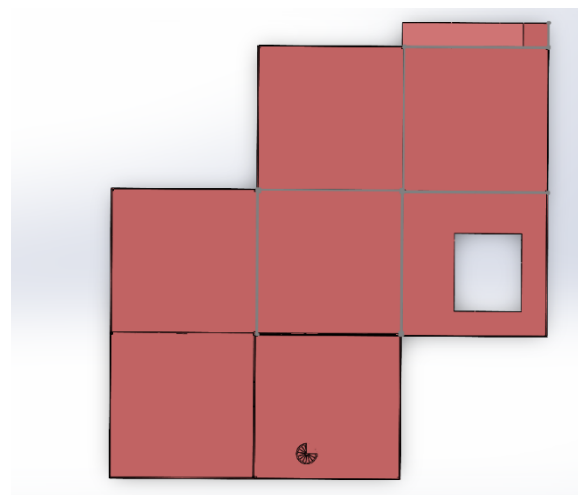


Figure 8.1.4.1: Top view

8.1.5 Cost Analysis

Detailed cost for the Mezzanine Floor is as follows:

Table 8.1.5.1: Cost Analysis of mezzanine Floor

SL No.	Description	Unit	Quantity	Rate (Tk.)	Amount (Tk)
	STEEL WORKS	Sq.ft	5854.00		
1	Main Members (Column, Beam, Rafter, Base Plate, Joint Plate, End Plate, Connection Plate, Purlin Cleat, Strut Member) etc. (40 Ksi Steel)	kg	27223.00	143.00	3892889.00
2	Decking Panel (0.70mm)	kg	4846.00	145.00	702670.00
3	Share Bolt	kg	636.00	180.00	114480.00
4	Edge Capping (0.70mm)	R.ft	197.00	95.00	18715.00
5	Chemical Bolt	Pcs	134.00	400.00	53600.00
6	Nut Bolt	kg	333.00	210.00	69930.00
7	Erection Charge	kg	32401.00	8.00	259208.00
8	Paint (2 Coat)	kg	32401.00	6.00	194406.00
9	Stair	kg	520.00	150.00	78000.00
10	Transportation	LS	32401.00	2.00	64802.00
Sub - Total Amount (Tk) =					5448700.00

8.1.6 Benefits of Installing Mezzanine Floor

Some benefits of mezzanine floor are given below:

- 1. Expanded Storage Capacity:** One of the primary advantages of installing a Mezzanine floor is the significant increase in storage capacity for the soap section. This additional space will cater to the department's growing need for storage, allowing for higher inventory levels. With more stock readily available, the production process can be bolstered, leading to increased output and ultimately, higher profits.
- 2. Enhanced Efficiency in Production Room:** Presently, a considerable portion of the production floor is occupied by storing finished products, leading to congestion and difficulty in maneuvering within the workspace. By implementing a Mezzanine floor, this challenge can be effectively addressed. The Mezzanine floor will provide designated storage areas, freeing up valuable space in the production room. This optimized layout will enhance workflow and overall efficiency, making the production processes smoother and more streamlined.
- 3. Improved Heat Release and Ventilation:** The limited room in the production floor hampers the heat-releasing effectiveness of the machines. This can negatively impact both the performance of the equipment and the working conditions for employees. However, with the installation of a Mezzanine floor, there will be increased airflow and ventilation in the production area. As a result, the machines' heat can be efficiently released into the outer environment, leading to better working conditions and prolonged machine lifespan.
- 4. Optimal Resource Utilization:** The implementation of a Mezzanine floor demonstrates a commitment to optimizing resource usage within the company. By utilizing the full potential of the inventory room through this strategic upgrade, the company ensures that every available square footage is effectively employed. This forward-thinking approach to resource management will undoubtedly yield long-term benefits, contributing to improved productivity and cost-efficiency.

Overall, the installation of a Mezzanine floor brings forth a range of benefits that positively impact various aspects of the company's operations. From increased storage capacity and improved production efficiency to better working conditions and enhanced resource utilization, this decision represents a smart investment that can significantly contribute to the company's growth and success in the competitive market.

Project 2: Incorporating Excel and Power BI

8.1.7 Introduction

During our industrial attachment at “Kohinoor Chemicals Company LTD”, we had the opportunity to visit and study their “Quality Control” Department. They conduct the necessary chemical experiments to figure out the necessary values of various parameters to check the quality of the soaps. The testing facility was top notch to maintain and test soaps quality. But there was a major problem in their system of handling test data.

8.1.8 Our finding

What we noticed; they write down the test data in a record book. And when we asked them about using a computer to store the data for further analysis, we got to know that they don't input their raw data in computer. This seemed like a huge drawback to us. Because in the stated method they will be able to check the quality of a soap, but they will not be able to check the stability or capability of their production process.

8.1.9 Motivation

In our academic education we studied quality management. There we learned that a capable and stable production process can make a production system or industry far more efficient with its present resources. And for a large number of products and a large number of parameters it is insane to analyze all these data manually. Moreover, with the stated method it was not possible for the company to know their process's capability or stability. Thus, it seemed like a serious problem to be solved.

8.1.10 Our proposed idea

The solution lay in integrated use of two software, Microsoft Excel, and Microsoft Power BI. At first, we built some excel forms for them, so that they can entry their data without any hassle. While creating the form our primary focus was to make it easy for them to entry the data in an efficient manner. After that, all the rest of the work will be handled by Power BI.

Power BI will read these files. And we have set it in such a manner that after reading the file it will generate the necessary graphs (CUSUM chart, Individual Measurement Graph, MR graph etc.) automatically, just with a click of a button.

Moreover, this system is very easy to use. All they will need to do is to enter their data, in the excel form and press “Refresh” in Power BI. And all the charts along with necessary insights will be updated automatically.

8.1.11 Features

8.1.11.1 Easy data entry

While creating the system one of the main focuses was given on the input method. Because the only work user has to do is to give the inputs. That is why we tried to make the data input process as easy as we could. Here is the form that is used for entering the data.

8.1.11.2 Data Input

The form presented in “Appendix A as figure A1” is used to enter the data which were collected during various quality tests. Here raw data can be given directly. The columns are:

- Date: Date of experiment
- Soap: Name of the soap
- Shift: Production shift of the tested soap
- M&V: Percentile of mass of moisture and volatile material
- FA: Percentile of mass of fatty acid material
- NaCL: Percentile of mass of salt (NaCl)
- TFM: Percentile of mass of total fatty material
- Net weight: Net weight of the soap

There are 4 more tables used for data entry. One is data entry for “CUSUSM Chart”. Next 3 are for the fixed values of various parameters set by BSTI and the company for quality assurance. Those forms are given below.

CUSUM input:

The form for “CUSUM” and “Data Input” might look the same. But their characteristics are different. We know that to draw a CUSUM chart we need data from 2 shifts of the same process. In “Data Input” form only the raw data which were collected during daily tests were given. But for “CUSUM” form dedicated data has to be collected for subsequent two shifts. The form is given below as “*figure 2*”. All the parameters of the form are the same as “Data Input” form.

BSTI Input:

This form is divided into two parts. “BSTI_A” and “BSTI_B”. “BSTI_A” is used for toilet soap parameters and “BSTI_B” is used for laundry soap parameters. These forms are given as “*figure 3*” and “*figure 4*”.

Company target:

The company also has a personal target for each parameter. They also try to maintain the amounts within a range for better quality of both the soap and the production process. And thus, the final form for setting the company targets for further analysis. This form is presented as “figure A2 in Appendix A”. USL stands for “Upper Specification Limit”, and LSL stands for “Lower Specification Limit”. The data in this table is kept hidden due to confidentiality.

	A	B	C	D	E	F	G
1			Requirements		Method of test		
	SL No.	Characterstics	Grade 1	Grade 2	Ref to clause no of BDS 406	Ref. to BDS ISO	Ref. to Annex
2							
3	1	Total fatty matter, percent by mass,Min	76	70	9	685	
4	2	Rosin acid, percent by mass of totall fatty matter, Max	2	3	14		
5	3	Free caustic alkali, as sodium hydroxide (NaOH), percent by mass, Max	0.05	0.05	6.2	456	
6	4	Matter insoluble in alcohol, percent by mass,Max	2	3	5		
7	5	Chlorides (as sodium chloride), percent by mass, Max	1	1.5	10		
8	6	Free carbonated alkali, percent by mass, Max	0.5	1	28		
9	7	Lather in ml,Min	280	240		696 A	

Figure 8.1.11.1: BSTI Value for toilet soap

	A	B	C	D	E	F
1		Sl. No.	Requirements			Methods of test
			Type 1 Pure	Type 2 Built*	Type 3 Filled	
2		Characterstics				
3	1	Total fatty matter, percent by mass,Min	61	52	40	9
4	2	Free caustic alkali, as sodium hydroxide (NaOH), percent by mass, Max	0.1	0.2	0.3	6
5	3	Unsaponiated fatty matter, percent by mass, Max	0.5	0.4	1	13
6	4	Chlorides (as sodium chloride), percent by mass, Max	1	1	3	10
7	5	Matter insoluble in alcohol, percent by mass,Max	2.5	15		8
8	6	Matter insoluble in water, percent by mass,Max		10		7
9	7	Titre of total fatty acis, C, Min	33			15
10						

Figure 8.1.11.2: BSTI value for laundry soap

	A	B	C	D	E	F	G	H	I	J	K
1		M & V		FA		NACL		TFM		Net Weight	
2	Soap	USL	LSL	USL	LSL	USL	LSL	USL	LSL	USL	LSL
3	Sandalina										
4	Sandalina Rose										
5	Bactrol Cool										
6	Bactrol Fresh										
7	Tibet Pink										
8	Tibet Blue										
9	Tibet 570										
10	Tibet Ball										
11	Laundry Green										
12	Laundry Blue										

Figure 8.1.11.3: Form for company targets for various parameter

8.1.11.3 Automatic Output

The whole system is made automatic. The only work the user has to do is to enter the necessary data. After that he will just open the “Power BI file and all the necessary analysis and charts will be generated automatically. The user does not have to do any further work to generate these charts or analysis. All he must do is to observe the charts of reports and come to a decision. An example report for Soap: Sandalina, Parameter: TFM is given below.

For example, the first page of the report (*figure 6*) gives the “MR-Chart”, “Individual Measurement Chart” and “CUSUM Chart” along with some basic analytical values like maximum and minimum values of TFM, slandered deviation of TFM and average of TFM.

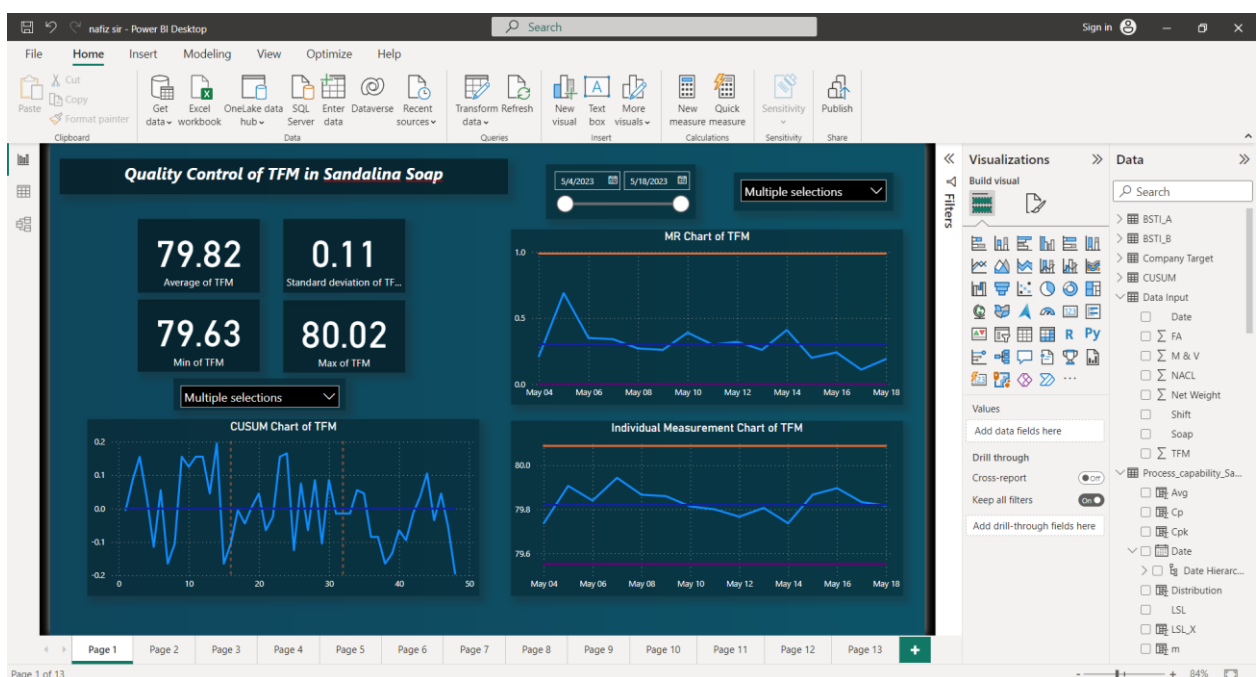


Figure 8.1.11.4: Report on TFM of Sandalina(charts)

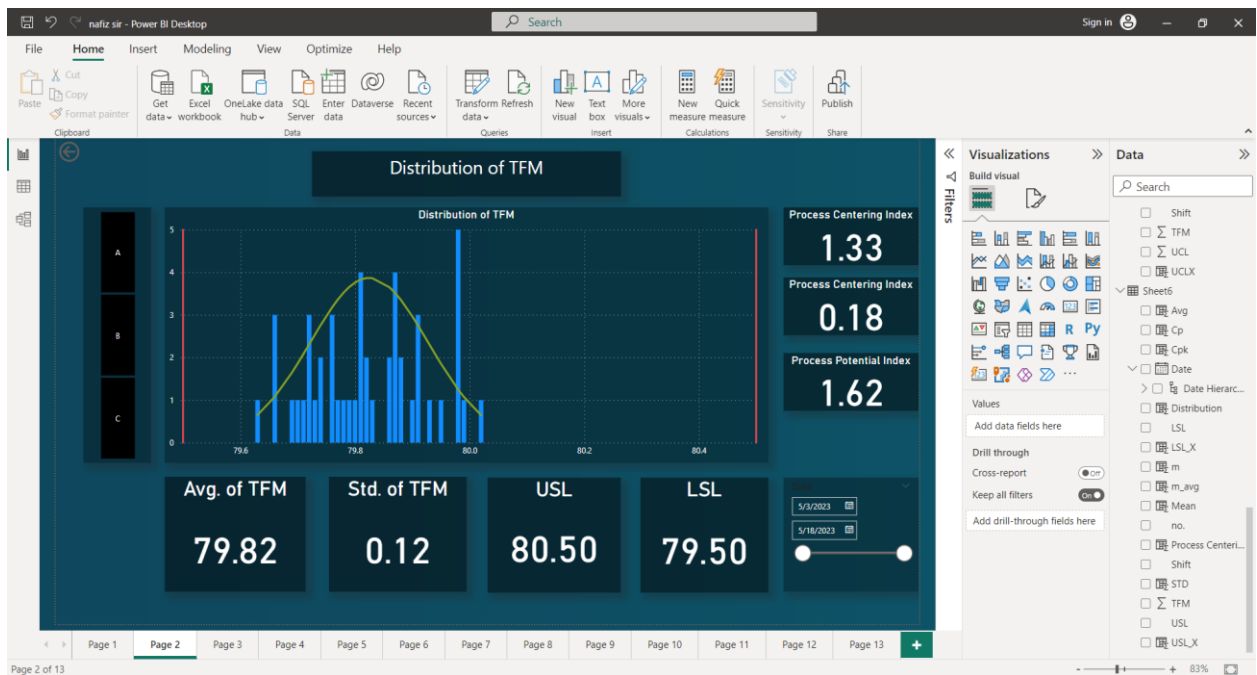


Figure 8.1.11.5: Report on TFM of Sandalina (Capability analysis)

And the second page (*figure 7*) gives the analytical report on process capability analysis.

All these reports of analysis will be done automatically. Because the system has been designed accordingly. How the system is working will be explained in the “Methodology” section.

8.1.11.4 Automatic Updates

The system will even update its report automatically when new data is provided. All the user has to do is to press on the “Refresh” button as indicated in “*figure 8*” and the reports will be updated accordingly.

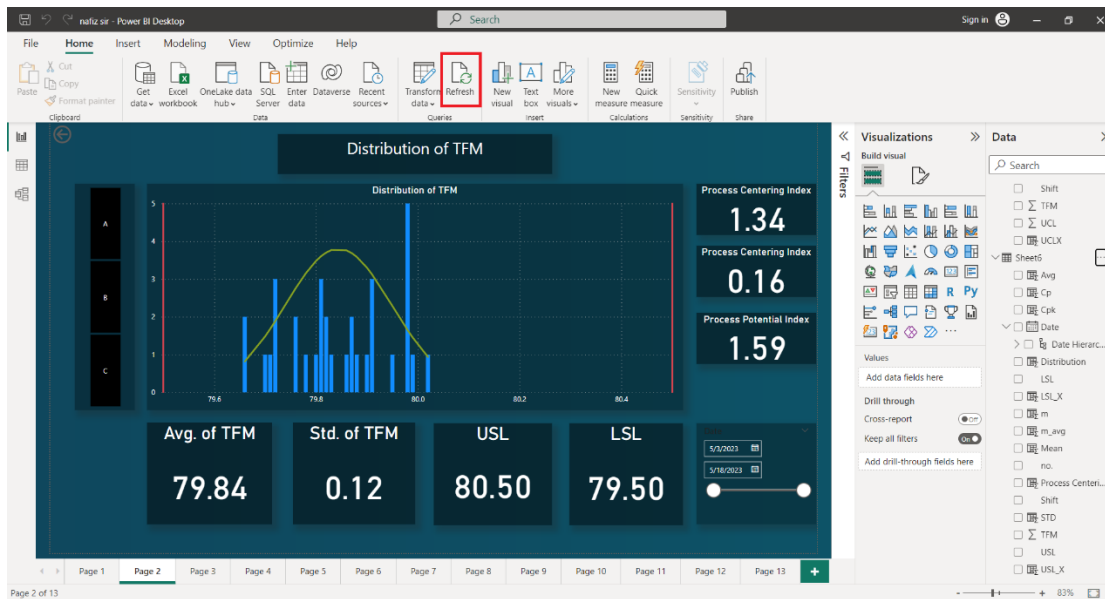


Figure 8.1.11.6: Report with old data

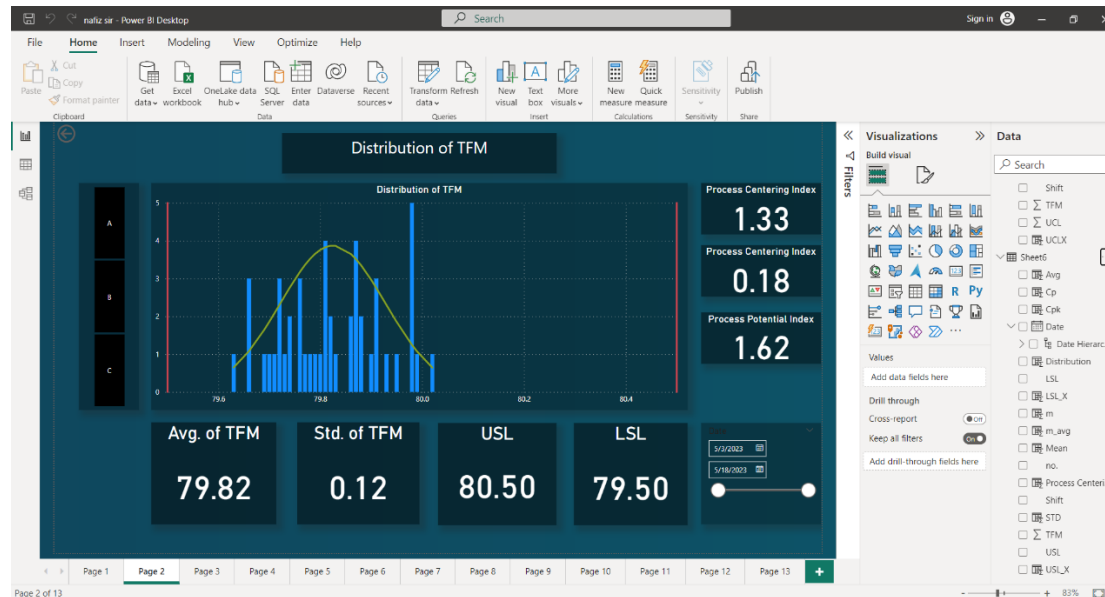


Figure 8.1.11.7: Automatic update with updated data

For example, the report on “figure 8” is generated with less data (36 data points). After updating more data, pressing “Refresh” a new report is generated using the updated data as can be seen in “figure 9”. As claimed the update is done automatically just by pressing “Refresh” once.

8.1.11.5 Easy filtering

All the reports generated in the “Power BI” can be filtered relative to “Date” and “Shift”. More precisely, we can set the range of dates and desired shifts for which we want the reports to be generated.

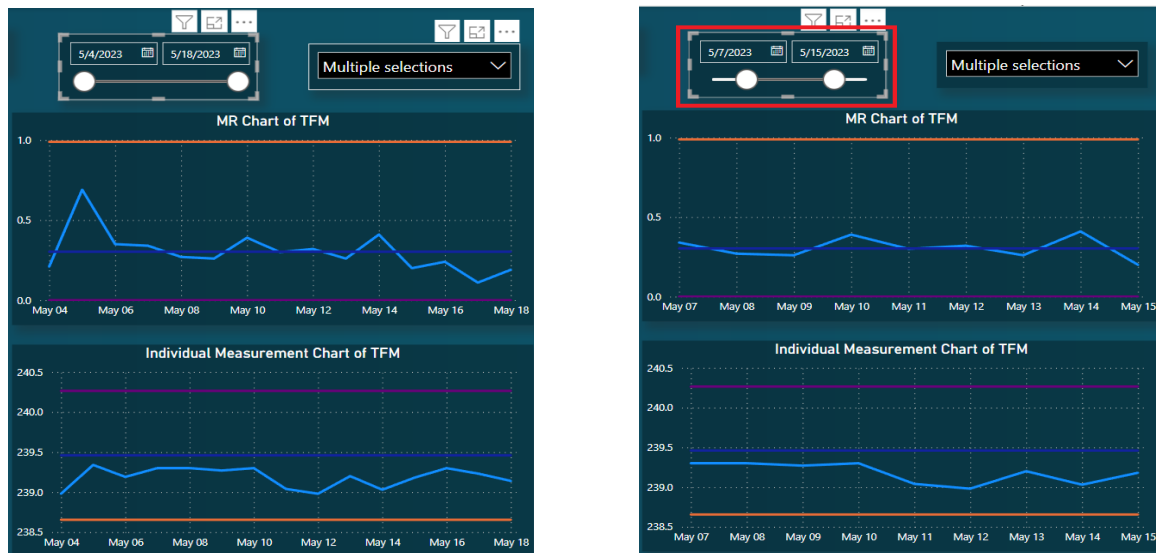


Figure 8.1.11.8: Data filtering with date

For example, in “figure 10” we can see that as the dates were changed in the date filter, at the top of the page, the charts changed according to the dates. And not only charts, but all the analytical results will also change along with that, which can be seen in the following figures (figure 11 and figure 12).

“Figure 11” gives the capability analysis and distribution chart with all the data, as the date filter is ranged for all the available dates. “Figure 12” gives the capability analysis and distribution chart according to the range set by the filter that can be seen in the figure.

In this manner the company will be able to compare their present process performance with their past process performances.

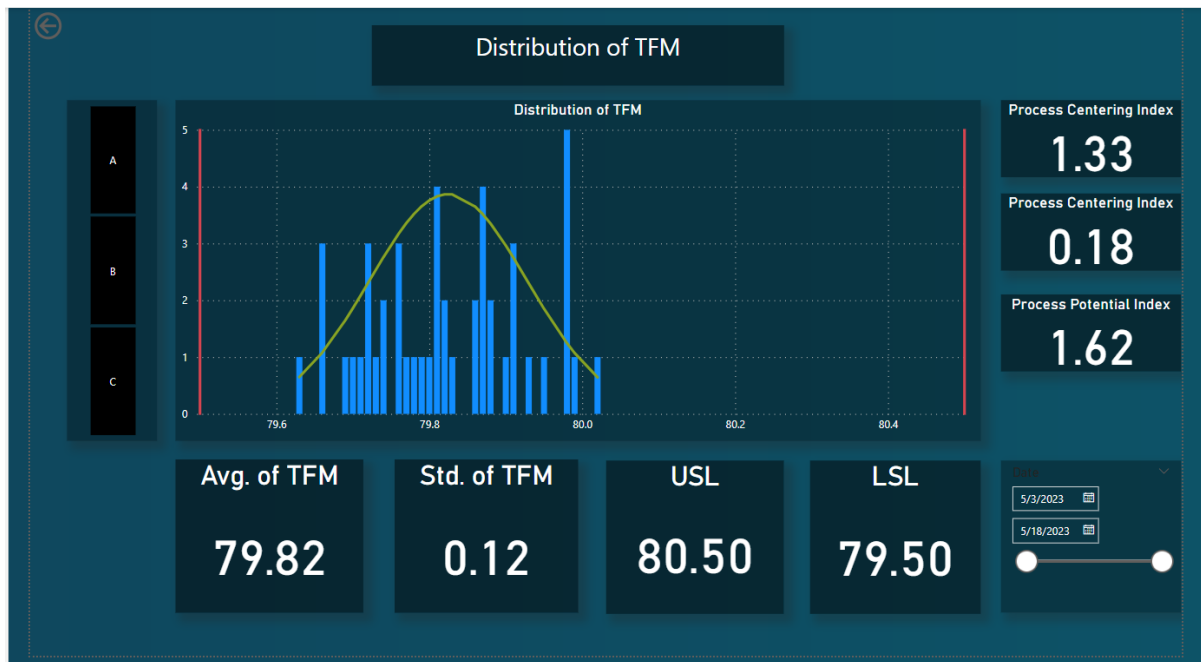


Figure 8.1.11.9: Comparison of filtered data with all the dates

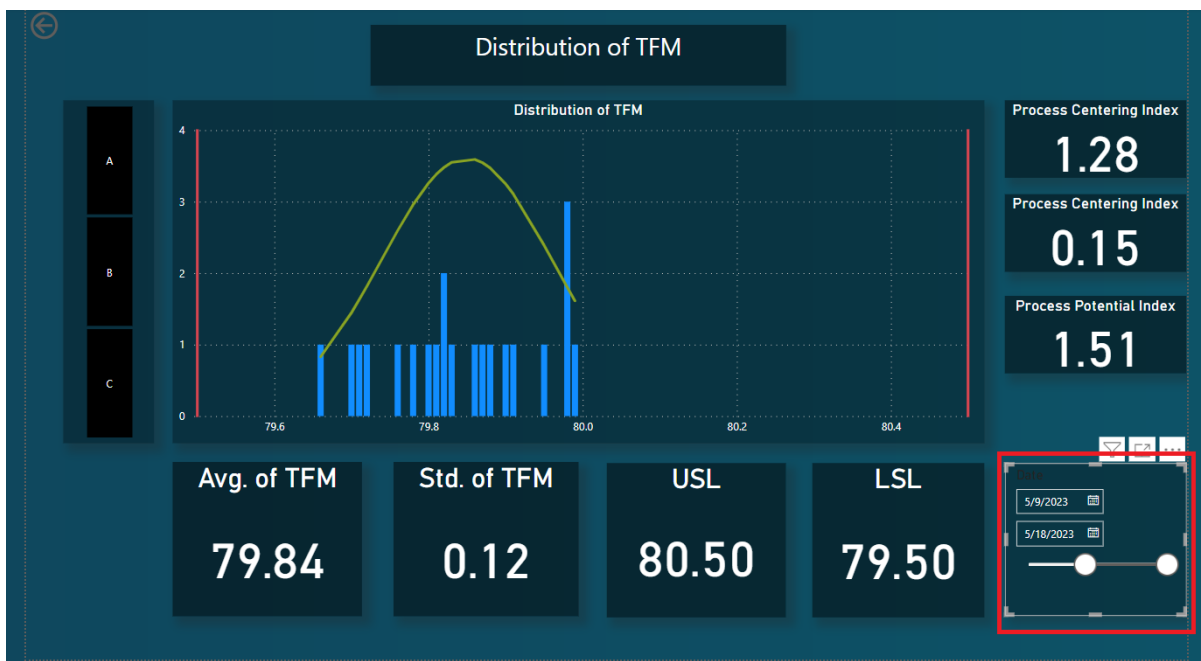


Figure 8.1.11.10: Comparison of filtered data after changing dates

The same attribute is present for shifts. We can filter the reports according to the shifts we want. The following figures will explain it more clearly.

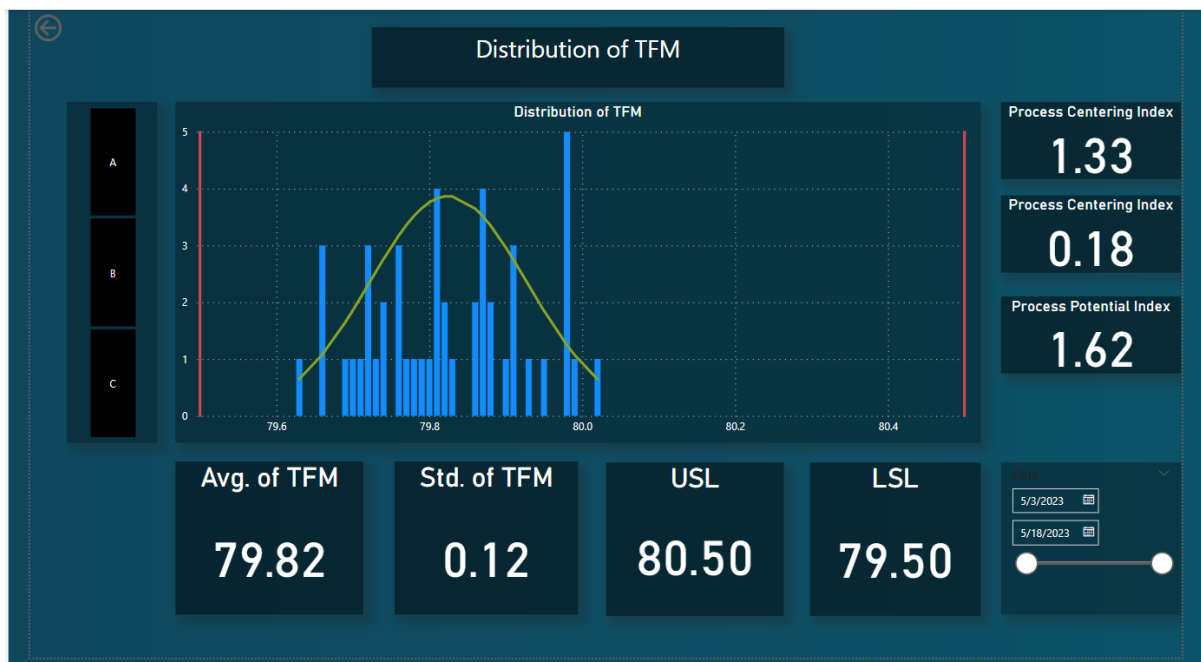


Figure 8.1.11.11: Report with all shifts activated

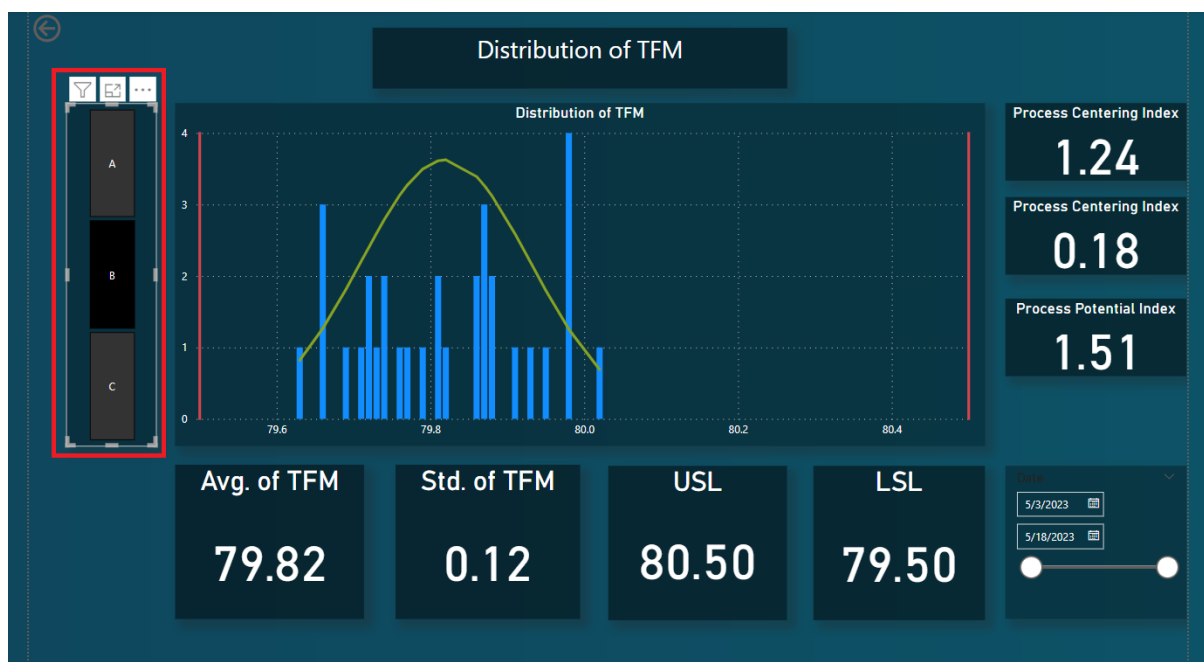


Figure 8.1.11.12: Report with shift filter activated

As we can see by comparing “figure 13” and “figure 14” when shift filter is activated the report is changed according to the shifts selected. In “figure 14” shift A and shift B are selected for

the reports. Same can be done for all the other charts and analysis. The filter position for the other charts are shown below.

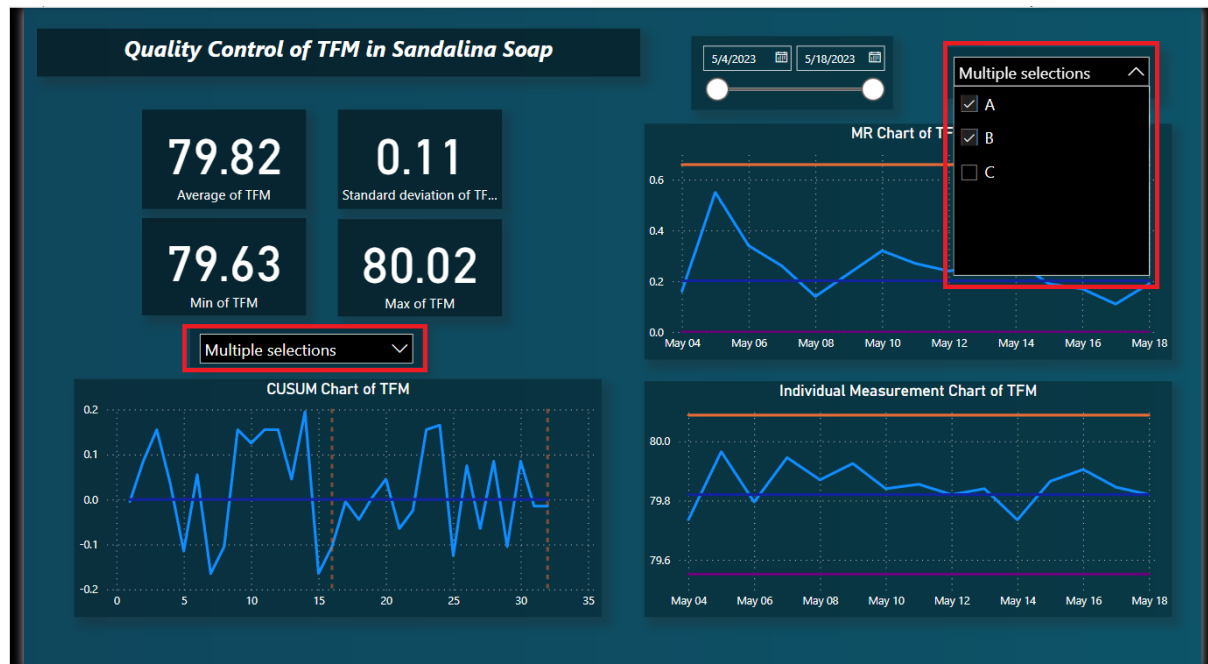


Figure 8.1.11.13: Position of shift filters along with updated charts

By using this filter, the company will be able to analyze whether any shift is facing any problem in the production process.

8.1.12 Methodology

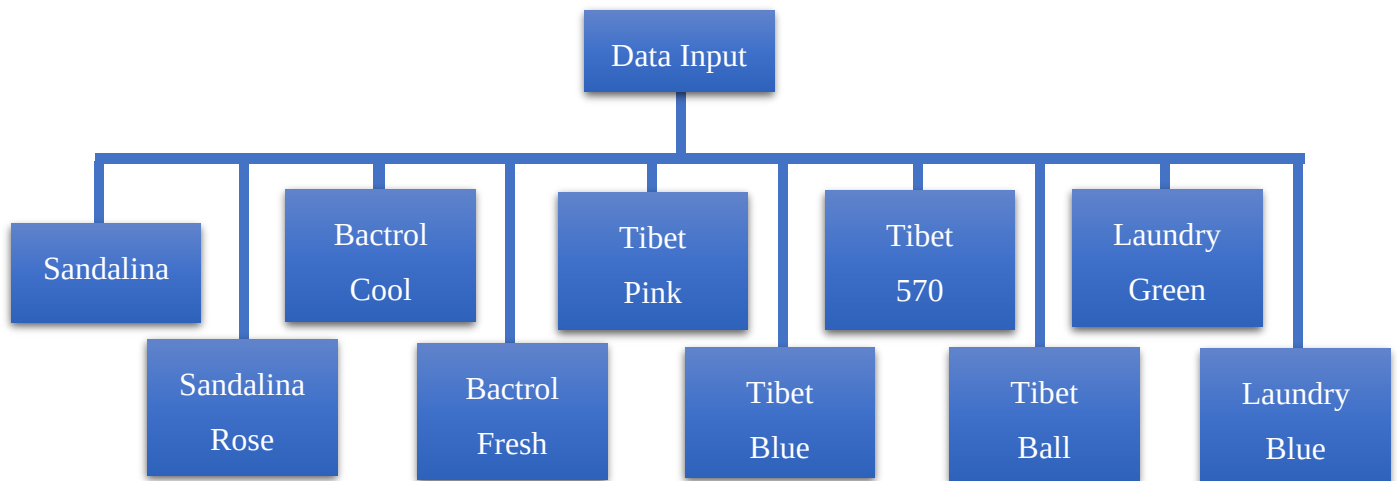


Figure 8.1.12.1: Segregating Soap data

At the very beginning of the “Data Input” table, raw data are segregated into 10 tables according to soap name. The hierarchy can be seen in “figure 16”. To form the soap tables, DAX (Data Analysis Expressions) is used. One example for Sandalina soap, along with DAX and soap table are given in Appendix A as “figure A3”.

After that for each soap, for each parameter, tables are formed to draw MR chart and IM chart using DAX codes.

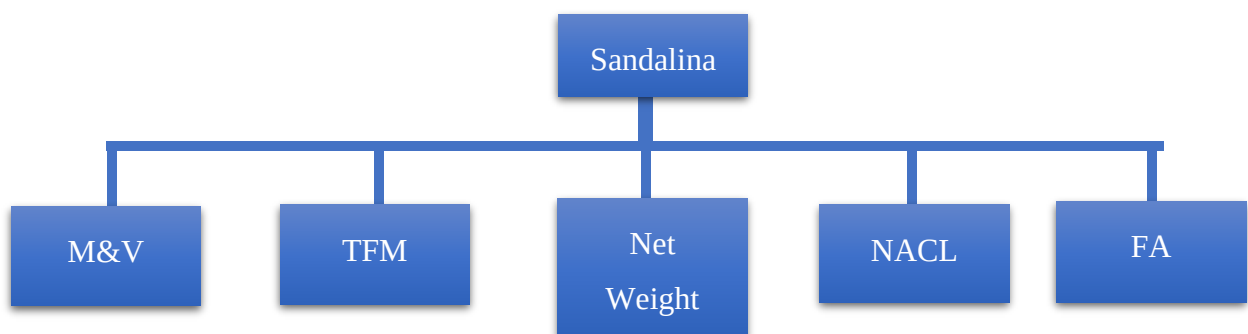


Figure 8.1.12.2: Creation of parameter table

One example of parameter table (TFM) for Sandalina soap is given below in Appendix A as “figure A4”.

In the parameter table all the necessary variables (UCL, LCL, UCLX, LCLX etc. for MR and IM charts) are calculated using DAX codes. In “Paragraph B1 ” in Appendix B code for UCLX (UCL for IM chart) is shown.

Using this table and the corresponding ones the MR and IM charts are drawn for all soaps and parameters.

Next for drawing the distribution and for doing capability analysis a table is formed for each soap’s each parameter fitting a normal curve to the data and calculation all the necessary process capability indexes. To calculate all the necessary indexes of process capability analysis, other necessary values are collected from “BSTI_A”, “BSTI_B”, “Company Target” forms. All the work is done through DAX codes. As an example, the DAX code for creating the “Normal Distribution” is given in “figure A5” in Appendix A.

The DAX code for the distribution is given in “Paragraph B2” in Appendix B.

The code for USLX is given in “Paragraph B3” in Appendix B.

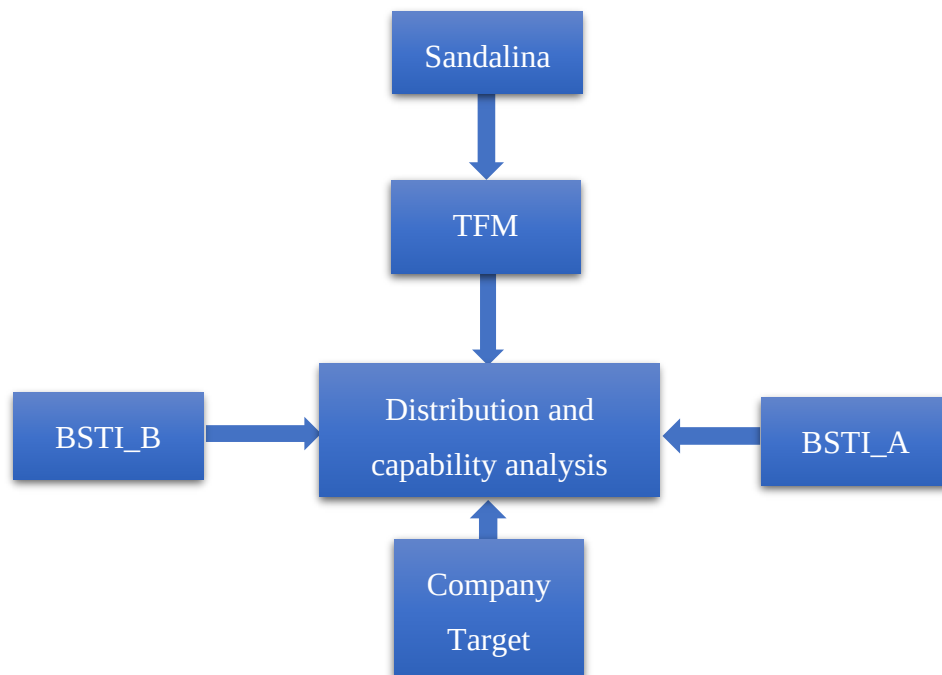


Figure 8.1.12.3: Creation of Distribution and Capability analysis table

Lastly the same is done for the CUSUM chart. From “CUSUM input” for each soap’s separate tables are formed. From each table formed separate tables for parameters are formed. All the other necessary variables are calculated using DAX afterwards. The final table generated from the “CUSUM input” is used to draw the CUSUM chart.

The flowchart of Sandalina’s TFM’s CUSUM is given below along with the final table.

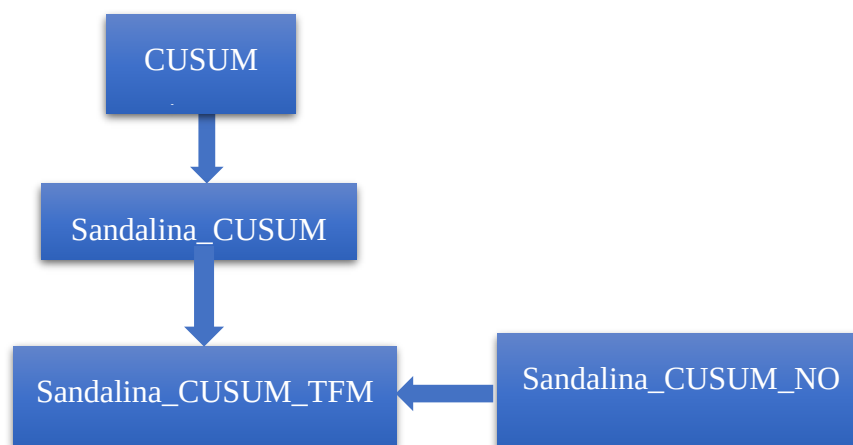


Figure 8.1.12.4: Creation of CUSUM tables

The DAX code used to produce “Sandalina_CUSUM” table is given in “Paragraph B4” in Appendix B.

Code for the table, deviation and CUSUM is given in “Paragraph B5” in Appendix B.

That is how the database is made for one parameter of one soap. The same is repeated for all the others.

Chapter 9: Conclusion

Our industrial attachment at Kohinoor Chemical Co (Bd.) Ltd. has been an invaluable learning experience that has enriched our understanding of the cosmetic manufacturing industry. Over the course of my time here, we have been exposed to various facets of the production process, quality control procedures, and the overall intricacies of creating cosmetic products.

During our attachment, we had the opportunity to work closely with experienced professionals who generously shared their knowledge and expertise. This exposure has not only allowed us to witness real-world applications of theoretical concepts but also provided us with practical insights into problem-solving and decision-making in a fast-paced industrial setting.

Being a part of a dynamic team within the factory has improved our interpersonal and communication skills. We have learned the importance of effective collaboration and the significance of clear communication in ensuring an efficient production process. Additionally, the emphasis on safety protocols and adherence to industry regulations has instilled in us a strong sense of responsibility and professionalism.

The factory environment has exposed us to the importance of sustainable practices in the cosmetics industry.

Throughout this attachment, we have also learned to adapt to various challenges and handle unforeseen situations. Whether it was dealing with production delays or addressing quality issues, these experiences have honed my problem-solving abilities and resilience in a professional setting.

During the internship, we were involved in various projects, from Inventory management to marketing strategies, allowing us to gain a comprehensive understanding of the entire cosmetic production process. Additionally, we had the opportunity to witness the company's commitment to sustainability and ethical practices, which left a lasting impression on us.

Overall, the industrial attachment in the cosmetic factory has been a transformative journey, equipping us with practical skills, industry knowledge, and a deeper appreciation for the intricacies of cosmetics manufacturing. We are grateful for the opportunity to be a part of such a dynamic and innovative industry, and we are confident that the lessons learned during this experience will play a pivotal role in shaping our future career endeavors in the sector.

Appendix A

	A	B	C	D	E	F	G	H
1	Date	Soap	Shift	M & V	FA	NACL	TFM	Net Weight
2	18.05.23	Sandalina`	A	12.44	0	0.56	79.82	75.08
3	18.05.23	Tibet Pink	A	15.34	0.01	0.49	74.46	50.13
4	18.05.23	Tibet 570	A	31.5	0.05	0.47	57.06	126.03
5	18.05.23	Tibel Ball	A	32.36	0.06	0.45	57.47	125.8
6	18.05.23	Sandalina`	C	12.48	0	0.53	79.81	75.36
7	18.05.23	Bactrol Cool	C	15.23	0.01	0.49	74.55	100.46
8	18.05.23	Tibet Pink	C	15.29	0.01	0.5	74.49	50.79
9	18.05.23	Tibet 570	C	32.39	0.07	0.47	56.23	127.79
10	18.05.23	Tibet Ball	C	33.28	0.08	0.46	56.59	126.32
11	17.05.23	Tibet Pink	A	15.2	0.01	0.5	74.57	26.1
12	17.05.23	Sandalina`	B	12.53	0	0.52	79.78	75
13	17.05.23	Bactrol Cool	B	15.19	0.01	0.8	77.35	100
14	17.05.23	Tibet Pink	B	15.2	0.01	0.51	77.26	50
15	17.05.23	Tibet 570	B	32.61	0.06	0.47	61.3	125
16	17.05.23	Tibet Ball	B	33.16	0.07	0.46	60.83	125
17	17.05.23	Sandalina`	C	12.4	0	0.52	79.88	75.1
18	17.05.23	Bactrol Fresh	C	15.36	0.01	0.48	74.37	30.16
19	17.05.23	Tibet Ball	C	33.43	0.07	0.47	55.24	125.89
20	16.05.23	Sandalina`	A	12.3	0	0.52	79.98	75.8
21	16.05.23	Bactrol Cool	A	15.3	0.01	0.49	74.49	100.1
22	16.05.23	Tibet Pink	A	15.4	0.01	0.48	74.41	50
23	16.05.23	Tibet 570	A	32.6	0.07	0.44	56.06	125
24	16.05.23	Tibet Ball	A	33.1	0.06	0.46	56.77	125.5
25	16.05.23	Laundry Green	A	32.36	0.06	0.4	54.21	125.36
26	16.05.23	Sandalina	B	12.46	0	0.52	79.83	75.3
27	16.05.23	Bactrol Cool	B	15.4	0.01	0.5	74.39	100
28	16.05.23	Tibet Pink	B	15.3	0.01	0.49	74.49	50
< > <u>Data Input</u>			CUSUM	BSTI_A	BSTI_B	Company Target		+
Ready  Accessibility: Investigate								

Figure A 1: Data entry form

	A	B	C	D	E	F	G	H
1	Date	Soap	Shift	M & V	FA	NACL	TFM	Net Weight
2	18.05.23	Sandalina	A	12.44	0	0.56	79.82	75.08
3	18.05.23	Sandalina	A	12.38	0	0.51	79.84	125
4	18.05.23	Sandalina	A	12.3	0	0.52	79.98	75.8
5	18.05.23	Sandalina	A	12.43	0	0.51	79.86	75.83
6	18.05.23	Sandalina	A	12.58	0	0.53	79.71	75.58
7	18.05.23	Sandalina	A	12.38	0	0.57	79.88	75
8	18.05.23	Sandalina	A	12.63	0	0.54	79.66	125.1
9	18.05.23	Sandalina	A	12.56	0	0.54	79.72	126.1
10	18.05.23	Sandalina	A	12.3	0	0.52	79.98	125
11	18.05.23	Sandalina	A	12.32	0	0.5	79.95	100.36
12	18.05.23	Sandalina	A	12.3	0	0.52	79.98	100
13	18.05.23	Sandalina	A	12.3	0	0.52	79.98	125.2
14	18.05.23	Sandalina	A	12.38	0	0.56	79.87	100.02
15	18.05.23	Sandalina	A	12.28	0	0.51	80.02	75.3
16	18.05.23	Sandalina	A	12.63	0	0.54	79.66	100.02
17	18.05.23	Sandalina	A	12.56	0	0.54	79.72	100.08
18	18.05.23	Sandalina	A	12.51	0	0.53	79.92	125.6
19	18.05.23	Sandalina	B	12.48	0	0.53	79.84	75
20	18.05.23	Sandalina	B	12.53	0	0.52	79.78	75.3
21	18.05.23	Sandalina	B	12.46	0	0.52	79.83	125.6
22	18.05.23	Sandalina	B	12.56	0	0.51	79.91	126.12
23	18.05.23	Sandalina	B	12.54	0	0.52	79.76	75.79
24	18.05.23	Sandalina	B	12.39	0	0.53	79.8	125.5
25	18.05.23	Sandalina	B	12.3	0	0.52	79.98	75.31
26	18.05.23	Sandalina	B	12.28	0	0.5	79.99	100.12
27	18.05.23	Sandalina	B	12.58	0	0.54	79.7	100
28	18.05.23	Sandalina	B	12.38	0	0.52	79.9	100.25
< > Data Input				CUSUM	BSTI_A	BSTI_B	Company Target	

Figure A 2: Data entry for CUSUM

		1 Sandalina Details = CALCULATE('Data Input', 2 'Data Input'[Soap]="Sandalina")]					
	Date	M&V	FA	NACL	TFM	NW	Shift
	Thursday, May 18, 2023	12.44	0	0.56	79.82	75.08	A
	Wednesday, May 17, 2023	12.4	0	0.547	79.91	75.8	A
	Tuesday, May 16, 2023	12.3	0	0.52	79.98	75.8	A
	Monday, May 15, 2023	12.43	0	0.51	79.86	75.83	A
	Sunday, May 14, 2023	12.58	0	0.53	79.71	75.58	A
	Saturday, May 13, 2023	12.38	0	0.57	79.88	75	A
	Friday, May 12, 2023	12.63	0	0.54	79.66	125.1	A
	Thursday, May 11, 2023	12.56	0	0.54	79.72	126.1	A
	Wednesday, May 10, 2023	12.3	0	0.52	79.98	125	A
	Tuesday, May 9, 2023	12.32	0	0.5	79.95	100.36	A
	Monday, May 8, 2023	12.3	0	0.52	79.98	100	A
	Sunday, May 7, 2023	12.3	0	0.52	79.98	125.2	A
	Saturday, May 6, 2023	12.38	0	0.56	79.87	100.02	A
	Friday, May 5, 2023	12.28	0	0.51	80.02	125.4	A
	Thursday, May 4, 2023	12.63	0	0.54	79.66	100.02	A
	Wednesday, May 3, 2023	12.56	0	0.54	79.72	100.08	A
	Thursday, May 18, 2023	12.5	0	0.53	79.82	125.3	B
	Wednesday, May 17, 2023	12.53	0	0.52	79.78	75	B
	Tuesday, May 16, 2023	12.46	0	0.52	79.83	75.3	B
	Monday, May 15, 2023	12.49	0	0.52	79.87	75.4	B
	Sunday, May 14, 2023	12.54	0	0.52	79.76	126.12	B
	Saturday, May 13, 2023	12.39	0	0.53	79.8	75.79	B
	Friday, May 12, 2023	12.3	0	0.52	79.98	125.5	B
	Thursday, May 11, 2023	12.28	0	0.5	79.99	75.31	B
	Wednesday, May 10, 2023	12.58	0	0.54	79.7	100.12	B
	Tuesday, May 9, 2023	12.38	0	0.52	79.9	100	B
	Monday, May 8, 2023	12.54	0	0.52	79.76	100.25	B

Figure A 3: Segregation of Soap table with its DAX code

1 UCLX = Sandalina_TFM[CLX]+(3*(Sandalina_TFM[MR_AVG]/1.128))													
No.	MR_TFM	MR_AVG	UCL	LC L	No.1	CLX	LCLX	TFM	Date	UCLX	Shift		
1	0.0900000000000034	0.1008888888888888	0.3296039999999998	0	1	79.82	79.5516784869976	79.82	Thursday, May 18, 2023	80.0883215130024	A		
2	0.0700000000000074	0.1008888888888888	0.3296039999999998	0	2	79.82	79.5516784869976	79.91	Wednesday, May 17, 2023	80.0883215130024	A		
3	0.1200000000000005	0.1008888888888888	0.3296039999999998	0	3	79.82	79.5516784869976	79.98	Tuesday, May 16, 2023	80.0883215130024	A		
4	0.1500000000000006	0.1008888888888888	0.3296039999999998	0	4	79.82	79.5516784869976	79.86	Monday, May 15, 2023	80.0883215130024	A		
5	0.1700000000000002	0.1008888888888888	0.3296039999999998	0	5	79.82	79.5516784869976	79.71	Sunday, May 14, 2023	80.0883215130024	A		
6	0.2199999999999999	0.1008888888888888	0.3296039999999998	0	6	79.82	79.5516784869976	79.88	Saturday, May 13, 2023	80.0883215130024	A		
7	0.0600000000000023	0.1008888888888888	0.3296039999999998	0	7	79.82	79.5516784869976	79.66	Friday, May 12, 2023	80.0883215130024	A		
8	0.2600000000000005	0.1008888888888888	0.3296039999999998	0	8	79.82	79.5516784869976	79.72	Thursday, May 11, 2023	80.0883215130024	A		
9	0.0300000000000011	0.1008888888888888	0.3296039999999998	0	9	79.82	79.5516784869976	79.98	Wednesday, May 10, 2023	80.0883215130024	A		
10	0.0300000000000011	0.1008888888888888	0.3296039999999998	0	10	79.82	79.5516784869976	79.95	Tuesday, May 9, 2023	80.0883215130024	A		
11	0	0.1008888888888888	0.3296039999999998	0	11	79.82	79.5516784869976	79.98	Monday, May 8, 2023	80.0883215130024	A		
12	0.1099999999999999	0.1008888888888888	0.3296039999999998	0	12	79.82	79.5516784869976	79.98	Sunday, May 7, 2023	80.0883215130024	A		
13	0.1499999999999991	0.1008888888888888	0.3296039999999998	0	13	79.82	79.5516784869976	79.87	Saturday, May 6, 2023	80.0883215130024	A		
14	0.3599999999999999	0.1008888888888888	0.3296039999999998	0	14	79.82	79.5516784869976	80.02	Friday, May 5, 2023	80.0883215130024	A		
15	0.0600000000000023	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.66	Thursday, May 4, 2023	80.0883215130024	A		
16	0.0999999999999943	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.82	Thursday, May 18, 2023	80.0883215130024	B		
17	0.0399999999999992	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.78	Wednesday, May 17, 2023	80.0883215130024	B		
18	0.0499999999999972	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.83	Tuesday, May 16, 2023	80.0883215130024	B		
19	0.0400000000000063	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.87	Monday, May 15, 2023	80.0883215130024	B		
20	0.1099999999999999	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.76	Sunday, May 14, 2023	80.0883215130024	B		
21	0.0399999999999992	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.8	Saturday, May 13, 2023	80.0883215130024	B		
22	0.1800000000000007	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.98	Friday, May 12, 2023	80.0883215130024	B		
23	0.00999999999999901	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.99	Thursday, May 11, 2023	80.0883215130024	B		
24	0.2899999999999992	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.7	Wednesday, May 10, 2023	80.0883215130024	B		
25	0.2000000000000003	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.9	Tuesday, May 9, 2023	80.0883215130024	B		
26	0.1400000000000001	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.76	Monday, May 8, 2023	80.0883215130024	B		
27	0.1499999999999991	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.91	Sunday, May 7, 2023	80.0883215130024	B		
28	0.1899999999999998	0.1008888888888888	0.3296039999999998	0	15	79.82	79.5516784869976	79.72	Saturday, May 6, 2023	80.0883215130024	B		

Figure A 4: TFM parameter table with internal DAX command

1 Distribution = NORM.DIST(Process_capability_Sandalina_TFM[TFM], 2 average(Process_capability_Sandalina_TFM[TFM]), 3 STDEV.S(Process_capability_Sandalina_TFM[TFM]), 4 false)													
no.	TFM	Distribution	USL_X	LSL_X	Cp	m	Avg	m_avg	Process Centerig index	Cpk	USL	LSL	Date
1	79.82	3.864496579719	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Thursday, May 18, 2023
2	79.91	2.74980880715397	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Wednesday, May 17, 2023
3	79.98	1.24637674468728	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Tuesday, May 16, 2023
4	79.86	3.64962699141489	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Monday, May 15, 2023
5	79.71	2.08201383205051	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Sunday, May 14, 2023
6	79.88	3.35213996322652	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Saturday, May 13, 2023
7	79.66	1.07901558187261	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Friday, May 12, 2023
8	79.72	2.30846201638101	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Thursday, May 11, 2023
9	79.98	1.24637674468728	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Wednesday, May 10, 2023
10	79.95	1.8511148977902	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Tuesday, May 9, 2023
11	79.98	1.24637674468728	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Monday, May 8, 2023
12	79.98	1.24637674468728	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Sunday, May 7, 2023
13	79.87	3.51420685119571	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Saturday, May 6, 2023
14	80.02	0.64480395039816	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Friday, May 5, 2023
15	79.66	1.07901558187261	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Thursday, May 4, 2023
16	79.72	2.30846201638101	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Wednesday, May 3, 2023
17	79.82	3.864496579719	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Thursday, May 18, 2023
18	79.78	3.52040861101011	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Wednesday, May 17, 2023
19	79.83	3.86373955063445	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Tuesday, May 16, 2023
20	79.87	3.51420685119571	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Monday, May 15, 2023
21	79.76	3.17569683180445	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Sunday, May 14, 2023
22	79.8	3.75846957055291	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Saturday, May 13, 2023
23	79.98	1.24637674468728	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Friday, May 12, 2023
24	79.99	1.07206212588478	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Thursday, May 11, 2023
25	79.7	1.86020355124612	80.5	79.5	1.61621992990255	80	79.8247916666667	0.175208333333333	0.175208333333333	1.33304472968421	85.5	75.5	Wednesday, May 10, 2023

Figure A 5: Process Capability and Distribution table for Sandalina's TFM

X ✓		1 Sandalina_CUSUM = CALCULATETABLE(CUSUM,CUSUM[Soap]="Sandalina")					
Date	Soap	Shift	M & V	FA	NACL	TFM	Net Weight
18.05.23	Sandalina	A	12.44	0	0.56	79.82	75.08
18.05.23	Sandalina	A	12.38	0	0.51	79.84	125
18.05.23	Sandalina	A	12.3	0	0.52	79.98	75.8
18.05.23	Sandalina	A	12.43	0	0.51	79.86	75.83
18.05.23	Sandalina	A	12.58	0	0.53	79.71	75.58
18.05.23	Sandalina	A	12.38	0	0.57	79.88	75
18.05.23	Sandalina	A	12.63	0	0.54	79.66	125.1
18.05.23	Sandalina	A	12.56	0	0.54	79.72	126.1
18.05.23	Sandalina	A	12.3	0	0.52	79.98	125
18.05.23	Sandalina	A	12.32	0	0.5	79.95	100.36
18.05.23	Sandalina	A	12.3	0	0.52	79.98	100
18.05.23	Sandalina	A	12.3	0	0.52	79.98	125.2
18.05.23	Sandalina	A	12.38	0	0.56	79.87	100.02
18.05.23	Sandalina	A	12.28	0	0.51	80.02	75.3
18.05.23	Sandalina	A	12.63	0	0.54	79.66	100.02
18.05.23	Sandalina	A	12.56	0	0.54	79.72	100.08
18.05.23	Sandalina	A	12.51	0	0.53	79.92	125.6
18.05.23	Sandalina	B	12.48	0	0.53	79.84	75
18.05.23	Sandalina	B	12.53	0	0.52	79.78	75.3
18.05.23	Sandalina	B	12.46	0	0.52	79.83	125.6
18.05.23	Sandalina	B	12.56	0	0.51	79.91	126.12
18.05.23	Sandalina	B	12.54	0	0.52	79.76	75.79
18.05.23	Sandalina	B	12.39	0	0.53	79.8	125.5
18.05.23	Sandalina	B	12.3	0	0.52	79.98	75.31
18.05.23	Sandalina	B	12.28	0	0.5	79.99	100.12
18.05.23	Sandalina	B	12.58	0	0.54	79.7	100
18.05.23	Sandalina	B	12.38	0	0.52	79.9	100.25
18.05.23	Sandalina	B	12.54	0	0.52	79.76	125.35

Figure A 6: Sandalina_CUSUM table along with DAX code

X ✓		1 Sandalina_CUSUM_TFM = SELECTCOLUMNS(Sandalina_CUSUM, 2 "DATE",[Date],"TFM",[TFM],"Shift",[Shift],"No.",[No.])					
TFM	no.	Dev	CUSUM	zero line	Shift	Date	
79.82	1	-0.00479166666666231	-0.00479166666666231	0	A	Thursday, May 18, 2023	
79.91	2	0.0852083333333411	0.0804166666666788	0	A	Thursday, May 18, 2023	
79.98	3	0.155208333333348	0.2356250000000027	0	A	Thursday, May 18, 2023	
79.86	4	0.0352083333333439	0.270833333333371	0	A	Thursday, May 18, 2023	
79.71	5	-0.114791666666662	0.156041666666709	0	A	Thursday, May 18, 2023	
79.88	6	0.055208333333334	0.2112500000000049	0	A	Thursday, May 18, 2023	
79.66	7	-0.164791666666659	0.0464583333333906	0	A	Thursday, May 18, 2023	
79.72	8	-0.104791666666657	-0.0583333333332661	0	A	Thursday, May 18, 2023	
79.98	9	0.155208333333348	0.09687500000000824	0	A	Thursday, May 18, 2023	
79.95	10	0.125208333333347	0.222083333333343	0	A	Thursday, May 18, 2023	
79.98	11	0.155208333333348	0.377291666666778	0	A	Thursday, May 18, 2023	
79.98	12	0.155208333333348	0.5325000000000127	0	A	Thursday, May 18, 2023	
79.87	13	0.0452083333333491	0.5777083333333476	0	A	Thursday, May 18, 2023	
80.02	14	0.195208333333341	0.772916666666816	0	A	Thursday, May 18, 2023	
79.66	15	-0.164791666666659	0.6081250000000157	0	A	Thursday, May 18, 2023	
79.72	16	-0.104791666666657	0.5033333333333501	0	A	Thursday, May 18, 2023	
79.82	17	-0.00479166666666231	0.498541666666839	0	B	Thursday, May 18, 2023	
79.78	18	-0.0447916666666544	0.4537500000000184	0	B	Thursday, May 18, 2023	
79.83	19	0.00520833333334281	0.4589583333333527	0	B	Thursday, May 18, 2023	
79.87	20	0.0452083333333491	0.504166666666876	0	B	Thursday, May 18, 2023	
79.76	21	-0.0647916666666504	0.4393750000000226	0	B	Thursday, May 18, 2023	
79.8	22	-0.0247916666666583	0.4145833333333567	0	B	Thursday, May 18, 2023	
79.98	23	0.155208333333348	0.569791666666916	0	B	Thursday, May 18, 2023	
79.99	24	0.165208333333339	0.7350000000000255	0	B	Thursday, May 18, 2023	
79.7	25	-0.124791666666653	0.6102083333333603	0	B	Thursday, May 18, 2023	
79.9	26	0.0752083333333502	0.685416666666953	0	B	Thursday, May 18, 2023	
79.76	27	-0.0647916666666504	0.6206250000000302	0	B	Thursday, May 18, 2023	

Figure A 7: Final CUSUM Table

Appendix B

Paragraph B1:

UCLX = Sandalina_TFM[CLX]+(3*(Sandalina_TFM[MR_AVG]/1.128))

Paragraph B2:

Distribution = NORM.DIST(Process_capability_Sandalina_TFM[TFM],
average(Process_capability_Sandalina_TFM[TFM]),
STDEV.S(Process_capability_Sandalina_TFM[TFM]),
false)

Paragraph B3:

USLX = SELECTCOLUMNS('Company Target',"USLX",[Sandalina TFM USL])

Paragraph B4:

Sandalina_CUSUM = CALCULATETABLE(CUSUM,CUSUM[Soap]="Sandalina")

Paragraph B5:

Sandalina_CUSUM_TFM = SELECTCOLUMNS(Sandalina_CUSUM,
"DATE",[Date],"TFM",[TFM],"Shift",[Shift],"No.",[No.])

Deviation = [TFM]-AVERAGE([TFM])

CUSUM =

CALCULATE(
SUM(Sandalina_CUSUM_TFM[Deviation]),
MAX(Sandalina_CUSUM_TFM_No.[No.])>=Sandalina_CUSUM_TFM_No.[No.]
)

